Development of Flight-Test Performance Estimation Techniques for Small Unmanned Aerial Systems

DISSERTATION

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By

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Abstract

This dissertation provides a flight-testing framework for assessing the performance of fixed-wing, small-scale unmanned aerial systems (sUAS) by leveraging sub-system models of components unique to these vehicles. The development of the sub-system models, and their links to broader impacts on sUAS performance, is the key contribution of this work. The sub-system modeling and analysis focuses on the vehicle's propulsion, navigation and guidance, and airframe components. Quantification of the uncertainty in the vehicle's power available and control states is essential for assessing the validity of both the methods and results obtained from flight-tests. Therefore, detailed propulsion and navigation system analyses are presented to validate the flight testing methodology.

Propulsion system analysis required the development of an analytic model of the propeller in order to predict the power available over a range of flight conditions. The model is based on the blade element momentum (BEM) method. Additional corrections are added to the basic model in order to capture the Reynolds-dependent scale effects unique to sUAS. The model was experimentally validated using a ground based testing apparatus. The BEM predictions and experimental analysis allow for a parameterized model relating the electrical power, measurable during flight, to the power available required for vehicle performance analysis.

Navigation system details are presented with a specific focus on the sensors used for state estimation, and the resulting uncertainty in vehicle state. Uncertainty quantification is provided by detailed calibration techniques validated using quasi-static and hardware-in-the-loop (HIL) ground based testing. The HIL methods introduced use a soft real-time flight simulator to provide inertial quality data for assessing overall system performance. Using this tool, the uncertainty in vehicle state estimation based on a range of sensors, and vehicle operational environments is presented.

The propulsion and navigation system models are used to evaluate flight-testing methods for evaluating fixed-wing sUAS performance. A brief airframe analysis is presented to provide a foundation for assessing the efficacy of the flight-test methods. The flight-testing presented in this work is focused on validating the aircraft drag polar, zero-lift drag coefficient, and span efficiency factor. Three methods are detailed and evaluated for estimating these design parameters. Specific focus is placed on the influence of propulsion and navigation system uncertainty on the resulting performance data.

Performance estimates are used in conjunction with the propulsion model to estimate the impact sensor and measurement uncertainty on the endurance and range of a fixed-wing sUAS. Endurance and range results for a simplistic power available model are compared to the Reynolds-dependent model presented in this work. Additional parameter sensitivity analysis related to state estimation uncertainties encountered in flight-testing are presented. Results from these analyses indicate that the sub-system models introduced in this work are of first-order importance, on the order of 5-10% change in range and endurance, in assessing the performance of a fixed-wing sUAS.

Dedication

This dissertation is dedicated to my family. Their unwavering love and support is the bedrock on which this work is built.

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Finishing a PhD is as much a soul-searching endeavor as it is blood, sweat, and tears. To that end, I would like to thank my committee Dr. Jim Gregory, Dr. Charles Toth, Dr. Cliff Whitfield, and Dr. Jeffery Bons for continually challenging and encouraging me during my tenure at the Ohio State University. I would like to give a special thanks to my advisor Dr. Jim Gregory. At the point my dissertation work literally caught fire and was destroyed, he was there providing exuberant encouragement as I was contemplating ritualistic harikari. Under his guidance we were able to grow from no central UAS capabilities to a fully functioning research venture. He allowed me the intellectual freedom to pursue my interests while subtly (sometimes) keeping me focused on larger milestones. Dr. Gregory also gave me a fundamental understanding of perfection in research. I now have a bar by which to judge any and all future accomplishments.

The mere fact that I was able to tackle a project like UAS from scratch is fundamentally related to my previous mentors at Boise State University. Don Plumlee and Amy Moll in particular provided an environment in which I was able to explore engineering to my heart's content. As a recurring theme, they allowed me to explore my interests while providing the guidance and challenges to be successful in later endeavors.

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Publications

Journal Publications

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Fields of Study

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Chapter 1: Introduction

1.1 Purpose of the proposed research

The purpose of this research is the development of a flight-test based performance estimation methodology for fixed-wing small-scale unmanned aerial systems (sUAS). Customary methods for characterizing aircraft performance are typically expensive, time-consuming endeavors and therefore do not fit well with the low-cost benefits of sUAS. Therefore, a systematic approach to flight-testing sUAS aircraft is proposed. This approach is aimed at validating aircraft performance estimates based on combining fundamental aircraft sub-system models, onboard sensor fusion techniques, and autonomous flight capabilities. General models for the sub-systems used in estimating the vehicle's performance are developed in this work. Specific focus is placed on the effect of measurement and estimation uncertainty on resulting performance predictions. It is anticipated that the models developed in this work will reduce the time and cost associated with the flight-test evaluation of new fixed-wing sUASs.

Broadly speaking, a fixed-wing sUAS may be divided into three major sub-systems. The first is the vehicle's propulsion system. For sUASs, these are typically either electric or internal combustion motors providing power to a propeller. For this work, an electric propulsion system model is developed which accounts for scale effects somewhat unique to sUAS applications. The second sub-system is the combined inertial navigation system (INS) and flight management computer (FMC). These systems are comprised of any number of sensors designed to estimate the vehicle's state and provide closed-loop control over its trajectory. The estimated flight performance is intrinsically linked to the accuracy of the INS and vehicle control during the test maneuver. Therefore, this system is characterized and its effects on the estimation of the vehicle performance are developed in detail. The third sub-system is the airframe, which includes the fuselage, wing, and any other aerodynamic stabilizing and control surfaces. Information on the design of a fixed-wing sUAS named the Peregrine is presented. This aircraft is used exclusively for evaluation of the proposed flight-testing methods. Each of these systems will be developed in detail through the remainder of this work, beginning with the propulsion system.

1.2 Propulsion System Overview

Development of a predictive propulsion model for small-scale unmanned aerial systems (sUAS) represents a continuing challenge for the designers of these vehicles. Typical components available for sUAS are not supported by robust performance data sets. This paper presents a predictive performance model for an electric propulsion system typical of sUAS. The model attempts to capture the effects of Reynolds scaling on the overall performance of the system at various operating conditions. The predictive model is based on the Blade Element Momentum (BEM) methodology, which has been widely used for full-scale wind turbines, helicopters and propellers[1-5]. Several corrections to the basic model are needed in order to capture the effects of compressibility, viscosity, and three-dimensional flows, and are presented in section 2.2.

The BEM modeling methodology has been applied to small-scale propeller testing by Uhlig and Selig[4] to predict the performance at extreme angles of attack, but without handling Reynolds scaling effects. Ol, Zeune, and Logan presented BEM predictions for a small scale propeller with experimental data.[6] Ol *et al.* demonstrated the large impact of geometry estimation errors on the resulting propulsive performance. No specific BEM correction factors were mentioned in these works. Several investigations report experimental data sets for sUAS propellers,[7-9] and acknowledge the lack of published performance information. The systematic development of a scale-specific model with experimental validation for small-scale propeller and propulsion system performance is still lacking.

The variation in propulsive efficiency over a range of Reynolds numbers was measured by Brandt and Selig.[7] In their work, the propeller RPM was held constant and the efficiency versus advance ratio was measured. As the Reynolds number was reduced, the overall propulsive efficiency decreased. Brandt and Selig concluded that there are significant Reynolds-based effects on small propellers that are not typically encountered on large scale, higher power propellers. Merchant and Miller measured the reduction in propulsive efficiency at different local Reynolds numbers for different sUAS propellers.[10] In their work, the reduction in propulsive efficiency was tied to the reduction in lift, and increase in drag at low Reynolds numbers. In a similar manner, Ol *et al.* measured a similar reduction in propulsive efficiency at low Renyolds numbers.[6] In all reported cases, the Reynolds number effects were measured by holding the propeller RPM constant as the advance ratio was increased. A reduction in local Reynolds number typically results in a decreased L/D for a given angle of attack. The reduced L/D translates to lower thrust for a similar power input. The lift and drag variation of immersed lifting surfaces at low Reynolds numbers has been well documented in various works.[11-14]

This work seeks to match the performance predictions of a BEM model to experimental data gathered on geometrically similar propellers over a range of Reynolds numbers. Emphasis is placed on the quantification of operational conditions and local Reynolds number effects on the overall propulsive efficiency. Reynolds effects are anticipated to be of first-order importance to the propeller performance, yet no BEM models applied to small-scale propellers have substantiated this hypothesis. Thus, carefully-controlled experiments were conducted in order to produce constant Reynolds number propeller data across a range of test conditions. A key contribution of this work is the incorporation of Reynolds effects in the BEM model, and experimental demonstration of the first-order importance of these effects when modeling small-scale propellers.

Experimental extensions to the propulsion model include capturing the electrical efficiency of the system. This will allow estimates of the electrical efficiency to be combined with the propeller efficiency, yielding a model relating the measured electrical power to the thrust power. The BEM model can be used along with real-time measurements of airspeed, propeller rotation speed, and electrical power to infer the power supplied to the flow under various flight test conditions. This estimate is valuable for rapid flight test determination of the aircraft drag polar and closed-loop control for real-time optimization of vehicle performance. This method of in-flight determination of power is necessary for production aircraft, which are typically not instrumented with load cells that can directly measure thrust and torque. The BEM model developed here is also valuable for propeller and aircraft design, since it enables rapid iteration on relevant design variables with accurate results.

1.3 Inertial Navigation System Overview

Vehicle performance estimation relies on the propulsive efficiency models previously discussed and vehicle state estimates provided by the onboard INS. INSs have been used for many years and taken many different forms.[15] The most recent developments focus on miniaturization and formulation of state models based on the capabilities of micro-electro-mechanical system (MEMS) sensors.[16-18] The INS used in this work relies on MEMS sensors for attitude, velocity, and position estimation. These systems are typically smaller and less expensive than the tactical grade systems found on full-scale aircraft, but feature correspondingly lower performance.[19] MEMS-based INS systems have been developed and characterized by numerous research efforts and represent an ongoing area of development.[18-21]

The INS used in this work utilizes a gyroscope and accelerometer as the core inertial measurement sensors. The performance of the INS is intrinsically related to the accuracy of these sensors. Therefore, calibration of the gyroscope and accelerometer sensors is a key component to extracting the maximum performance from a MEMS based INS. Classical calibration methods rely on exciting the sensor triads using precisely known rates or orientations.[22, 23] Extensions to these methods include gyrocompassing to align the computational frame to the Earth frame using specially formulated Kalman filters. These filters typically rely on resolving the Earth's rotation rate vector to estimate bias and scale factor errors in the gyroscope and accelerometers.[23] MEMS-based sensors typically do not have the resolution or bias stability to resolve the Earth rate. Therefore, other calibration techniques are needed which match the capabilities of the sensors. Least-squares techniques have been proposed for identifying scale factor and bias errors using inputs from precision rate tables or known angular displacements.[21, 24] Other near real-time calibration techniques rely on additional states included in the onboard sensor fusion algorithms.[20, 25]

Precise positioning, filter-based, and least-squares calibration methods provide the parameters necessary for accurate short-term inertial navigation. However, these methods typically require precise calibration equipment or computationally expensive filters. An alternative calibration method is proposed in which a numerical optimization scheme is used to estimate the required correction parameters. Optimization techniques have previously been used for INS calibrations.[26, 27] These methods rely on aligning the sensors axes to a known rate or excitation vector (Earth rate vector, gravity, magnetic field, etc.). The method proposed in chapter 3 relies on minimizing the difference between estimated and measured gravity measurements over a series of small rotations. The proposed optimization algorithm does not rely on specific knowledge of the rates or orientation. It only requires that the three orthogonal gyroscopes be at rest between successive small rotations. This allows for any number of low cost positioning systems to be used for calibration.

Details of the sensor fusion algorithm used to correct the INS state estimates is then presented. This work follows from the error-state INS mechanizations developed in detail by numerous sources.[23, 28, 29] The mechanization of the error-state filter requires uncertainty information from the aiding sensors used to provide correction data. For the INS system used in this work, the aiding sensors include a GPS, magnetometer, and barometer. Errors in each of these devices are well characterized.[18, 30] However, additional magnetometer errors can be encountered in an electric sUAS due to the proximity of high current wires. A novel magnetometer filter is proposed based on high-fidelity current measurements provided by the onboard power supply. This filter reduces the noise in the magnetic field measurements without injecting a phase lag inherent when using a low-order low pass filter.

Characterizing INS performance typically relies on either covariance analysis from the filter, or comparison with other high-precision aiding sources.[23] Numerical simulations have been used to estimate the performance of error-state algorithms.[31-33] These simulations typically rely on an exact solution to the attitude and navigation equations.[28, 34] Many of these methods are designed for tactical navigation systems, which require higher numerical precision than many MEMS-based INSs. Recent trends have pointed towards using commercially available flight simulators such as X-Plane® for hardware-in-the-loop (HIL) control and systems simulations.[35-37] Extensions to the methods presented in these works are outlined in chapter 4. The extensions allow for the near real-time output of inertial data to the INS for state estimation and closed-loop control analysis. This provides a unique capability in terms of assessing the performance of both the INS algorithms and sensors used for flight control. With this capability, the effects of sensor noise and bias are quantified. The resulting INS errors are then used to quantify the uncertainty in the flight performance analysis.

1.4 Flight-test Performance Overview

Flight-testing of conventional aircraft is a well-defined discipline. Recent developments have attempted to incorporate the unique operational characteristics of sUAS into this well established framework. These include maneuvers taking advantage of a limited degree of autonomy, and using existing sensors for parameter estimation.[38, 39] The final chapter of this work seeks to combine the Reynolds dependent propulsion and INS error models to demonstrate conventional methods for estimating aircraft performance. The proposed propulsion and INS models are specialized to sUAS, and their influence on flight-test performance measurements will be detailed. Application of the Reynolds dependent propulsion and INS error models to compare conventional methods for estimating aircraft performance seeks to fill a gap not addressed in the current literature.

The specific flight-tests considered in this work are aimed at determining the aircraft's drag polar and Oswald efficiency factor, e. Three methods for determining the drag polar are tested. These methods follow from standard flight-test procedures outlined by Kimberlin.[40] These include full power rate-of-climb and level acceleration tests. Each test is detailed in chapter 5. The resulting drag polar and Oswald efficiency are then used to assess the range and endurance for the vehicle presented. The Reynolds-based propulsion model is incorporated into this work to provide a range and endurance estimation tool unique to fixed-wing electrically powered sUAS.

Previous research efforts have focused on passive data gathering and analysis for fixed-wing sUAS maneuvers. Wypyszynski detailed both the sensors and measurements necessary to identify sUAS aerodynamic parameters including the drag polar and stability margins.[41] For their tests, the aircraft was flown in a guided rather than autonomous mode. Data reduction focused on GPS and specific force measurements rather than fused state estimates. Uncertainty was addressed in this work, but a systematic buildup of error was not presented. Ostler et al. presents a full description of the flight-testing methodology necessary for determination of a sUAS drag polar and stability margins using the methods outlined by Kimberlin.[38, 42] Application of Kimberlin's performance estimation techniques with commercially available hardware allowed for generation of an aircraft drag polar by averaging numerous rate-of-climb and glide tests. It was found that the low speed and often low wing loading of sUAS aircraft drives the uncertainty in flight-test measurements. This is primarily related to inaccuracies in the attitude estimation and limitations of the flight computer controlling the vehicle. In their work, the resolution of the airspeed sensor produced large uncertainty over the majority of the aircraft's flight envelope. Low wing loading led to large errors due to wind and day-to-day variability in atmospheric conditions. As in Wypyszynski's work, GPS and airspeed data were largely used in place of fused sensor estimates for performance estimation. In both cases, the drag polars presented were generally in close agreement with their baseline performance models. Neither the Wypysznski and Ostler et al. flight-testing campaigns attempted to account for the Reynolds dependency of the power available models, and limited uncertainty data was presented to quantify the quality of the resulting flight-test data. In both cases, the airspeed was a prime driver of the resulting performance uncertainty. Methods of limiting the uncertainty in airspeed measurement by integrating GPS readings have been developed.[43] Extensions to these methods are proposed by incorporating INS measurements to provide *in situ* calibration information.

Gross et al. detailed flight-testing methods based on fused attitude solutions provided by an onboard INS.[44] In that work, comparisons between flight-test reconstruction using different INS data fusion methods were presented. For the test cases considered, the sensor fusion algorithms provided significantly lower uncertainty in estimated states than the standalone GPS. The reduction in uncertainty with an INS points towards the necessity of using an INS-based system for limiting the uncertainty in performance estimations. This conclusion has been reiterated by others conducting INS/GPS studies.[39, 45, 46] However, these tests did not document the influence of INS error on performance estimation techniques for sUAS type applications. Giacomo attempted to validate a mathematical model of a sUAS using a commercially available autopilot system.[47] This flight-testing centered on adapting the gains of the controller to match model predictions of short period, long period, and Dutch roll modes. The input model utilized a basic drag polar estimated using classical aircraft design techniques. This model was developed in conjunction with autopilot integration studies performed by Ausserer.[48] In this work, a flight-testing plan for assessing vehicle performance was developed in detail, but not implemented in practice. Instead, modeling results were presented to assess the feasibility of the proposed flight-testing methods. Difficulties were encountered in both studies in interfacing with a “black box” autopilot system. It was determined that integrating specific test airspeeds and attitudes was difficult using a closed source autopilot not specifically developed for flight-testing. Analysis of the uncertainty in performing fixed-wing flight-tests were mentioned, but not implemented in this work.

Lundstrom and Krus performed testing on a manually flown micro air vehicle to determine the lift and drag characteristics of the airframe.[49] In this work, a propulsion model was developed which allowed for post flight reconstruction of the power available. They introduced an intriguing methodology that involved attachment of the propulsion system to a car in order to simulate forward flight. Data from these road tests resulted in a propulsion system model that exhibited no discernable Reynolds effects for all measured freestream velocities and advance ratios. This is likely due to uncertainty buildup from the thrust and moment measurements. The propulsion system model was used in conjunction with an onboard accelerometer and GPS to estimate the aircraft’s drag polar. This was accomplished by flying the aircraft in a racetrack pattern at arbitrarily chosen throttle settings. During the test, the vehicle was held visually at a set altitude by the pilot. Their test flights were not aimed at validating performance predictions, but rather at generating *L/D* curves. From their analysis, the effect of wind was shown to be a strong contributor to the overall uncertainty in the *L/D* measurements. A basic uncertainty analysis was presented which pointed towards the inadequacy of using GPS in highly dynamic (windy) environments. Their conclusions point towards the possible performance estimation improvements gained by using a fused attitude/velocity/position solution.

Chaney presented a systematic approach to assessing the impact of using empirical aircraft design constraints on sUAS design.[50] In this work, a complete vehicle design was presented and validation was attempted though flight-testing. A propulsion system model was developed following the work presented by Lundstrom and Krus.[49] In this model, no specific Reynolds effects were noted. An Arudpilot autopilot system was used in place of manual flying for initial testing of the vehicle. The first tests were aimed at reconstructing the drag polar using rate-of-climb measurements and unpowered glides. No assessment of the impact of sensor error or fused state error is presented in the findings. While the autopilot was used for initial flights, difficulty in programming specific flight test maneuvers led to the remainder of the test flights being flown manually. In the manual flight mode, relevant data for determining the lift and drag characteristics of the vehicle were extracted using windowing functions determined by the maneuver. Their results indicate reasonable agreement between predicted and measured rate-of-climb at moderate airspeeds. Large estimation errors were noted at low airspeeds due in part to suspected noise in the sensor data and difficulty in piloting the aircraft.

No known studies have addressed the influence of Reynolds-based propulsion characteristics and few have addressed the effects of INS/sensor uncertainty on fixed-wing sUAS flight-test performance estimations. Therefore, the purpose of this research is the development of a flight-test based performance estimation methodology for determining aircraft performance based on a rigorous analysis of the capabilities of the onboard systems. A specific focus is on the effects of incorporating a Reynolds-based propulsion and INS error model on the resulting vehicle performance estimates.

1.5 Key Contributions

The key contribution of this work is the development of the modeling and analysis framework necessary for quantifying sUAS performance via flight testing. This framework relies on system models of unique sUAS components and detailed error analysis of the onboard state estimation systems. The application of a Reynolds dependent BEM propulsion model to flight-test performance estimation for sUAS is a unique contribution of this work.

A novel optimization-based calibration method for MEMS INS systems is developed. This method allows for determination of scale factor, bias, and other error sources needed to provide an accurate inertial state estimate. An advantage of the proposed calibration method is its reliance on readily available low-cost positioning hardware. Demonstration of the INS accuracy after calibration is presented and used as the basis of an error-state Kalman filter. The performance of the INS is quantified using HIL testing of the error-state filter. This testing methodology is uniquely suited for sub-tactical grade INSs and allows for the quantification of state errors based on measured sensor characteristics. The development of an interface to the commercially available flight simulator Xplane® for INS state estimation and system validation is a unique contribution of this work.

The propulsion system and INS error models are then used to estimate the flight performance of a fixed-wing sUAS. Three different flight-test methods for determining the aircraft's drag polar are compared. Specific focus is placed on Reynolds-based propulsion effects and propagation of state uncertainty to the flight-test performance estimates. The flight testing results are combined with the Reynolds-based propulsion model to provide a unique insight into the range and endurance characteristics of a fixed-wing electrically powered sUAS. By quantifying the buildup of uncertainty in the performance estimates, the proposed framework provides insight useful in flight testing of current sUAS and potential feedback to the initial design phase of new aircraft.

Chapter 2: Propulsion System Model

2.1 Propulsion Model Overview

This chapter details the development of a propulsion model using the blade element momentum (BEM) modeling methodology. This model is used for inferring the propulsive efficiency, and therefore thrust available during flight. Contributions of this work include an analysis of Reynolds effects unique to the scale size of the propellers tested. These effects are of first order importance in predicting aircraft range and endurance as is demonstrated in chapter 5. The BEM-predicted Reynolds performance is compared to wind tunnel experiments. Good agreement is obtained when the proposed corrections to the BEM model are incorporated. A discussion of these corrections and their overall impact on the results of the model is presented. Finally, extensions to measuring system performance are presented. This includes a surface fit correlation between the Reynolds number, advance ratio, and propulsive efficiency. This allows for estimation of the Reynolds-based in-flight thrust, detailed in subsequent chapters.

2.2 Blade Element Momentum Model

The Blade Element Momentum (BEM) model attempts to incorporate the geometric properties of the propeller system into a momentum-transfer-based model of the flow. Determination of the thrust production and torque required derives from individually analyzing small elemental sections along the span. In this work, data on elemental sections is generated computationally, creating a look-up table for input to the BEM model. The individual forces acting on each section are integrated to find the total force in the direction of motion, the thrust, and the torque opposing the rotation of the blade. Figure 1 shows the streamtube encapsulating the propeller, and Figure 2 shows the basic forces acting on an elemental blade section.

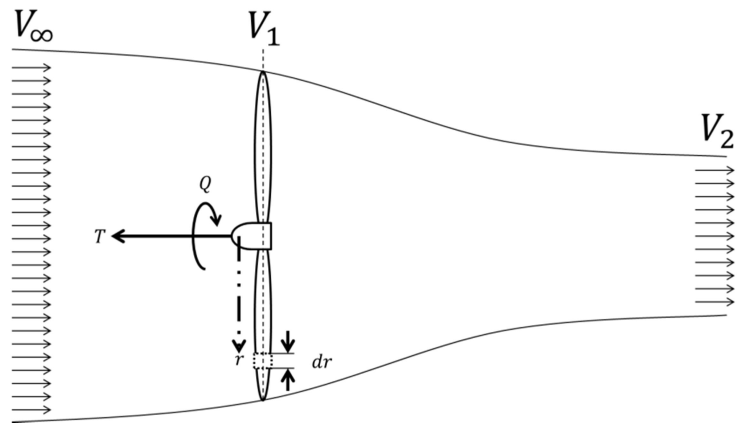


Figure 1: Farfield streamtube for BEM model development showing positive elemental propeller force and torque conventions

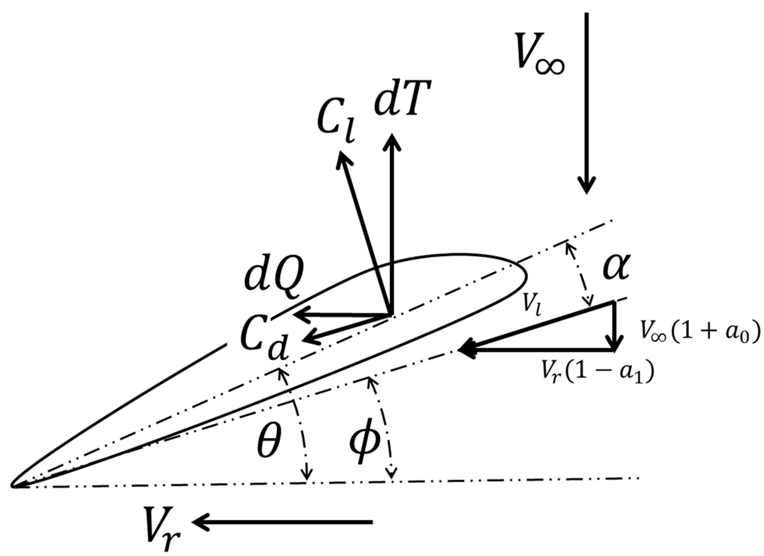


Figure 2: Local blade section properties for BEM model development. Forces and torque represent those acting on an infinitesimal blade element.

The differential thrust and torque components are integrated over the surface of the blade to determine the total force and torque produced by the propeller. A detailed derivation of the BEM model can be found in Madsen *et al*.[1] A basic overview of the BEM model as it applies to the current construct is presented here. The primary difficulty in BEM modeling is the estimation of the induced axial and radial velocity components created by the rotating propeller. As the propeller is operating in a predominantly subsonic regime, the pressure field created by the propeller induces a rotational flow component upstream of the blade. The magnitude and direction of the induced velocity components vary in the spanwise direction as a function of the propeller aerodynamic, geometric, and operational conditions. To date, there is no closed form solution for the induction factors. Therefore, an iterative approach is used to approximate them over a finite section of the propeller blade. The induced velocity factors, a_0 and a_1, are formulated by equating the local aerodynamic forces acting on the blade with the bulk momentum transfer of the blade to the streamtube surrounding the propeller. The thrust and torque components acting on each blade element are given by[1]

$$\begin{aligned} dT &= \frac{1}{2}\rho V_\infty^2 cB \frac{\left(1+a_0\right)^2}{\sin^2\phi} C_T dr \\ dQ &= \frac{1}{2}\rho V_\infty cB\omega r^2 \frac{\left(1+a_0\right)\left(1-a_1\right)}{\sin\phi\cos\phi} C_Q dr \end{aligned}, \tag{2.1}$$

where C_T and C_Q are related to the local lift and drag coefficients by the rotation matrix given as

$$\begin{bmatrix} C_T \\ C_Q \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} C_l \\ C_d \end{bmatrix}, \tag{2.2}$$

and the lift and drag coefficients can be determined across a range of conditions by experimental, analytical, or computational methods.

From Figure 1, the total thrust can be determined via momentum balance between the upstream farfield and a location far downstream (shown as location 2 in Figure 1). Applying the conservation of momentum to the streamtube gives the following relationship for the thrust and torque components across the propeller disk

$$\begin{aligned} dT &= 4\pi r\rho V_\infty^2 (1+a_0)a_0 dr \\ dQ &= 4\pi r^3 \rho V_\infty \omega (1+a_0)a_1 dr \end{aligned}. \tag{2.3}$$

Equating (2.3) with the geometric thrust and torque terms in (2.1) gives the following implicit relationship between the induced velocity components,

$$
\begin{aligned}
a_0 &= \frac{1}{\dfrac{4\sin^2\phi}{\sigma C_T} - 1} \\
a_1 &= \frac{1}{\dfrac{4\sin\phi\cos\phi}{\sigma C_Q} + 1}
\end{aligned}
, \tag{2.4}
$$

where σ is the local solidity of the disk, given as the ratio of the local chord divided by the area of the annular disk formed by the infinitesimal radial blade element, $cB/2\pi r$.[51, 52] Using the relations provided by (2.3) and (2.4), an implicit iterative solver can be set up to approximate the axial and radial induction factors.

2.2.1 Corrections

The BEM model is nominally a 2-D estimation of local blade performance and therefore provides no connection between adjacent infinitesimal blade elements. For blades of moderate aspect ratio, there will be an appreciable spanwise variation in velocity, angle of attack, and Reynolds number. Under some conditions, the radial flow components can affect the propulsive efficiency of the propeller.[3] To account for the spanwise aerodynamic coupling, correction factors are introduced. These include allowances for tip losses, Mach effects, and pressure-based radial flow components. The following sections detail these corrections.

2.2.1.1 Tip Loss Factor

Due to pressure equalization, the lift goes to zero as the radial location approaches the tip. Prandtl addressed this effect by incorporation of a tip loss coefficient.[53] The tip loss factor is given as

$$F = \left(\frac{2}{\pi}\right)\cos^{-1}\left(e^{-f}\right), \tag{2.5}$$

and the localized correction factor f is given by

$$f = \frac{B}{2}\left(\frac{1-r}{r\phi}\right), \tag{2.6}$$

where B is the number of blades, and ϕ is the local inflow angle ($\theta - \alpha$). Since the inflow angle is not known *a priori*, it must be determined iteratively in the BEM algorithm. These corrections represent a reduction in lift occurring near the tip of a blade element due to pressure equalization. The tip loss coefficient is incorporated into the BEM model by modifying the inflow coefficients a_0 and a_1,[52]

$$\begin{aligned} a_0 &= \frac{1}{\dfrac{F 4\sin^2\phi}{\sigma C_T} - 1} \\ a_1 &= \frac{1}{\dfrac{F 4\sin\phi\cos\phi}{\sigma C_Q} + 1} \end{aligned} \quad . \tag{2.7}$$

2.2.1.2 Mach Number Correction

The tip Mach number on low pitch propellers can reach up to 0.6. At these Mach numbers, the 2-D airfoil sections will experience an increase in the lift curve slope approximated by Glauert's Mach correction factor,[51] given as

$$C_l = \frac{C_{l,\,M=0}}{\sqrt{1-M^2}} \,. \tag{2.8}$$

The Mach correction is applied to the lift coefficient and drag coefficient for the cases considered in this work.

2.2.1.3 Three-Dimensional Flow Correction

Blade element momentum theory fundamentally assumes quasi-two-dimensional flow, with no interaction between adjacent radial locations. However, spanwise flows are often present and cannot be accounted for with the baseline BEM technique. Thus, a semi-empirical correction must be used to address three-dimensional flow effects induced by the rotation of the blade.[3, 54, 55] The correction model used in this work follows the work of Snel *et al*.[3] and Liu & Janajreh,[55] and will be briefly summarized here within the context of propeller performance. The basic BEM model assumes that the blade sections are decoupled in the spanwise direction. Spanwise (radial) flows occur due to the presence of locally separated regions near the training edge that extend over some portion of the blade surface. The complex pressure field resulting from these separated regions promotes a radial flow component which tends to move from root to tip, as the spanwise pressure gradient is stronger than the chordwise gradient in these separated regions. The bulk movement of fluid requires additional energy from the motor. The additional energy is formulated similarly to a mechanical pumping loss term. Semi-empirical corrections for the "pumping" loss were proposed by Snel *et al*. and Liu & Janajreh after detailed theoretical, computational, and experimental investigations. The correction relies on modification of the local lift coefficient based on its angle of attack and proximity to other stalled blade regions.

A reduction in the radial flow component occurs near the tip due to a lower angle of attack and localized pressure equalization. To match the extents of the three-dimensional effects, the centrifugal pumping correction is applied between the root and 85% of the span.[3] The semi-empirical 3-D correction is formulated by augmenting the uncorrected 2-D normal force components, given by

$$C_{n_{3-D}} = C_{n_{2-D}} + 1.5\left(\frac{c}{r}\right)^2 \left(C_{l,pot} - C_{l_{2-D}}\right)\left(\frac{\omega r}{V_l}\right)^2 . \tag{2.9}$$

where the normal force coefficient is normal to the local airfoil chord. $C_{l,pot}$ is the lift coefficient from thin airfoil theory (potential flow) for a symmetric section evaluated at the local angle of attack. Estimation of the aerodynamic parameters required for the BEM model and correction factors are addressed in following sections. The results from the BEM model will be compared to the experimental measurements detailed in the next section.

2.3 BEM Model and Experimental Parameters

This section details the performance parameters used in the BEM model and experimental testing. A discussion of a visual technique used for estimation of the propeller geometry is presented. A description of the aerodynamic characteristics of the propeller blade sections is also presented.

2.3.1 Propeller Performance Metrics

This section details the terminology and metrics associated with the analysis of the propeller-based propulsion system. These include the non-dimensional parameters commonly associated with propeller design such as the advance ratio, thrust and power coefficients, and the propulsive efficiency.

The advance ratio is a dimensionless term used to quantify the effects of forward motion and angular velocity and is given as

$$J = \frac{V_\infty}{nD}, \tag{2.10}$$

where J is the advance ratio, n is the rotational frequency in rev/s, and D is the diameter of the propeller. The thrust and power coefficients are non-dimensional quantities relating the thrust and power producing capability of a propeller to its rotational velocity and diameter. From dimensional analysis, the thrust coefficient is given as

$$C_T = \frac{T}{\rho n^2 D^4}, \tag{2.11}$$

where T is the thrust, ρ the freestream density, n the rotation rate of the propeller, and D is the diameter. Following a similar development, the torque and power coefficients are expressed as

$$C_Q = \frac{Q}{\rho n^2 D^5},\; C_P = \frac{P}{\rho n^3 D^5}, \tag{2.12}$$

where Q is the torque and P is the motor power required to drive the propeller.[52]

The propulsive efficiency is given as the ratio of the power transferred to the air mass moving through the propeller disk to the mechanical power required to drive the propeller. This is expressed as

$$\eta_{propulsive} = \frac{TV_\infty}{2\pi nQ} = \frac{1}{2\pi}\frac{C_T}{C_Q}J \,. \tag{2.13}$$

The propulsive efficiency is a key metric for comparison of propellers of different geometric configurations or operational conditions. For the electrical propulsion system used in this work, there is an additional electrical efficiency term used to correlate the electrical power delivered from the battery to the mechanical power produced by the motor. This efficiency term is given by

$$\eta_{electrical} = \frac{2\pi nQ}{EI} \,, \tag{2.14}$$

where E is the battery voltage and I is the supplied current to the electronic speed controller. In practice, the speed controller and motor efficiencies are lumped into a single term. The total system efficiency, a ratio of flow power to supplied electrical power, is therefore the product of the electrical and propulsive efficiencies:

$$\eta_{total} = \eta_{electrical}\eta_{propulsive} \,. \tag{2.15}$$

2.3.2 Geometric and Aerodynamic Parameters

2.3.2.1 Propeller Geometry Estimation

The APC propellers used in this work are hobby-grade models commonly used on sUAS, and do not have published tolerances on the geometry. Furthermore, specific information on the variation of chord or geometric angle of attack (θ) across the span of the propeller is unknown. Thus, a visual integration method was developed in order to determine the blade geometry, following the method suggested by Deters *et al.* and Ol *et al.*[6, 14] The integration method correlates two perpendicular images to estimate the spanwise variation of chord and angle of attack. A third perpendicular image is used in difficult-to-resolve regions, typically occurring near the root and tip. The specific propellers investigated with the BEM model are the APC thin electric 10×5E, 10×6E, and 10×7E. The first number in the nomenclature represents the diameter of the blade (in inches). The second number represents the theoretical forward advance (pitch) in one revolution of the blade (again, in inches). The propellers use the Clark Y airfoil section from 0 to 5% span, and the NACA 4412 for the remainder of the span. The airfoil distribution was determined by cutting the blades in 10% increments and imaging the cross cut sections. Ol *et al.* proposed a similar method in which the cross cut sections were automatically fit with splines. These splines were then matched to airfoil data accessed through the UIUC airfoil database. In this work, a similar manual version of this method was used. Splines were fit to the cross cut sections and compared to standard NACA profiles.

For each propeller, two perpendicular images were collected. The first is in the streamwise direction allowing for identification of the hub region. The hub contains features of known sizes, which provide datums for scaling and rotation. The second image is perpendicular to the streamwise direction and span. A black line was drawn on the leading edge to allow for proper edge identification. Figures 3 and 4 show representative images for each blade. The images were collected with a 10 MP Canon DSLR camera taken from 0.5 m, giving an effective resolution of 0.1 mm per pixel at a distance of 0.5 m. From these images, negatives were formed using the edge detection and correction capabilities of the GIMP image editing software package, the results of which are shown in Figure 5. The negative images were derotated and scaled using the MATLAB image processing toolbox. Scaling relies on correlating known dimensions with pixel counts. The images taken in the streamwise direction show the hub and mounting hole, the dimensions of which were directly measured with calipers. After scaling based on hub size, the blade is derotated using a known overall length.

Figure 3: Top view of APC 10×5E, 10×6E, and 10×7E propellers, from top to bottom.

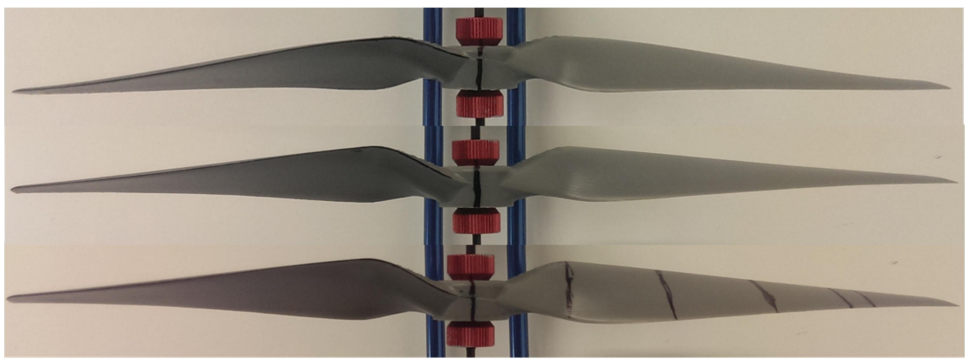


Figure 4: Side view of APC 10×5E, 10×6E, and 10×7E propellers, from top to bottom. Spanwise black lines used for leading edge detection.

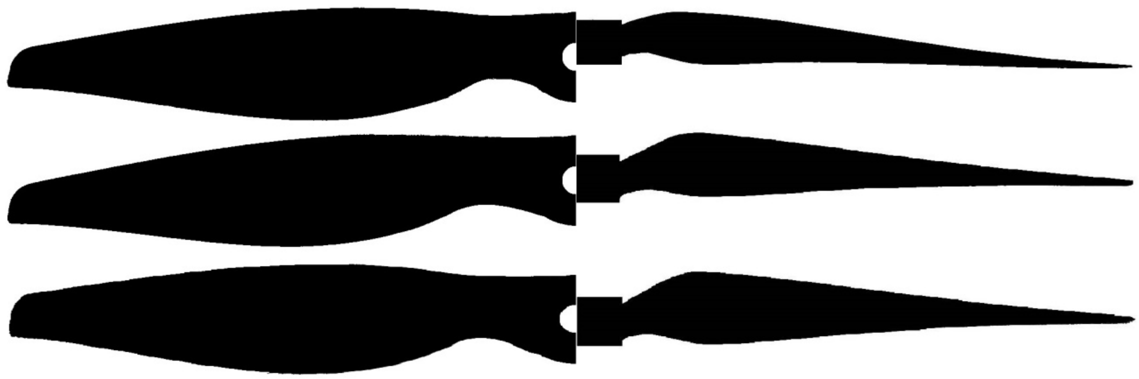


Figure 5: APC propeller top (left) and side (right) negative images for the APC 10×5E, 10×6E, and 10×7E (top to bottom).

Alignment and registration of the top and side negatives occurs by finding the maximum cross-correlation between the top and side images. The chord is estimated by integrating the pixel count in the streamwise direction across the top image and the chordwise direction of the side image. The local chord is then the square root of the sum of the squares of the top and side lengths estimated by pixel integration. The inverse tangent of the top and side pixel integral values gives the geometric angle of attack. Figure 6 shows the resulting angle of attack and chord variation for the blades used in this work. The BEM model calculations start immediately outside the hub region. The hub is shown in Figure 6 and included only for clarity.

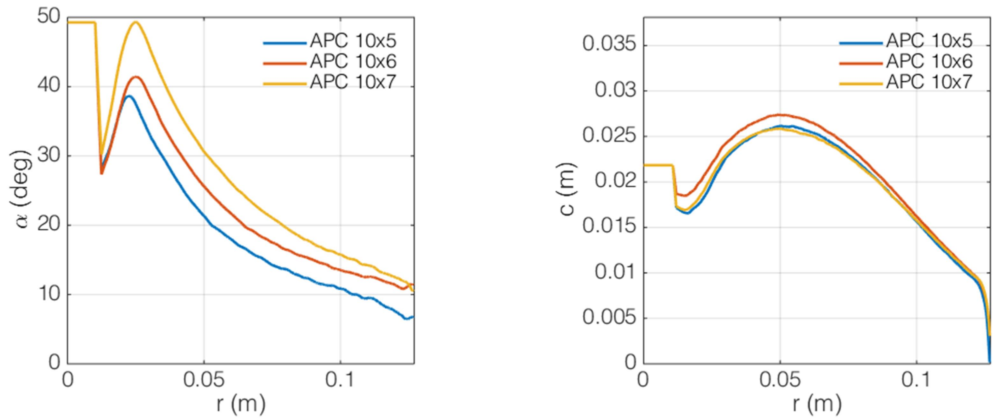


Figure 6: Angle of attack (left) and chord length (right) for APC 10×5E, 10×6E, and 10×7E propellers.

The resulting propeller geometry was verified using a NextEngine 3-D scanner. This system generates a precise map of three-dimensional surfaces. However, it is unable to simultaneously map the underside of the propeller, making it unsuitable for complete definition of the propeller geometry. The accuracy of the sensor is limited to 0.5 mm with an overall point spacing of 2 mm. The results are a coarse estimate of the propeller geometry bounding the estimates provided by the imaging technique previously outlined. A representative dataset for the APC 10×6E is shown in Figure 7, which confirms the validity of the imaging technique.

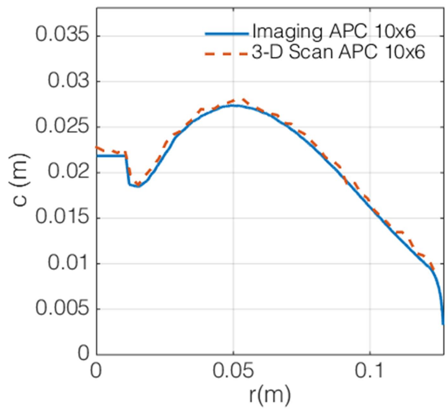


Figure 7: APC 10×6E chord distribution estimated by imaging and 3-D scan.

2.3.2.2 Aerodynamic Coefficients

The aerodynamic coefficients for the BEM model were generated using XFOIL and augmented with experimental data. XFOIL is a panel code designed to model the effects of viscosity (to first order), and hence Reynolds number on various aerodynamic bodies.[56] XFOIL was used to analyze 2-D sections over a range of Reynolds numbers from 10^4 to 10^6, and angles of attack from -10 to 20 degrees in 0.5 degree increments (see Figures 8 and 9) at a Mach number of zero. For angles of attack beyond stall, high angle of attack experimental data from Ostowari & Naik[57] and Tangler & Kocurek[58] were used. Representative results for a NACA 4412 operating at a Reynolds number of 1×10^6 are shown in Figure 8. From this merged set of analytical and experimental data, maps of sectional lift and drag coefficients across a range of angle of attack and Reynolds number were created. Sectional lift and drag coefficients for the BEM model at a particular condition can then be obtained from this data using the MATLAB bi-linear interpolation function (Figure 9).

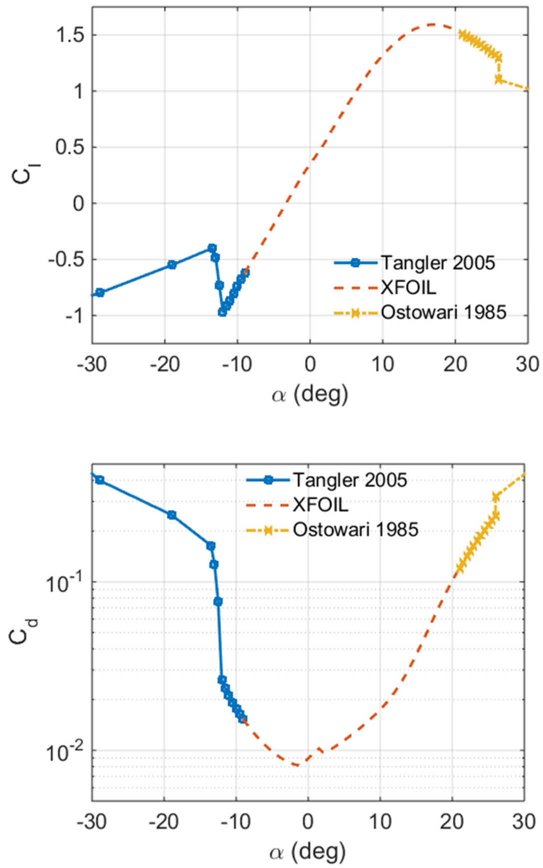


Figure 8: Lift (top) and drag (bottom) coefficients for the NACA 4412 at *Re* = 1×10^6.[57, 58]

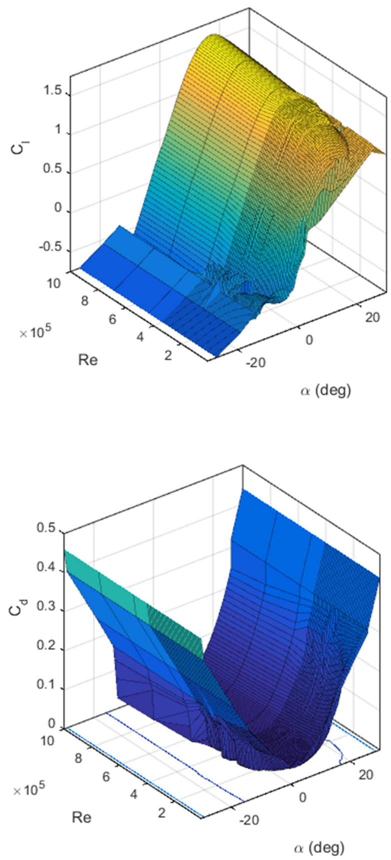


Figure 9: Lift (top) and drag (bottom) coefficients versus angle of attack and Reynolds number for NACA 4412.

2.3.3 Operating Conditions

The BEM model was applied to three propeller geometries discussed previously. The performance of each propeller was modeled over a range of advance ratios and Reynolds numbers, the limits of which were selected from previous flight and wind tunnel testing of the propulsion system (summarized in Table 1). For this testing, the Reynolds number is defined based on blade diameter and the resultant blade-relative velocity evaluated at 70% of the blade radius. This radial position represents the approximate location of the average force acting on the blade. It should be noted that the BEM model uses the local chord to determine the Reynolds number for aerodynamic parameter estimation.

Table 1: BEM model operational bounds.

Parameter	Minimum	Maximum
J	0.01	1
n	4000 RPM	15,000 RPM
V_∞	0.01 m/s	45 m/s
Re	1×10^5	1×10^6
M	0.16	0.6

2.4 Experimental Testing

Thrust and power data for common hobby grade motor and propeller systems are typically highly dependent on the specific components tested.[59] Therefore, a successive buildup method is proposed to quantify the individual component efficiencies for the propeller and motor, along with the propulsion system thrust and torque coefficients. A test stand in a wind tunnel was used for measurement of the thrust and torque over a range of airspeeds and motor speeds in order to generate a map of propeller efficiency versus advance ratio. In addition to the thrust and torque data, engine RPM, motor voltage, and current were monitored. Figure 10 shows a schematic of the engine testing apparatus.

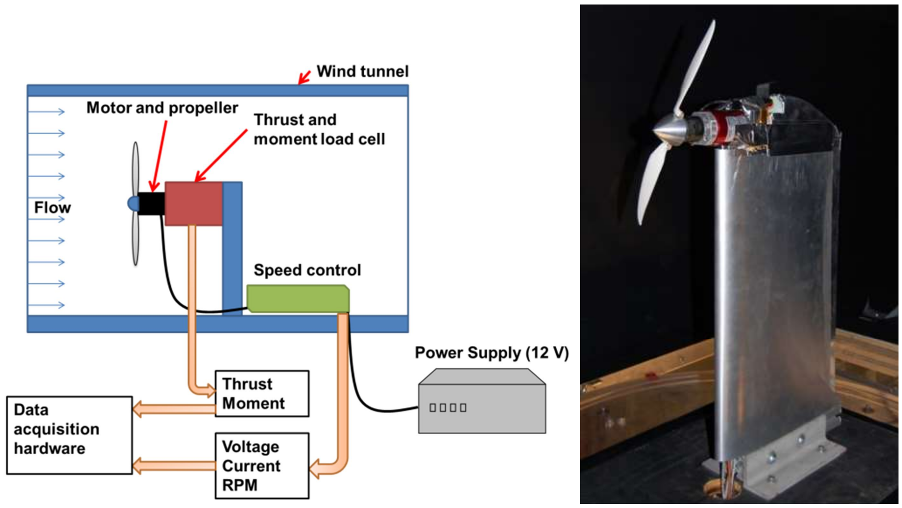


Figure 10: Schematic of propulsion system test setup (left) and experimental test stand (right).

2.4.1 Experimental Setup

The thrust and torque measurements were made using a Futek MBA500 with a thrust rating of 222 N and a torque capacity of 5.64 N-m. Measurements of rotational speed were made based on the back EMF produced by the commuting motor. The back EMF signal contains a voltage spike each time the coils in the motor are energized. The back EMF signal was passed through a Schmitt trigger, which provided a square wave output proportional to the motor speed with six pulses per revolution. An inline Allegro ACS758 100-A current sensor allowed for measurement of the electrical power. A bank of six HP Proliant 406421-001 1300-W, 12-V server power supplies provided the electrical power necessary to run the motor. Each power supply was capable of providing 40 A and was placed in parallel to buffer the current output capability.

The wind tunnel is an open return type with a 22”×22” test section and a maximum velocity of 65 m/s. The tunnel features a 3” flow straightening honeycomb structure on the inlet, followed by three turbulence-reducing screens. Tunnel velocity was measured using two static rings, with one placed in the plenum after the turbulence screens, and the other immediately before the test section. Test section dynamic pressure was measured across the two static rings using an Omega PX2650 differential pressure transducer. The test section is sealed; therefore, the local static pressure was measured at both the upstream and downstream ends of the test section, in order to account for buoyancy effects. Test section static pressure was measured using two Omega PX2650 differential pressure transducers. The static temperature was measured near the test article using a K-type thermocouple and a National Instruments NI-9213 thermocouple amplifier. A model-following controller designed in LabVIEW controls the wind tunnel velocity. The controller allows for fast ramping and precise control over the test section velocity.

The motor revolution speed, current, voltage, torque, and thrust signals were sampled using a National Instruments NI-6009 data acquisition board. The DAQ gathers analog and digital information from all sensors at 10 kHz. The sampling rate was sufficiently higher than the vibrational modes of the thrust stand (2 kHz), which limited aliasing and allowed for adaptive filtering in the data reduction. The motor was mounted directly to the load cell for force and torque measurement, and the motor/load cell assembly was positioned in the center of the test section on a faired mount. A vibration isolation mount held the motor and increased the natural resonant frequency of the motor/propeller combination beyond the upper RPM limit. The vibration mount made the calibration of the thrust and torque sensor more difficult by introducing elasticity into the system. However, the calibrations were repeatable over the course of the experimental campaign.

Testing of the propellers at a constant Reynolds number across a range of advance ratios requires a reduction of the propeller RPM while the tunnel speed is increased. For this testing, the Reynolds number is evaluated based on the resultant freestream and rotational velocity components evaluated at 70% of the blade span. If the density and viscosity are treated as constants, the constant Reynolds number testing reduces to maintaining a resultant velocity evaluated at 70% of the blade span. The resulting variation in chord based Reynolds number is shown in Figure 11 for a blade based Reynolds number of 5×10^5. Coordinated control of the tunnel/propeller combination for constant-*Re* is complicated by the fact that the propeller induces significant flow in the tunnel. Furthermore, for the fixed-pitch propellers tested in this work, as the freestream velocity changes the angular velocity of the propeller also changes (even when applied power is held constant). This is due to a decreased angle of attack at high advance ratios, resulting in a drag reduction. The drag reduction decreases the power required to drive the propeller, resulting in a higher rotational velocity.[60]

$$n = \sqrt{\frac{V_l^2 - V_\infty^2}{\left(2\pi 0.7r\right)^2}}; V_l = \frac{\mathrm{Re}\,\mu}{\rho D} \tag{2.16}$$

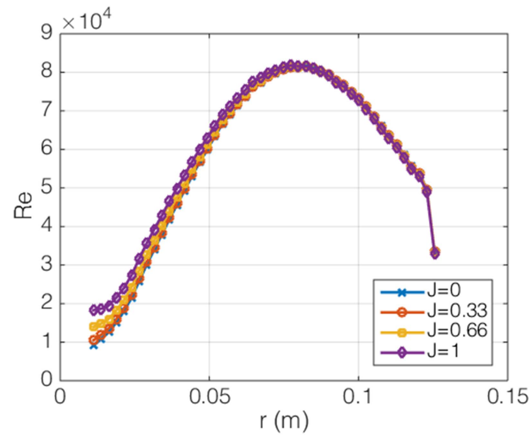


Figure 11: Variation in chord based Reynolds number during constant Reynolds testing at Re=1×10^5 for different advance ratios.

For all of these reasons, careful real-time closed-loop control of the propeller speed and tunnel velocity must be done in order to maintain the desired Reynolds number. The is controlled via LabVIEW and a standalone RPM and throttle controller based on a Parallax Propeller microcontroller. Using these devices, the propeller Reynolds number varied less than 3% from the desired set point over the full range of advance ratios.

2.4.2 Wind Tunnel Correction Factors

Two correction factors were used to account for the influence of the wind tunnel on the performance of the system. The first accounts for the tunnel wall effects which can increase the static pressure in the propeller wake.[61] Following the work of Barlow *et al.*[61] and Brandt and Selig,[7] the following corrected velocity is used in place of the measured tunnel velocity,

$$\frac{V_{\infty,\,corr}}{V_\infty} = 1 - \frac{TA/qC}{2\sqrt{1+2\left(T/q\right)}}\,. \tag{2.17}$$

The second correction addresses the local velocity increase caused by the presence of a three-dimensional body in the wind tunnel (solid blockage). To correct the artificial increase in velocity, the method proposed by Brandt and Selig was used.[7] Briefly, the increase in velocity is related to the relative volume of the test piece and the overall wind tunnel geometry. The relationship for the relative increase in velocity is given by[61]

$$\frac{V_{\infty,\,corr}}{V_{\infty}} = \frac{K\tau\boldsymbol{\upsilon}}{A^{3/2}}, \tag{2.18}$$

where $K = 1.045$, $\tau = 0.92$, and the volume of the fairing ($\boldsymbol{\upsilon}$) is 3.28e^{-3} m^3.

2.5 Results

This section compares the predictions generated using the BEM model and data gathered during experimental testing of the three propellers considered in this work. The first section details the full power, high RPM performance of the propeller via modeling and experimental results. Using this data, the effects of the three proposed corrections to the BEM are each illustrated separately. The effects of varying the Reynolds number are then demonstrated and compared to experimental data gathered using the constant Reynolds testing method outlined previously. Finally, the complete system performance is detailed using measurements of the electrical efficiencies of the supporting propulsion hardware.

Uncertainty in the experimental data was estimated via a Monte Carlo simulation of the complete data acquisition and reduction process.[62] The resulting uncertainty varies with the particular experimental conditions (for example, due to the quoted accuracy of the pressure transducers used for wind tunnel freestream velocity measurements, low speed data points will have higher uncertainty). Thus, the following experimental data sets are presented with error bars on each data point that express the 95% confidence interval for advance ratio, thrust coefficient, torque coefficient, and propulsive efficiency.

2.5.1 Full Power Performance

2.5.1.1 Thrust and Torque Coefficients

The initial set of data presented here is for the propeller operating at full power (fixed voltage) across a range of advance ratios. This experimental data, and the corresponding BEM analytical prediction, serves to quantify the uncertainty associated with the experimental data and to validate the model. Furthermore, full power tests demonstrate the upper performance limit for each propeller and establish bounds for the mapping between $\eta_{propulsive}$, J, and the Reynolds number.

For these initial validation tests at full power, the advance ratio was controlled by varying the test section velocity and the Reynolds number was not strictly controlled. In this type of testing, the blade Reynolds number will vary on the order of 10% as flow conditions change. This is because an increase in freestream velocity reduces the blade angle of attack, thus reducing the induced drag and the motor torque requirement (for a fixed-pitch propeller). The reduction in motor torque required, while full power is applied, allows the motor to rotate at a higher speed. An increase in rotational speed results in changes to the local Reynolds number and tip Mach number in a parabolic fashion. The Reynolds number used for the BEM model prediction was selected as the average propeller Reynolds number encountered during the full power testing.

Figure 12 shows the BEM model results for the predicted and measured thrust and torque coefficients. The overall trend of the BEM model data matches the experimental data very well, with any differences between the two being less than the experimental uncertainty for all three propellers studied in this work. The decrease in the experimental C_T uncertainty at high advance ratios is due primarily to the increase in RPM. The uncertainty associated with density and diameter are small and do not change appreciably with increasing J. The variation in J at low advance ratios is due primarily to the dominant influence of the static and dynamic pressure uncertainties. As the advance ratio increases, this effect is somewhat offset by the increasing rotational velocity.

The accuracy of the BEM model thrust prediction is largely dependent on the fidelity of the local lift coefficient estimates. Within the advance ratios considered, the lift coefficient varies appreciably from –1 to 1.25 with an angle of attack variation between –30 and 25 degrees, as shown in Figure 14. The variation in angle of attack and lift coefficient is much more significant near the hub, which indicates that this region of the blade could drive the accuracy of the model under certain conditions. The data presented compares favorably with Brandt and Selig's investigation of the APC thin electric 10×5E and 10×7E at lower angular velocities (Figure 13).[63] This comparison is based on experimental testing performed at similar operating conditions to those outlined by Brandt and Selig and involved holding the propeller RPM constant as the freestream velocity was increased.

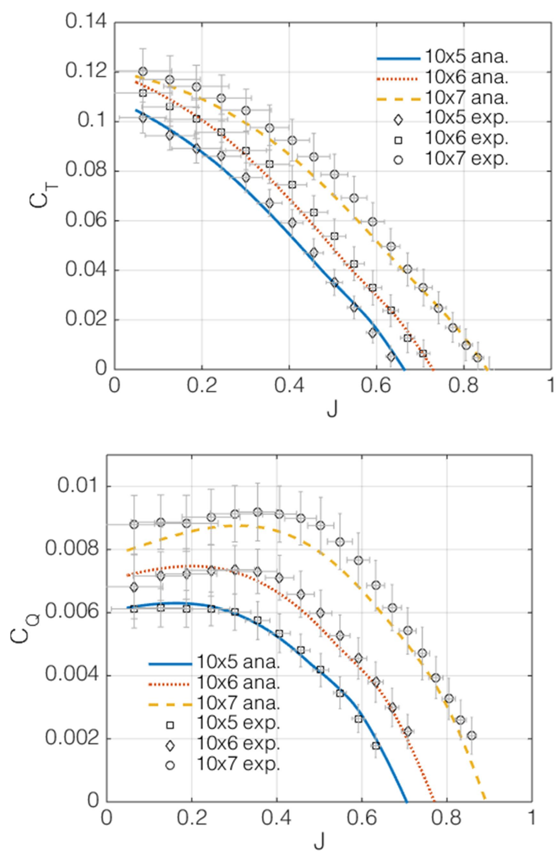


Figure 12: Analytic and experimental thrust (top) and torque (bottom) coefficients for the APC 10×5E, 10×6E, and 10×7E at Re ≈ 1.5×10^6. Error bars depict uncertainty in experimental data.

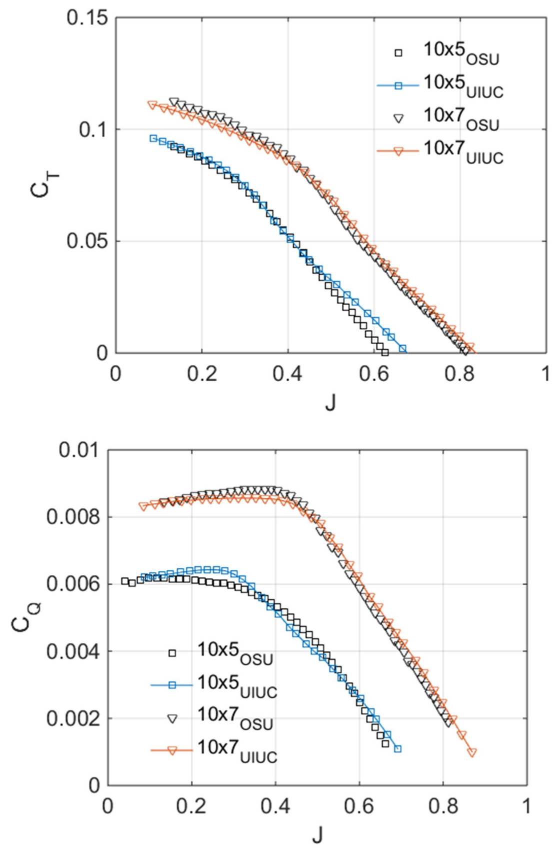
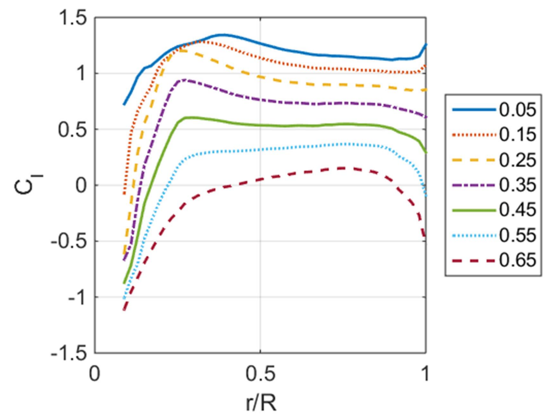


Figure 13: Thrust coefficient (top) and torque coefficient (bottom) versus advance ratio for APC 10×5E, and 10×7E. Experimental data presented in this work and from Brandt, published in the UIUC database at ~6000 RPM.[63]



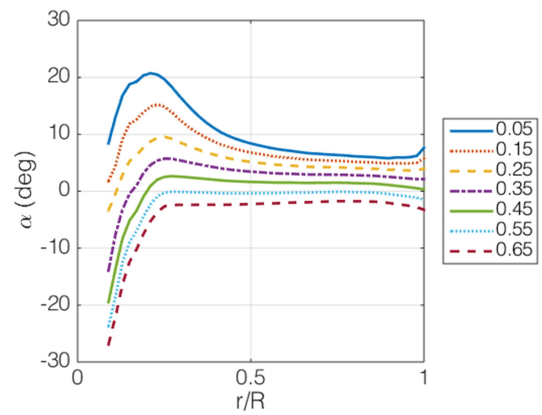


Figure 14: Spanwise C_l (top) and angle of attack (bottom) variation at different advance ratios for APC 10×5E at Re ≈ 1.5×10^6.

The difference between experimental and BEM torque results (Figure 12) at low advance ratios for the APC 10×7E is likely due to the angle of attack being beyond stall over the majority of the blade. In this condition, the predicted torque is sensitive to the magnitude of the drag coefficient which increases parabolically with angle of attack. At low advance ratios, the inboard sections of the blade are stalled as shown in Figure 15. These stalled regions largely govern the required torque. As the advance ratio is increased, the torque required is relatively constant until the local angle of attack drops below the stall angle. Below the stall angle, the required torque drops dramatically and the blade efficiency will increase.

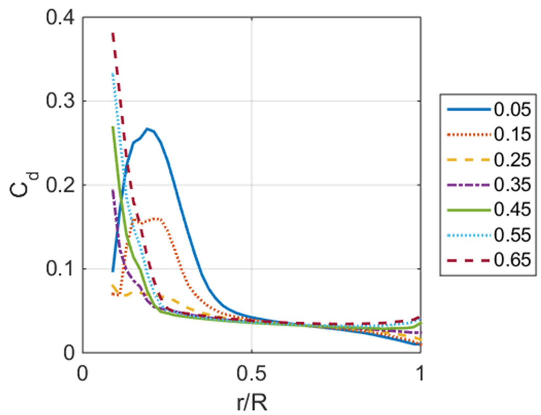


Figure 15: C_d versus spanwise location for different advance ratios for APC 10×5E at Re ≈ 1.5×10^6.

2.5.1.2 Propulsive Efficiency

Figure 16 shows the BEM model predictions and experimental measurements of the propulsive efficiency for the three propellers considered in this work. The experimental measurements closely match the predicted performance provided by the BEM model. The uncertainty in propulsive efficiency at low advance ratios is dictated primarily by the uncertainty in the airspeed. As the advance ratio increases, the ratio of uncertainty in the thrust and moment is relatively constant. Therefore, increasing RPM at higher advance ratios decreases the total uncertainty.

The design pitch of the propeller can be determined by finding the upper advance ratio limit at which the efficiency is zero. At this point, the propeller is essentially acting as an airscrew and the pitch is therefore equal to the advance ratio multiplied by the diameter of the propeller. Table 2 summarizes the pitch determined from BEM theory and from experimental data, along with the pitch specified by the propeller manufacturer. These results show that the model and experimental data are in close agreement with one another. The manufacturer's reported pitch for all three propellers is lower than the experimental and analytic estimates, which is consistent with other investigations of similar propellers.[7, 8] The performance variation is more dramatic when comparing propellers of the same diameter and pitch from different manufacturers.[7, 8, 59]

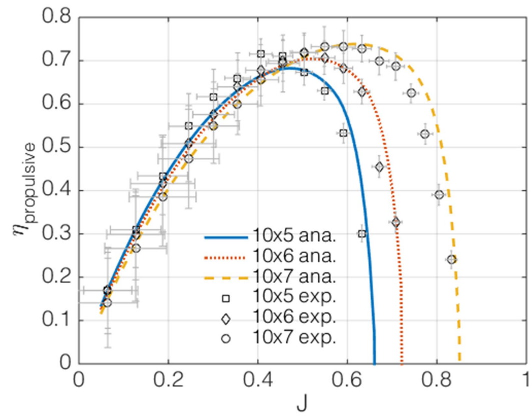


Figure 16: Analytic and experimental propulsive efficiency for the APC 10×5E, 10×6E, and 10×7E at Re ≈ 1.5×10^6, error bars depict uncertainty in experimental data.

Table 2: Analytic advance ratio

Propeller	Published pitch (inches)	Analytic pitch (inches)	Experimental pitch (inches)	Percent Difference Pub./Ana.	Percent Difference Pub./Exp.	Percent Difference Ana./Exp.
APC 10×5E	5	6.4	6.5	28%	30%	1.5%
APC 10×6E	6	7.2	7.4	20%	23%	2.7%
APC 10×7E	7	7.9	8.2	13%	17%	3.8%

2.5.2 BEM Model Correction Factors

Having established the fidelity of the BEM model to the experimental data, the significance of the contribution from each correction factor will be evaluated in turn. The correction factors applied to the BEM model are a tip loss model, Mach correction, and quasi-three-dimensional spanwise flow correction. To demonstrate the effect of each correction individually, the BEM model was run with each switched off in isolation. A comparison between the BEM model with no corrections is also presented for reference. The resulting model data is then shown relative to the experimental data, with thrust and torque data serving as the basis for comparison.

2.5.2.1 No Correction Factors

For comparative purposes, the BEM model is presented with no corrections, as shown in Figure 17. The fully corrected BEM model results and experimental data are also shown. The basic trend of the uncorrected model is in reasonable agreement with the experimental data at moderate advance ratios, but there is a significant offset across all conditions. Furthermore, at low advance ratios, the model fails to capture the trend present in the experimental data. Thus, the uncorrected BEM model is generally insufficient for predicting the propeller performance.

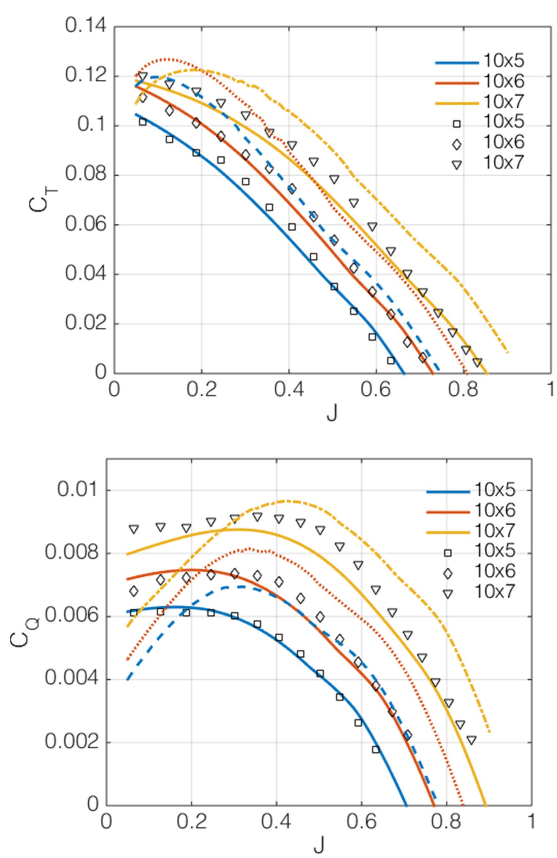


Figure 17: BEM model results with no corrections (dashed lines). Fully corrected BEM model (solid lines) and experimental results (symbols) shown for comparison at Re ≈ 1.5×10^{6}.

2.5.2.2 Tip Loss Correction

The Prandtl tip loss correction model forces the lift to approach zero at radial locations approaching the blade tip, in an effort to capture the pressure equalization occurring near the tip of finite lifting surfaces. Tip losses are most pronounced at low advance ratios where the blade is highly loaded and, if not stalled, operating with high local lift and drag coefficients. The effect of neglecting the tip loss model is shown in Figure 18. In this figure, the largest differences between experimental data and the BEM model occur below advance ratios of approximately 0.5. Below J=0.5, the higher lift and drag near the tip (without tip-loss correction) results in higher predicted torque and thrust.

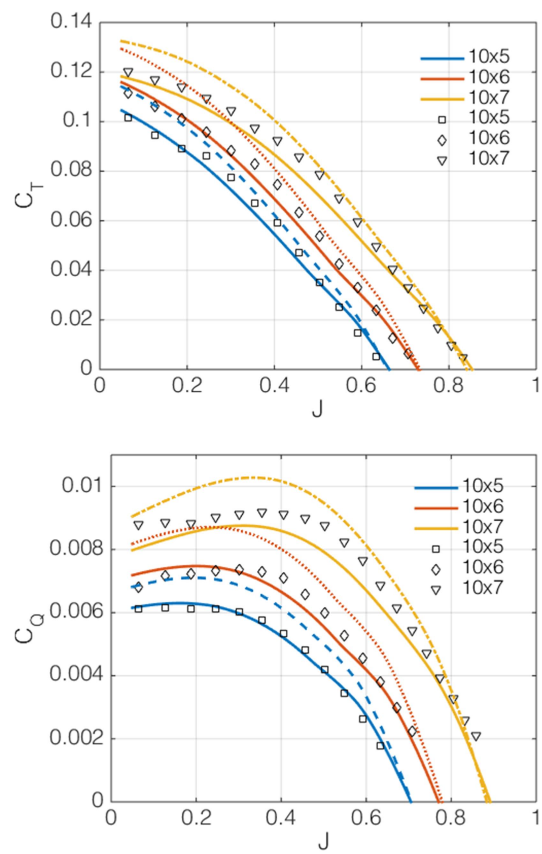


Figure 18: Thrust coefficient (top) and torque coefficient (bottom) for BEM model predictions without tip loss correction (dashed lines) and fully corrected BEM model (solid lines). Experimental results (symbols) shown for $Re \approx 1.5\times10^6$.

2.5.2.3 Mach Correction

The Mach correction increases the lift curve slope as a function of local Mach number. This effect is pronounced when high lift coefficients and high rotation rates are simultaneously present. These conditions occur at low advance ratios as evidenced in Figure 19. The BEM model without Mach correction under predicts the thrust at low advance ratios, particularly for the more highly loaded 10×7E propeller. The lower lift coefficients associated with the BEM model without Mach correction result in lower drag, and under predicted power (torque) to drive the propeller.

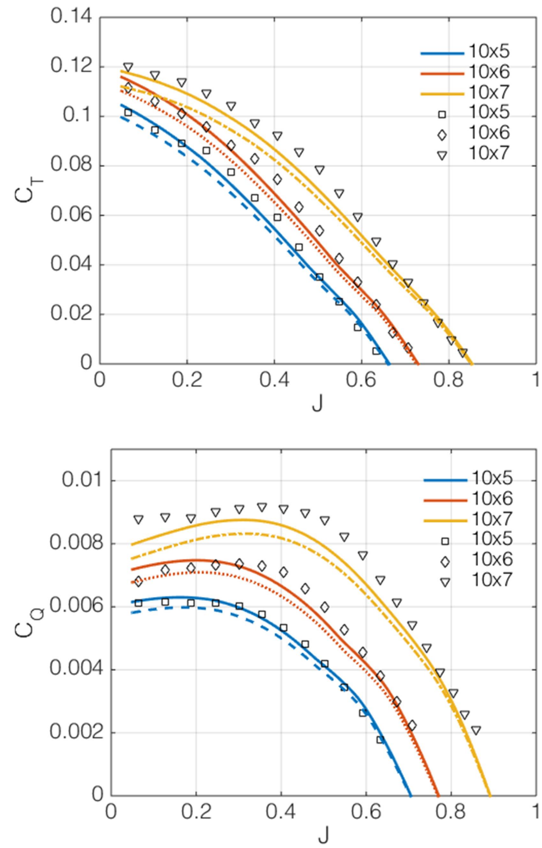


Figure 19: Thrust coefficient (top) and torque coefficient (bottom) for BEM model predictions without Mach correction (dashed lines) and fully corrected BEM model (solid lines). Experimental results (symbols) shown for $Re \approx 1.5 \times 10^6$.

2.5.2.4 Three-Dimensional Flow Correction

The three-dimensional correction modifies the local lift coefficient parabolically in the spanwise direction. The correction concentrates the largest effect near the tip region. In this region, the local lift coefficient is typically lower than the lift coefficient from potential flow theory, thus providing an amplification to the normal force coefficient. At low advance ratios this increases the lift and drag components. This translates to an increase in thrust and torque. As the advance ratio increases, the ratio of the rotational velocity to the local velocity decreases, thus diminishing the effect of the correction. The effect increases with the difference between the potential flow lift coefficient and the local 2-D lift coefficient. For a highly loaded blade, such as the 10×7, the reduction in thrust and torque (shown in Figure 20) at low advance ratios caused by omitting the correction is clearly visible. The advance ratios over which the correction is effective is complicated by the inclusion of the local velocity component. The BEM model approximates the axial and radial induction factors iteratively, making the analytic application of the correction difficult.

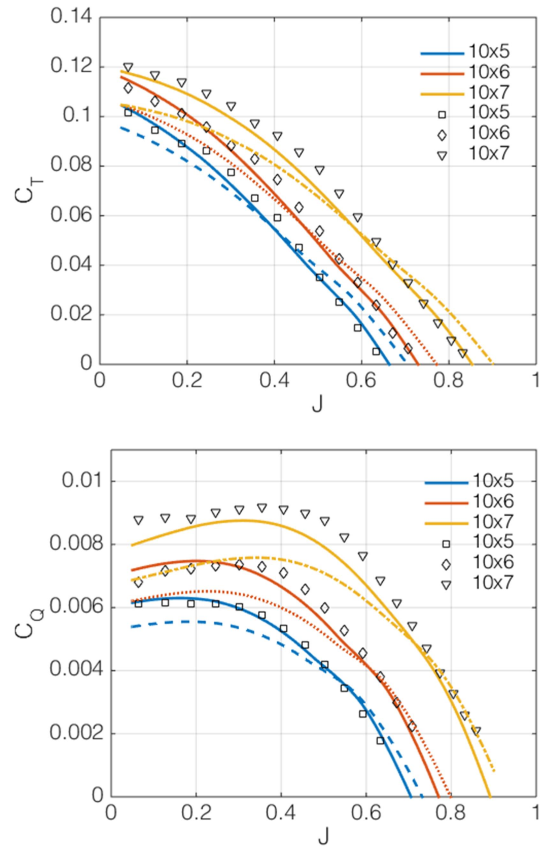


Figure 20: Thrust coefficient (top) and torque coefficient (bottom) for BEM model predictions without three-dimensional correction (dashed lines) and fully corrected BEM model (solid lines). Experimental results (symbols) shown for $\mathrm{Re} \approx 1.5{\times}10^6$.

2.5.3 Reynolds Number Effects

The BEM model was used to predict the performance of propellers at varying Reynolds numbers. The inclusion of the BEM model is critical to this effort as the local variation in spanwise Reynolds number is appreciable on the scale of the propellers tested, and is in the range where viscous effects become significant. The effects of Reynolds number have been explored by others for sUAS propellers.[7, 14] Both Deters *et al.* and Brandt & Selig presented experimental data for similar propellers tested at a constant RPM, showing a reduction in propulsive efficiency at lower chord based Reynolds number. The present work seeks to build on these studies by tightly controlling the propeller diameter based Reynolds number over the range of positive thrust advance ratios. The predictions generated by the BEM model are then compared to wind tunnel measurements using the same experimental apparatus as the full power tests. For the experimental data presented in this sub-section, the diameter based Reynolds number was tightly controlled via the closed-loop testing method described in the experimental setup. Here, the Reynolds number for the experimental data is defined based on the propeller diameter and blade-relative velocity occurring at 70% of the span. The same operating conditions are applied to the BEM model results, where sectional aerodynamic coefficients were determined based on local Reynolds number.

Figure 22 shows the variation in propulsive efficiency for the APC 10×7E at different Reynolds numbers. For low advance ratios, the overall trend is unaffected by the change in Reynolds number. At high advance ratios, however, there is a significant impact of Reynolds number. Efficiency curves for fixed-pitch propellers are characterized by an rapid drop in efficiency near the maximum advance ratio. As seen in Figure 22, the rapid drop in propeller efficiency occurs at lower advance ratios when the Reynolds number is decreased. This is because the NACA 4412 airfoil section used on these propellers experiences a dramatic increase in drag and decrease in lift over a relatively small decrease in Reynolds number (Figure 21). It is important to note that, in the current modeling environment, the Reynolds number is evaluated on the quasi-2D BEM model before the three-dimensional correction is applied (a one-way coupling of viscous effects). However, the spanwise variation in local velocity and local Reynolds effects are expected to be tightly coupled (two-way). Accurate modeling of these coupled viscous effects would require much higher fidelity modeling, which would be prohibitively complex and expensive for the current purposes. Despite the low fidelity of this viscous modeling approach, the BEM theoretical curves still show good agreement with the experimental data and clearly show the appropriate changes in the propeller efficiency curves as Reynolds number changes.

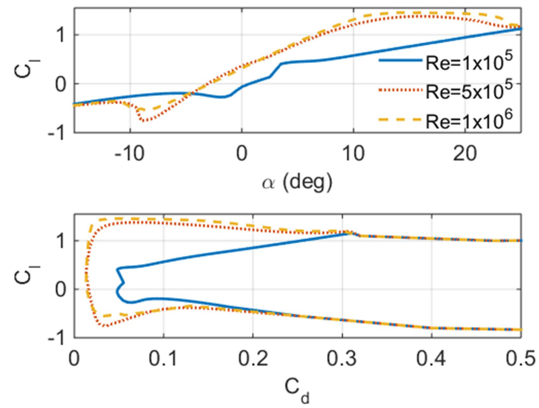


Figure 21: C_l versus α and C_l versus C_d for NACA 4412 at different Reynolds numbers, generated using XFOIL[56]

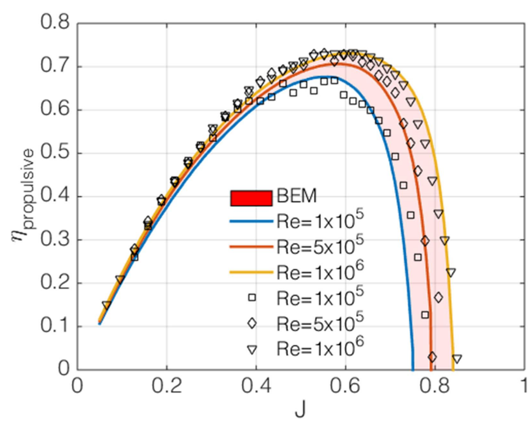


Figure 22: BEM model (lines and shaded region) and experimental (symbols) propulsive efficiency for the APC 10×7E at varying Reynolds numbers.

The performance difference at the three representative Reynolds numbers can be explained by a spanwise variation in aerodynamic characteristics. For the constant Reynolds number tests, the ratio of the freestream velocity and rotational velocity remains constant. Therefore, the angle of attack is relatively constant between Reynolds cases. Subtle variations are present due to the induced velocity components that change as a function of the local lift and drag coefficients. Figure 23 shows the BEM model predicted angle of attack for the three Reynolds number test cases at the same advance ratio. Although the angle of attack changes very little with Reynolds number, the lift and drag coefficients vary significantly, as shown in Figure 24.

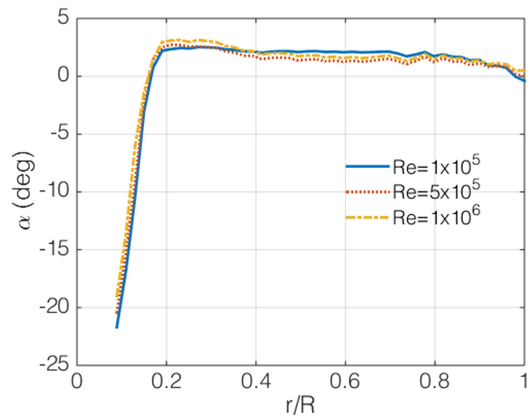


Figure 23: Spanwise angle of attack variation for different Reynolds numbers, APC 10×7E at *J* = 0.55.

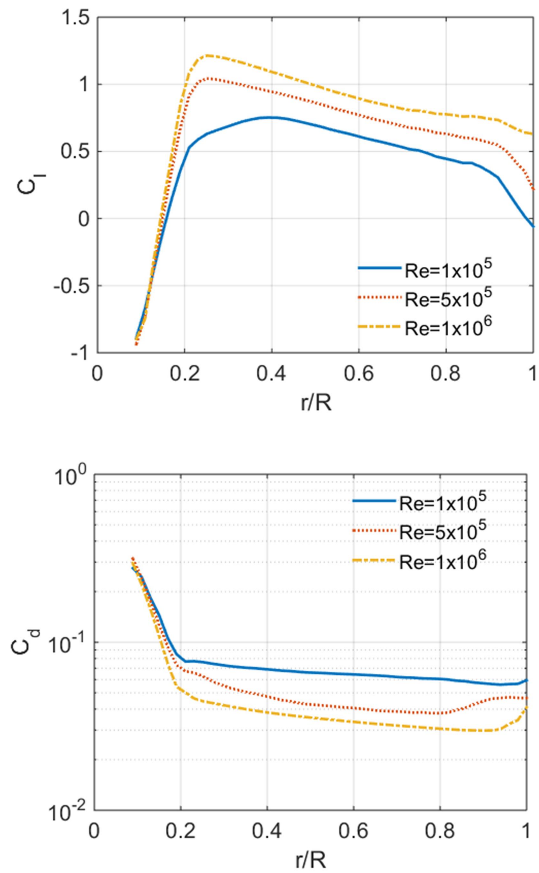


Figure 24: Spanwise C_l (top) and C_d (bottom) for different Reynolds numbers, APC 10×7E at J = 0.55.

Figure 25 shows the predicted variation in performance over a range of Reynolds numbers for the three propellers tested in this work. From this figure it is evident that the effect of changing Reynolds numbers is equivalent in some cases to operating with a different propeller pitch. It is clear from this plot that incorporating the Reynolds number into the basic BEM model is of first-order importance for propellers on this physical scale. In some cases, incorporation of Reynolds-varying aerodynamic parameters has a larger effect on the predicted results than the three other correction factors combined (Prandtl tip loss, compressibility, and 3D corrections). Incorporating the Reynolds number in to the BEM is increasingly important at low rotation rates where the local variation in lift and drag is much larger. At higher Reynolds numbers, the performance shows a much smaller overall variation as the change in aerodynamic parameters is not as severe.

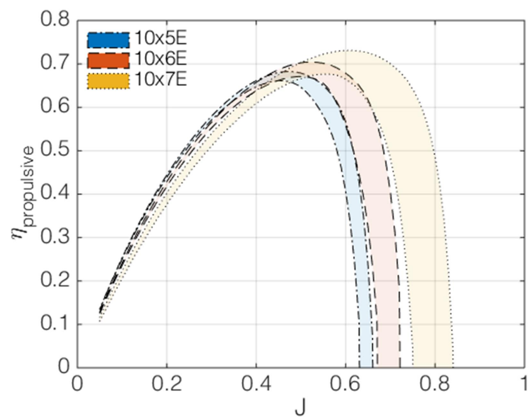


Figure 25: Variation in performance for the APC 10×5E, 10×6E, and 10×7E operating with Reynolds numbers between 1×10^5 and 1×10^6.

2.5.4 System Efficiency

The final characterized component is the electrical efficiency of the power system. The electrical efficiency term allows for the correlation between measured electrical power and propulsive efficiency. The electrical efficiency is given as the ratio of motor and applied electrical power. This encompasses the power delivery system and motor mechanical efficiency. Typical efficiencies for the power system components range from 70% to 90% and depend on operating temperature and electrical load.[8] The battery represents another efficiency term, but as the power in this work is measured downstream of the battery terminals, it does not impact the overall measured system efficiency.

Figure 26 shows the electrical efficiency for the three representative propellers tested at full power. This figure shows an appreciable and non-linear variation in electrical efficiency parameterized by advance ratio. To utilize this data in a predictive model, a surface fit is necessary. From Figure 26, approaching the full power design pitch, the electrical efficiency decreases by up to 10%. The speed controller efficiency decreases at high angular speeds and low power output, likely due to internal switching inefficiencies and timing delays. The variation in electrical efficiency versus Reynolds number shows similar trends. At low Reynolds numbers, the efficiency decreases as the advance ratio approaches the design pitch. As the Reynolds number increases, the efficiency at low advance ratios plateaus before sharply decreasing close to the design pitch. The trend is monotonic and similar between propellers operating at the same Reynolds number. The similarity in trends allows for the generation of a parameterized surface fit. Figure 27 shows an example of the surface fit for the electrical efficiency versus advance ratio. From this plot, the monotonic trend is fit by a third-order polynomial surface. The resulting fit has an adjusted R-squared value of 0.89. The electrical efficiency is combined with the propulsive efficiency to determine the overall system efficiency as a function of Reynolds number and advance ratio. The surface fit, shown in Figure 27, allows for inflight estimation of the power available and thrust based on measured airspeed, electrical power, and propeller rotation speed. The parameterized surface fit is a critical component necessary for future vehicle performance estimation. This surface fit could have a significant overall impact on the range and/or endurance of an electric powered sUAS.

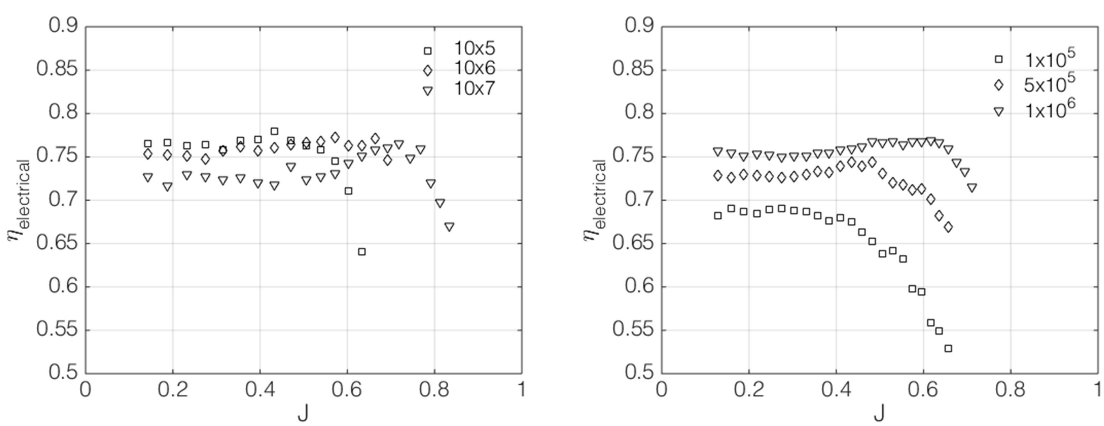


Figure 26: Electrical efficiency for the APC 10×5E, 10×6E, and 10×7E tested at full power (left). Electrical efficiency for the APC 10×6E tested at different Reynolds numbers (right).

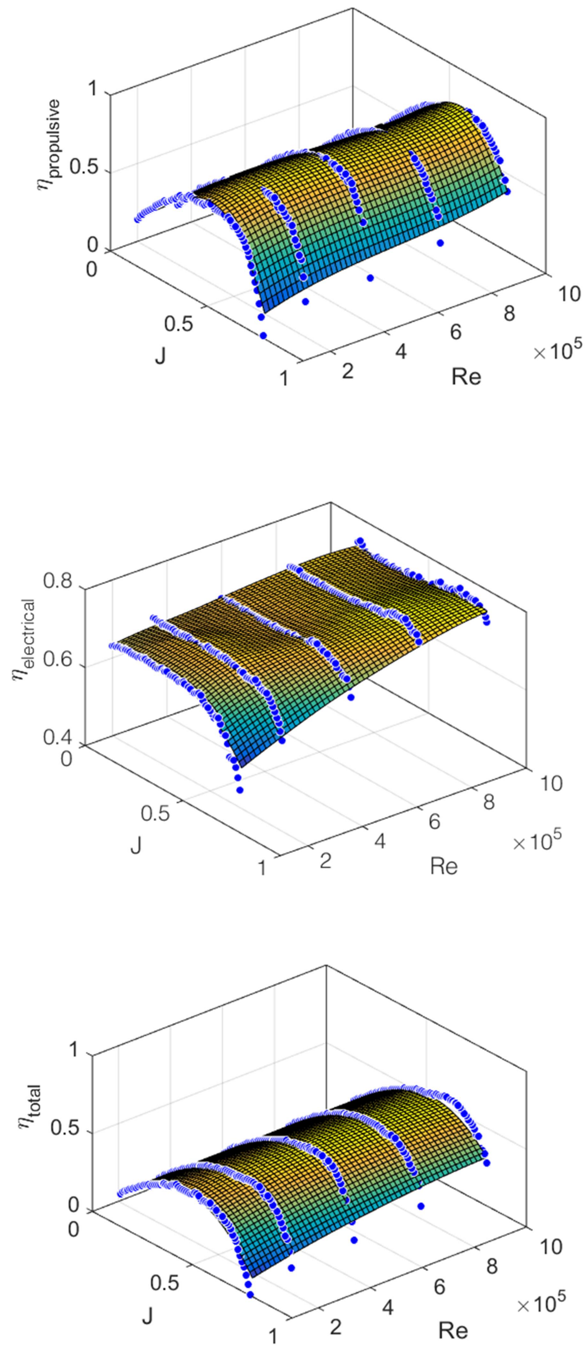


Figure 27: Surface fit of propulsive efficiency versus *Re* and J (top), electrical efficiency versus *Re* and *J* (middle), and the total system efficiency versus *Re* and *J* for the APC 10×7E (bottom).

2.6 Conclusion

A BEM model is presented and used for performance predictions on sUAS propellers. Several corrections were proposed to the BEM model to capture features unique to the rotational flow around finite low-Reynolds number propellers. Namely, the use of an XFOIL-generated aerodynamic database, tip loss corrections, Mach corrections, and the incorporation of spanwise flow components were added to the model. For the specific propeller geometries considered in this work, the BEM model predictions follow the general trends expected for fixed pitch propellers.

The BEM model was validated by a series of wind tunnel tests. From these tests, favorable comparisons between the predicted and measured theoretical pitch were noted. Full power testing yielded results similar to those published in previous studies of small-scale propellers. A novel constant Reynolds number test was presented to demonstrate the effects of scale on the propulsive efficiency. The favorable comparison between experimental and model-based performance metrics such as propulsive efficiency and thrust/power coefficients point towards the importance of incorporating Reynolds dependency in the analysis of small-scale propulsion systems. The methodology for determining the performance of the remaining individual propulsion components such as the motor and electronic speed controller were presented. Using these estimates, the system efficiency neglecting the battery was determined. These estimates allow for a direct correlation between measured electrical power, advance ratio, and propulsive efficiency. From the propulsive efficiency, the thrust and power available can be estimated.

The BEM model presented and validated here is highly useful for propeller design for sUAS, particularly since the operating Reynolds numbers of these vehicles is low enough where viscous effects are of first-order importance. Furthermore, the overall power available model for the sUAS propulsion system enables high fidelity vehicle performance estimates for sUAS and inflight determination of vehicle performance for flight testing and routine operations.

Chapter 3: Navigation Systems

3.1 Navigation Systems Overview

This chapter details the development of the onboard navigation system used to estimate the attitude, velocity, and position of the sUAS during flight. This system is comprised of an inertial measurement unit (IMU), flight management computer (FMC), and external aiding sensors including a global navigation satellite system (GNSS, or GPS in this work), magnetometer, barometer, and air data system. These components are combined to create an inertial navigation system (INS). The accuracy of the state estimates provided by the INS are critical to the vehicle control and performance estimations presented in later chapters. Therefore, details of the fundamental relations used for state estimation, the sensors used to propagate the states, and the methods for controlling the accuracy of the states are presented.

The INS used for state estimation onboard the vehicle uses three reference coordinate frames for the calculation of its attitude, velocity, and position. The sensor-fusion filter uses these same reference frames to determine the propagation of error in the predicted vehicle states. The choice of reference frame depends on the exact form of the navigation equations used, but in principle can be classified as either an inertial or non-inertial reference frame.[23, 64, 65] The principle coordinate system used in the development of the navigation equations are the Earth-Centered, Earth-Fixed (ECEF), navigation (NAV), and body frames. Subsequent sections detail these reference frames and the relations between them.

Using these coordinate frames, the equations of motion for the vehicle's attitude, velocity, and position are developed. The error in these estimates form the basis for the sensor fusion filter presented in section 3.3. The equations of motion and the resulting error equations form the mathematical framework on which the INS is based. The mechanization of these relations uses sensors mounted on the vehicle to measure angular rates and accelerations in the body frame. Sensor data combined with the equations of motion allow sensed rotations and accelerations in the body frame to be translated into velocities and displacements in the NAV and ECEF frames. The sensors used in the state estimation and correction are detailed with specific focus on their individual error sources. Calibration and bias estimation routines are presented which reduce the overall uncertainty in state estimation. Finally, the sensor fusion algorithm used to correct the state estimates based on the proposed error equations is presented. The performance of this algorithm in controlling the growth of error in the estimated vehicle states is then quantified.

3.1.1 Coordinate Frame Definitions

The principle coordinate system used in the development of the navigation equations is the ECEF frame. This reference frame has its Z-axis aligned with the Earth's poles and the X-axis aligned with the intersection of the Prime Meridian and equator (Figure 28).[15] The Y-axis then completes the right-handed coordinate system. The ECEF coordinate frame rotates with the Earth and is therefore a non-inertial reference frame. The second coordinate frame is the local NAV frame. This frame is tangent to the Earth's surface at the location of the vehicle. It is oriented such that the north-axis points toward the Earth's north pole, and the down-axis points towards the center of the Earth. The East axis then completes the right-handed coordinate system (shown in Figure 28). The final coordinate frame is the body-fixed frame. This coordinate frame has the x-axis aligned with the longitudinal axis of the aircraft, and the y-axis pointing towards the right wing tip (Figure 29). The z-axis points down through the body, completing the right-handed coordinate system.

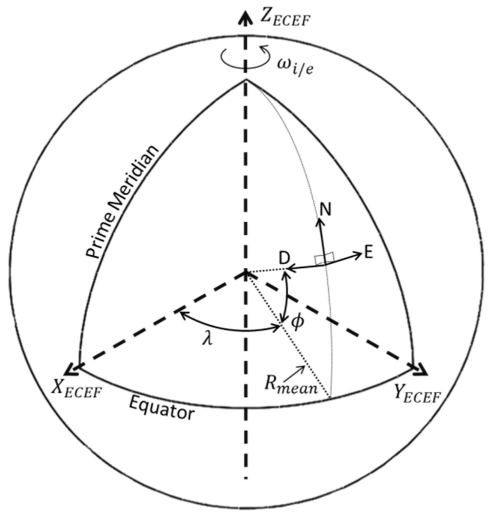


Figure 28: ECEF and NAV coordinate frames.

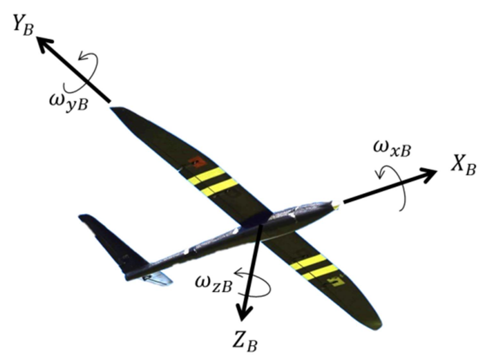


Figure 29: Body frame axes with positive rotation directions indicated, view from bottom of vehicle.

3.1.2 Attitude Parameterization

The attitude of the vehicle is specified uniquely by three Euler angles, pitch, roll, and yaw. However, if the attitude is formulated using the three Euler angles, singularities exist at pitch angles nearing 90 degrees. Therefore, the vehicle attitude is formulated and propagated using a four-element quaternion. The quaternion is a singularity free, compact method for storing and updating the coordinate frame orientations.[23, 64, 65] The quaternion is comprised of a single axis and a corresponding rotation given as

$$q = \begin{bmatrix} \cos\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) u\hat{\boldsymbol{i}} \\ \sin\left(\frac{\theta}{2}\right) u\hat{\boldsymbol{j}} \\ \sin\left(\frac{\theta}{2}\right) u\hat{\boldsymbol{k}} \end{bmatrix}. \tag{3.1}$$

From this definition, θ represents the angle by which the body is rotated about the axis u with orthogonal components given by the unit vector $\left[\hat{\boldsymbol{i}}, \hat{\boldsymbol{j}}, \hat{\boldsymbol{k}}\right]$. The axis and rotation angles are shown in Figure 30 for a right- handed coordinate system. In addition to the quaternion parameterization, equivalent direction cosine matrices are used in the attitude, velocity, and position equations denoted by C_a^b where a and b refer to the starting and ending coordinate frame. Direction cosine matrices are typically more computationally efficient for rotating vectors between reference frames. Mathematically the quaternion and direction cosine matrix are equivalent in three-dimensional space. Euler angles are not used in the development of the attitude, velocity, or position equations due to their well-known singularities and extensive use of transcendental relations.[22, 64, 66]

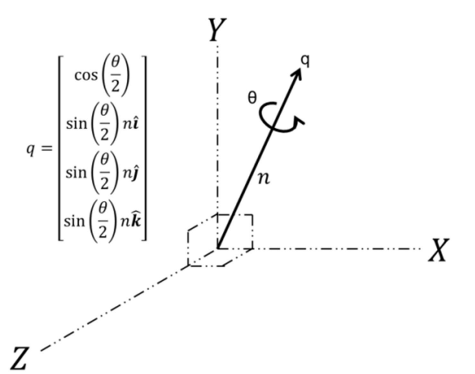


Figure 30: Quaternion representation

3.2 Navigation Equations

This section details the navigation equations used to transform body-referenced measurements into the states necessary for attitude, velocity, and position estimation.

3.2.1 Attitude Equations

The attitude of the vehicle is used to transform the specific force measured by the accelerometers to the NAV frame for velocity and position estimation. This necessitates a precise and bias-free attitude estimate to limit the buildup of error through the integration of the specific force measurements. Propagation of the attitude solution uses a strapdown gyroscope system. Strapdown sensors provide an estimate of the vehicle's angular rate in inertial space. These measurements include body frame and inertial, i.e. Earth rotational rate components. The measured rotational rate provided by the strapdown sensors is integrated to propagate the attitude of the vehicle. This mechanization differs from mechanical gyroscope mechanizations, which typically provide a direct indication of vehicle attitude. Mechanical systems have been used widely for inertial applications, but are typically larger and therefore not directly applicable to small-scale UAV guidance systems. MEMS sensors are used exclusively in this work. These sensors typically only provide rate output information, necessitating attitude determination via integration rather than observation. Methods for accurate rate integration and attitude propagation are therefore the focus of this section.

The vehicle attitude is parameterized as a quaternion. This parameterization provides an efficient and singularity-free means of tracking the vehicle's attitude. The time dependent attitude of the vehicle is expressed in quaternion form as

$$\dot{q} = \frac{1}{2}\omega_t q\,, \tag{3.2}$$

where

$$\vec{\omega}_t = \vec{\omega}_B + C_B^{NAV}\vec{\omega}_{in}\,. \tag{3.3}$$

In these expressions, q is the vehicle quaternion, ω_t is the total vehicle rotation rate, $\vec{\omega}_B$ is the body referenced gyroscope output, C_B^{NAV} is the body to NAV rotation matrix, and $\vec{\omega}_{in}$ is the combined Earth and NAV frame rotation rate vector given as[65]

$$\bar{\omega}_{in} = \begin{bmatrix} \omega_{i/e}\cos(\phi) \\ 0 \\ -\omega_{i/e}\sin(\phi) \end{bmatrix} + \begin{bmatrix} \dfrac{V_E}{R_{mean}+h} \\ -\dfrac{V_N}{R_{mean}+h} \\ -\dfrac{V_E}{R_{mean}+h}\tan(\phi) \end{bmatrix}, \tag{3.4}$$

where R_{mean} is the average Earth radius, h is the height above the mean radius, ϕ is the latitude, λ the longitude, and $\omega_{i/e}$ is the Earth's rotation rate given by the WGS84 Earth model.[67] Given the low speed and range of the vehicle considered in this work, the contribution of $\vec{\omega}_{in}$ to the vehicle attitude is typically very small, approximately 10^{-4} rad/s. When the system is initialized, the bias is removed from the gyroscope signals, including signal components linked to the Earth's rotation rate. After initialization, the in-situ resolution of the gyroscopes do not allow for direct measurement of the Earth rate. Therefore, the NAV frame rotation computed using equation (3.4) is used to periodically update the attitude solution based on position and velocity information provided by the INS and GPS sensors.

Integration of the body referenced angular rates provides the means for propagating the attitude solution in flight. The discrete time version of equation (3.2) is given as

$$q_{t+1} = q_t + \int_t^{t+1} \dot{q}_t dt = q^t q_t^{t+1} \tag{3.5}$$

In this expression, the q^t term represents the attitude quaternion at the current time step. q_t^{t+1} represents the quaternion update between time steps t and $t+1$. The update quaternion is given by[28]

$$q_t^{t+1} = \begin{bmatrix} \cos\left(\frac{\|\Phi\|}{2}\right) \\ \sin\left(\frac{\|\Phi\|}{2}\right)\frac{\Phi}{\|\Phi\|} \end{bmatrix}, \tag{3.6}$$

where Φ is the total body rotation between time t, and t+1. Incremental adjustments to the attitude due to coordinate frame rotation are accomplished by adding the contributions from equation (3.4) to the body sensed angular rates approximately once per minute. The evaluation of Φ is performed in a high-speed loop at close to the gyroscope's maximum bandwidth. This helps to eliminate aliasing and phase shifting which can occur if there is an appreciable high frequency rotation or vibration during the update interval.[28] Finite rotations in three-dimensional space are non-commutative and therefore any estimation of Φ must include corrections for the rotation of the body axis during the update interval.[68] The full form of Φ, including the body-axis rotation is expressed by Savage as follows[28]

$$\Phi(t) = \int_{t-1}^{t} \left(\bar{\omega}_B + \frac{1}{2}\Phi \times \bar{\omega}_B + \frac{1}{\Phi^2}\left(1 - \frac{\Phi \sin(\Phi)}{2(1 - cos\Phi)}\right)\Phi \times \left(\Phi \times \bar{\omega}_B\right)\right) dt \tag{3.7}$$

In this expression, a typical low cost inertial system would truncate the attitude update after the first angular rate term. For systems with higher accuracy gyroscopes and capable onboard processors, the remaining terms can be incorporated to improve the update accuracy. The second right-hand-side term represents a pseudo-Coriolis force which captures the steady state rotation of the coordinate frame within the update cycle. The third right-hand-side term in equation (3.7) is a coning correction. Coning motion occurs when there is an angular acceleration perpendicular to the mean axis of rotation. The resulting motion describes a path around a conic section. This motion represents one of the standard accuracy tests for attitude estimation algorithms.[19, 23] The above expression is known as the Bortz equation and is derived in detail for inertial applications by Savage.[28] Approximation of the transcendental expressions in equation (3.7) with a constant angular acceleration is expressed via power-series expansion as

$$\frac{1}{\Phi^2}\left(1-\frac{\Phi\sin(\Phi)}{2(1-\cos(\Phi))}\right)=\frac{1}{12}\left(1+\frac{1}{60}\Phi^2+\frac{1}{180}\Phi^3\ldots\right)\approx\frac{1}{12} \tag{3.8}$$

Savage estimated the preceding constant via experimentation and regression analysis.[28] This constant reduces equation (3.7) to

$$\Phi(t+1)=\int_t^{t+1}\left(\bar{\omega}_B+\frac{1}{2}\Phi\times\bar{\omega}_B+\frac{1}{12}\Phi\times(\Phi\times\bar{\omega}_B)\right)dt \tag{3.9}$$

Equation (3.9) is second-order accurate for small rotations and accounts for rotation of the body axis during the update interval. Equation (3.9) requires an unbiased estimate of the angular rate vector. In practice, the bias in each gyroscope is not static and must be estimated by the onboard sensor fusion algorithm, detailed in subsequent sections. Equation (3.9) is a convenient expression for high-speed, real-time integration in an onboard INS system. However, further improvements are possible to better account for errors introduced via coning motion. This results in a more computationally expensive algorithm and is therefore only used during periods of high angular and linear acceleration. The algorithm development begins with a Taylor expansion for the rotation between attitude update cycles

$$\Phi(\mathrm{t}+1)=\Phi(t)+\dot{\Phi}(t)h+\ddot{\Phi}(t)\left(\frac{h^2}{2}\right)+\ldots \tag{3.10}$$

Using three sequential gyroscope readings, equation (3.9) and (3.10) are combined via differentiation of the angular update vector to form the following

$$\Phi(t+1)=\left(\bar{\omega}_1+\bar{\omega}_2+\bar{\omega}_3+a(\bar{\omega}_1\times\bar{\omega}_3)+b\bar{\omega}_2\times(\bar{\omega}_3-\bar{\omega}_1)\right)dt\ , \tag{3.11}$$

where $\bar{\omega}_{1,2,3}$ corresponds to gyroscope signals sampled at times $t+1/3$, $t+2/3$, and $t+1$. The X and Y coefficients are determined experimentally to be $a=9/20$ and $b=27/40$.[28] This approximation is third-order accurate and provides an improved correction for coning motion occurring in periods of high angular acceleration.[69, 70] The computational and storage requirements are somewhat more burdensome than the succinct expression in equation (3.9), but the increase in accuracy for high angular and linear accelerations can help offset the computational investment.

A test of the attitude estimation during coning motion was undertaken. The resulting attitude error is shown in Figure 31 with an integration time step of 128 Hz. In this figure, the second-order method is equation (3.9) using only the first two right-hand-side terms. The second-order coning model is the full form of equation (3.9). The third-order model is given by equation (3.11). The test used a synthetic dataset simulating coning motion around a single axis at close to the gyroscope's maximum rotation rate. The total angular error is formulated as the error between the reference vector and the attitude estimate provided by the noted integration scheme. This motion represents an uncorrected estimate of the vehicles attitude after the initial gyroscope bias is removed. Sensor noise is simulated as a CMS390 gyroscope using the statistics presented in Table 4. The advantage of the higher-order method given in equation (3.11) is clearly visible for the highly dynamic coning motion. If the base integration level is increased from 128 Hz, the effectiveness of the additional higher order terms decreases owing to the smaller coordinate frame rotation during the update interval. However, it should be noted that simply increasing the integration rate is prohibitively computationally expensive for low-power embedded systems.

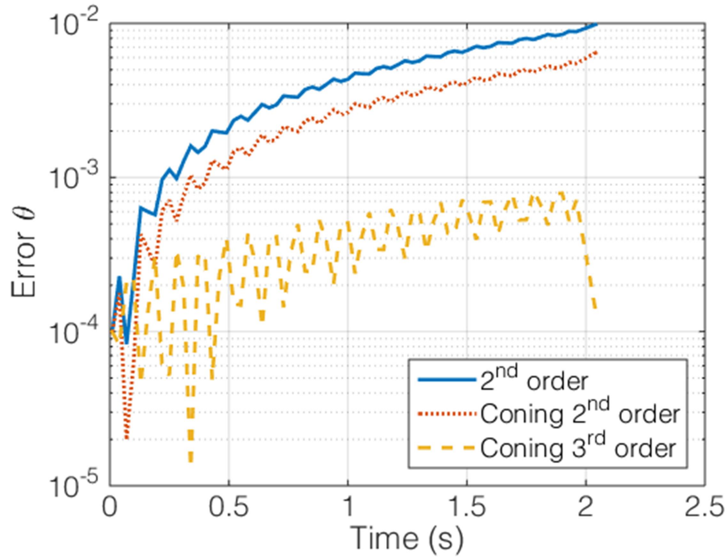


Figure 31: Angular error in degrees for different attitude integration schemes integrated at 128 Hz.

Determination of the vehicles attitude via the presented integration methods allows for estimation of the vehicles velocity and position in the NAV frame. The methods and integration schemes for determining the remaining vehicle states are the subject of the next sections.

3.2.2 Velocity Equations

Velocity and position information is provided by integration of the specific force provided by the body-frame accelerometers. For navigation purposes and error estimation, the velocity and position are calculated in the NAV frame. This requires the measured specific force in the body frame to be rotated to the NAV frame using the attitude estimate furnished by the gyroscopes and attitude estimation algorithm presented in section 3.2.1. The general form of the velocity update equation is given by [34]

$$\dot{V}^{NAV} = C_B^{NAV}\vec{a}_B + \vec{g}^{NAV} - (\vec{\omega}_{EN}^{NAV} + 2\vec{\omega}_{IE}^{NAV}) \times V^{NAV} , \tag{3.12}$$

where B represents the body frame, NAV is the locally level frame, and IE is the inertial to ECEF frame. C_B^{NAV} represents the body to NAV frame rotation matrix estimated using the attitude quaternion presented in section 3.2.1. $\vec{a}_B$ is the body measured specific force, and V^{NAV} is the computed NED velocity components in the NAV frame. The gravity term, $\vec{g}^{NAV}$, appearing in equation (3.12) is expressed in the NAV frame. For this work, the gravity vector is assumed constant given the low altitude, range, and velocity of the vehicle. The magnitude of the local gravitational field is given in the WGS84 Earth model.[67] The angular rates appearing in equation (3.12) are developed as follows. Transformation between the inertial, ECEF, and NAV frames is accomplished by updating their respective direction cosine matrices. $\vec{\omega}_{IE}^{NAV}$ represents the Earth's rotation rate expressed in the NAV frame. The Earth rotation rate in the ECEF frame is given as

$$\vec{\omega}_i^e = \begin{bmatrix} 0 \\ 0 \\ \omega_{i/e} \end{bmatrix}. \tag{3.13}$$

Transformation to the NAV frame is accomplished using the known latitude

$$\vec{\omega}_{IE}^{NAV} = \begin{bmatrix} \omega_{i/e}\cos(\phi) \\ 0 \\ -\omega_{i/e}\sin(\phi) \end{bmatrix}. \tag{3.14}$$

The NAV frame is defined as being locally tangent to the Earth's surface. This requires a forced rotation as the NAV frame is displaced from its starting location. The rotation required is formulated based on the vehicle velocity, latitude, and radius of the Earth.[64]

$$\vec{\omega}_{EN}^{NAV} = \begin{bmatrix} \dfrac{V_E}{R_{mean}+h} \\ -\dfrac{V_N}{R_{mean}+h} \\ -\dfrac{V_E}{R_{mean}+h}\tan(\phi) \end{bmatrix} \tag{3.15}$$

The specific force measured in the body frame is subject to errors similar to those affecting the attitude solution outlined in section 3.2.1. For attitude estimation, the corrupting motion is termed coning. For specific force integration, the motion is similar in nature and referred to as sculling. In this case, the error results from a motion similar to that traced out by the paddles on a rowboat.[34] Coning and sculling errors both arise due to an incomplete representation of coordinate frame rotation and acceleration during the finite update interval of the integration algorithm. To address this, an algorithm similar to the coning correction is given by Savage [34]

$$\Delta\overline{V}_t^{t+1} = \overline{V}_t + \frac{1}{2}\left(\int_t^{t+1} \overline{\omega}_B \, dt \times \overline{V}_t\right) + \frac{1}{2}\int_t^{t+1}\left(\int_t^{t+1} \overline{\omega}_B \, dt \times \overline{a}_B + \overline{V}_t \times \overline{\omega}_B\right) dt \quad (3.16)$$

In this equation, the second term on the right-hand-side represents a Coriolis type rotation of the body axis. The third term represents the sculling correction for the acceleration of the body axis during the velocity update interval.[34, 65] This relation is computed in series with the attitude algorithm presented previously. The computational burden is somewhat reduced by recognizing the incremental attitude estimate appearing inside the integral is also used in the attitude algorithm. Equation (3.16) gives the velocity increment occurring in the body frame over a single update iteration. The resulting velocity increment is then transformed to the NAV frame using the C_B^{NAV} rotation matrix. The NAV frame velocity increment then replaces the specific force appearing in equation (3.12).

3.2.3 Position Equations

The NAV frame position is obtained from integration of the velocity components using a fourth-order Runge-Kutta technique.[29] The position equation derives from elementary rigid body dynamics

$$S^{NAV} = \int \dot{S}^{NAV} dt = \int V^{NAV} dt \quad (3.17)$$

Incremental position updates are used to update the rotation matrix between the inertial, ECEF, and local NAV frames, given as

$$C_{NAV}^{IE(t+1)} = C_{NAV}^{IE(t)} + \int_t^{t+1} \zeta_{IE}^{NAV} dt \, , \quad (3.18)$$

where ζ for slow moving vehicles is given by [34]

$$\zeta_{IE}^{NAV} = \rho_Z + F_C \left(u_Z^{NAV} \times \sum_{m=1}^{i} \Delta S^{NAV} \right), \tag{3.19}$$

where ρ_z is the vertical transport rate, F_C is the local Earth curvature matrix, and u_z is a unit vector oriented normal to the Earth's surface. In this expression, the summation represents the displacement over an update cycle in the NAV frame. This expression applies for situations in which the rotation of the NAV frame relative to the Earth is small over a single update. For vehicles moving at relatively low speeds, this expression provides sufficient accuracy over other more computationally expensive models. For example, with a vehicle moving 30 m/s, the rotational component ζ is on the order 5×10^{-6} rad/s, much smaller than the body angular rates and position error introduced by vibration and other sensor errors. The rotation matrix C_{NAV}^{IE} is propagated using a second-order technique detailed by Phillips.[71]

3.3 Error Equations

Methods for estimating the vehicle attitude, velocity, and position are outlined in section 3.2. Due to sensor imperfections and computational limitations, the estimates provided by these equations will diverge over time. There are typically two methods of addressing this divergence. The first requires combining the vehicle state estimates provided by the navigation equations with reference sources provided by low-bias aiding sensors such as GPS systems supplemented with information provided by barometers and magnetometers. It should be noted that the performance of the INS is implicitly linked to the performance of the low-bias aiding sensors as will be demonstrated in subsequent sections. This formulation is commonly referred to as direct state estimation.[18] Depending on which states are estimated, the resulting implementation typically requires a computationally expensive multi-state non-linear filter. An alternative is to estimate the error in the states, rather than the states directly. Depending on the formulation of the error equations, the resulting filter can often be easily linearized when compared to the direct state estimation method. This dramatically reduces the filter complexity and allows for the inclusion of additional states for the same computational load. These states might include sensor scale factors, bias, and alignment errors, which can all exhibit a temporally random variation. Error-state formulations require additional relations to model the growth of error in the inertial solution. This error is then corrected using the same low-bias sensors used in the direct state estimation. This section details the attitude, velocity, and position error models used in the error-state filter presented later in this chapter.

The specific implementation of the error-state equations follow from the ψ -angle formulation.[18, 29, 72] This is known to produce less computationally intensive dynamic equations compared to the ϕ angle formulation.[17, 66, 73]

3.3.1 Attitude Error

The attitude error model follows the development presented by Rogers.[23] In this formulation, the attitude error is expressed as the combination of a body-NAV and NAV-ECEF tilt error,

$$\psi \equiv \phi - \delta\theta , \tag{3.20}$$

where ψ is the attitude error, ϕ represents a local tilt error between the NAV and body frames, and $\delta\theta$ is the alignment error between the NAV and ECEF frames. The resulting attitude error equation relates the time-rate-of-change of the body-NAV tilt error to the sensed Earth rotation in the NAV frame

$$\dot{\psi} = \psi \times \left(\vec{\omega}_{IE}^{NAV} + \vec{\omega}_{EN}^{NAV} \right) + \varepsilon^{NAV} , \tag{3.21}$$

where ε^{NAV} represents an additional Gaussian noise term.

3.3.2 Velocity Error

The velocity error model is formulated by comparing the true and estimated velocities in the NAV frame. The resulting error is of the form

$$V_{error} \equiv \left[I - \left(skew(\delta\theta) \right) \right] V^{NAV} + \delta V , \tag{3.22}$$

where δV is an additional error in the NAV velocity caused by errors in integration or sensor imperfections. The velocity error, V_{error}, can be related to errors appearing in the navigation attitude and velocity equations. The algebra is omitted here for clarity, but can be found in Rogers.[23] The final mechanized form of the velocity error equation is given by

$$\delta \dot{V}^{NAV} = -\left(\omega_{EN}^{NAV} + 2\omega_{IE}^{NAV}\right) \times \delta V^{NAV} + a^{NAV} \times \psi + \delta a^{NAV} , \tag{3.23}$$

where $\delta \dot{V}^{NAV}$ is the NAV frame velocity error increment and δa^{NAV} is an additional specific-force Gaussian noise viewed in the NAV frame.

3.3.3 Position Error

The position error model is formulated in a manner similar to the velocity error model. This model relates the true and estimated positions. For errors in the NAV frame, the position error is simply the integration of the velocity error components

$$\delta \dot{S}^{NAV} = \delta V^{NAV} . \tag{3.24}$$

This represents a simplification of the classical ψ angle formulation presented by Rogers.[23] For the approximation presented in equation (3.24), the angular error of the NAV frame with respect to the ECEF frame is assumed to be negligible. This implies that the NAV frame velocity is much smaller than the radius of the Earth. For the vehicle considered in this work, the ratio of the maximum velocity of the vehicle to the Earth radius is on the order of 7×10^{-6} rad/s. This is much smaller than the integrated error in velocity, thus allowing the additional terms to be neglected.

3.4 INS Sensors

A suite of inertial sensors provides the real-time estimate of the aircraft's state. These include three-axis gyroscopes and accelerometers with integrated temperature measurement capabilities. If sensor errors were not present, these would be sufficient to navigate within a local inertial frame for an extended period.

However, small errors can render a purely inertial solution invalid in a matter of seconds. This section details the sensors used in the attitude, velocity, and position estimation as well as the additional sensors necessary to bound the growth of the state estimation error. The additional sensors include a GPS for localization information, barometer for altitude estimation, and pressure sensors for airspeed and wind estimation, and a magnetometer for heading estimation. Due to the unbounded growth of state error when using a purely inertial solution, the accuracy of the inertial solution is fundamentally limited by the accuracy of the GPS sensor. Sensor error models and calibration techniques are detailed in the following sections.

3.4.1 Gyroscope

Gyroscopes are the fundamental component of the inertial system. This sensor detects rotations about its sensitive axes which includes the effects of body and inertial frame rotations, i.e. the Earth rotation rate. By integration of the gyroscope output, the body-NAV and NAV-ECEF frames are propagated using the attitude equations outlined in section 3.2.1. For a purely inertial solution, the estimated position and velocity components rely entirely on the accurate attitude estimation provided by the gyroscopes.

For this work, strapdown MEMS gyroscopes are used. These differ from classical mechanical gyroscopes in that they provide angular rates in place of angular displacements. The advantage of these systems is their small size, low power, and low cost. Drawbacks include higher noise, larger drift, and lower precision than standard rate integrating mechanical gyroscopes. The strapdown attitude equations rely on an unbiased estimate of the body-referenced angular rates. However, due to imperfections and non-linearities in the individual sensors, the raw output data can quickly corrupt the attitude estimate.

Gyroscopes are typically divided into performance classes based on their in-situ bias stability. This term refers to the shift in zero-level output over a prescribed period. Details on the classification of gyroscopes are presented by Kempe and are repeated for reference in Table 3.[19] Accurate state estimation relies on the ability to quantify and correct for scale factor, orthogonalization, alignment and in-situ bias terms. Additional complications include noise, vibration, and quantization effects. These error sources cause the sensor output to deviate from the ideal model considered in the attitude update. In general, the gyroscope error model is of the form[21, 24]

$$\begin{bmatrix} \omega_{xB} \\ \omega_{yB} \\ \omega_{zB} \end{bmatrix} = \begin{bmatrix} S_x & A_{xy} & A_{xz} \\ A_{yx} & S_y & A_{yz} \\ A_{zx} & A_{zy} & S_z \end{bmatrix} \begin{bmatrix} \omega_{xB_{raw}} \\ \omega_{yB_{raw}} \\ \omega_{zB_{raw}} \end{bmatrix} + \begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix} + \begin{bmatrix} C_x a_{xB} \\ C_y a_{yB} \\ C_z a_{zB} \end{bmatrix} + \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \varepsilon_z \end{bmatrix}, \tag{3.25}$$

where S is the scale factor error, A is the misalignment error, B is a random bias offset, C is correlated acceleration errors, and ε is a Gaussian white noise component. In general, each of these terms must be estimated and corrected before attitude integration. Section 3.5 details the calibration of the gyroscope. The onboard sensors include three redundant gyroscopes with different performance specifications as outlined in Table 4. The total error component refers to the combined scale factor, orthogonality, bias, temperature, and linearity errors. These sensors represent the currently available performance range of cost-effective MEMS sensors. Their impact on flight vehicle performance is detailed in the following chapter.

Table 3: Gyroscope performance classes[19]

Type	Full scale range (deg/s)	Bias stability (deg/hr)	Scale factor accuracy (deg/s)
Consumer	1000	>10	>0.5
Tactical	400	0.1-10	0.01-0.1
Inertial	400	<0.1	<0.01

Table 4: Gyroscope performance data

Manufacturer	Axes	Full scale range (deg/s)	Bias stability (deg/hr)	Resolution (deg/s/bit)	Noise Density (deg/s/√Hz)	Total error (deg/s)
Silicon Sensing CMS390	1	300	10.0	0.005	0.0055	1.5
VTI SCC1300	1	300	12.0	0.056	0.014	5.0
InvenSense MPU-9150	3	500	20.0-50.0	0.030	0.065	15.0

3.4.2 Accelerometer

Accelerometers provide the body-frame specific-force measurements necessary to determine the vehicle's translational states. The body frame measurements are rotated to the local level NAV frame for integration into velocity and position states as outlined in section 3.2.2 and 3.2.3. For a sensor attached to a moving vehicle, the sensed acceleration is a linear combination of that generated by the motion of the inertial reference body (Earth) and that generated by the vehicle. For this reason, an accelerometer at rest will experience $1g$ on the Earth's surface, and $0g$ in free-fall. If the sensor is in motion, the sensed acceleration will be a combination of the gravitational acceleration and body-frame specific force, given as

$$F = ma = m\left(g + a_B\right),\tag{3.26}$$

where the specific force is a linear combination of the gravitational component, g , and body force component, a_B . Error in the accelerometers will lead to a linear velocity and a quadratic position error. The form of the error equation is similar to the gyroscope error model and is given by[21, 24]

$$\begin{bmatrix} a_{xB} \\ a_{yB} \\ a_{zB} \end{bmatrix} = \begin{bmatrix} S_x & A_{xy} & A_{xz} \\ A_{yx} & S_y & A_{yz} \\ A_{zx} & A_{zy} & S_z \end{bmatrix} \begin{bmatrix} a_{xB_{raw}} \\ a_{yB_{raw}} \\ a_{zB_{raw}} \end{bmatrix} + \begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix} + \begin{bmatrix} C_x\omega_{xB} \\ C_y\omega_{yB} \\ C_z\omega_{zB} \end{bmatrix} + \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \varepsilon_z \end{bmatrix}.\tag{3.27}$$

In this expression, there are no g correlated errors as appear in the gyroscope model. Instead, a rate-dependent error is introduced for highly dynamic motions (third right-hand-side term in equation (3.27)), which closely approximates the effect of a phase delay on the signal. For MEMS gyroscopes, the resonant frequency of the sensors is on the order of 10 kHz, which makes this error term difficult to quantify and ultimately a small contributor to the accuracy of the inertial solution. Table 5 and Table 6 present the details for the classification of accelerometers and the sensors used in this work.

Table 5: Accelerometer performance classes[19]

Type	Full scale range (g)	Bias stability (μg)	Fixed bias (μg)	Bandwidth (Hz)
Consumer	±16	>100	>0.5	~
Rate	±1-100	1-100	0.1-1	100-1k
Tactical	±1-100	0.1-1	0.01-0.1	1k-10k
Inertial	±1-100	<0.1	<0.01	1k-10k

Table 6: Accelerometer performance data

Manufacturer	Axes	Full scale range (g)	Bias stability (mg)	Total error (mg)	Resolution (mg/bit)	Noise Density (mg/√Hz)
Silicon Sensing CMS390	2	±2.5/10	30.0	±13.0	0.08	1.0
VTI SCC-1300	3	±6	70.0	±40.0	1.5	5.0
InvenSense MPU-9150	3	±2/4/8/16	103.0	150.0 (4g mode)	0.4	4.0

3.4.3 Magnetometer

The magnetometer provides the azimuthal reference for alignment of the NAV frame with the NED coordinate axes. Localized distortions and geographic changes in the Earth's magnetic field can strongly affect magnetic heading estimation. Typically, *a priori* information regarding the strength and direction of the local magnetic field is required to compensate the sensor readings for proper NAV frame alignment. Local magnetic disturbances can significantly affect and overwhelm the sensors output range. These noise sources are typically linked to high frequency and high current devices such as motors and communications equipment. For small-scale aircraft, it can be difficult to eliminate these sources of error as physically moving the sensor to a remote location is not feasible.

The sensor error model for the magnetometer is the same as the accelerometer. However, given that the output of the magnetometer is not numerically integrated, the effect of bias, scale factor, and alignment on the inertial solution is more difficult to quantify. The overall effect of an azimuthal alignment error on the velocity and position error propagation is non-linear. The sensor fusion algorithms detailed in the subsequent sections are necessary to correct this error. Calibration of the magnetometer differs from the gyroscope and accelerometer as significant bias shifts can occur during flight and are often not known beforehand. This necessitates feedback between the inertial solution and the magnetometer output. Section

3.5.3 details the calibration process for this sensor while section 3.5.3.1 details a novel in-flight bias estimation technique. The aiding, rather than integrating function of the magnetometer means that precise classifications for magnetometers are typically not available. Table 7 outlines the performance of the specific sensors used.

Table 7: Magnetometer performance data

Manufacturer	Axes	Full scale range (Gauss)	Resolution (mGauss/bit)	Noise density (mGauss/√Hz)	Total error (mGauss)
Honeywell HMC-2003	1	±2	2.0	40.0	10.0
Honeywell HMC-5883L	3	±1-8	7.0	2.0	2.0
InvenSense MPU-9150	3	±12	3.0	4.0	10.0

3.4.4 Air Data Systems

For the performance evaluation of the aircraft discussed in chapter 5, accurate estimation of the aircraft's airspeed is required. Section 3.6 details an inertial estimation technique to determine the local airspeed without an installed pitot probe. The inertial airspeed estimate is not a high-fidelity signal and lags the actual airspeed by up to a second. However, it can be used when no pitot or other airspeed sensor is available. To provide better dynamic response, a pitot probe and differential pressure transducer is used (detailed in Table 8). This sensor allows for airspeed estimation at update rates similar to the inertial solution at 128 Hz. Additional in situ calibration routines are necessary to account for mounting induced errors and local pressure distortions caused by the wing and fuselage. Section 3.5.4 details these corrections.

Table 8: Pressure and barometric sensor performance data

Manufacturer	Static/ differential	Full scale range	Total error	Resolution $\left(/bit\right)$	Noise Density $\left(\sqrt{Hz}\right)$	Total Error (at STP)
Measurement Specialties MS4515DP 20” H_2O	D	±4981 Pa	±99 Pa	2.2 Pa	3.7 Pa	12.4 m/s
Measurement Specialties MS4515DP 2” H_2O	D	±498 Pa	±10 Pa	0.2 Pa	0.5 Pa	3.9 m/s
Bosch BMP085	S	8000 Pa	±10 Pa	0.03 Pa	0.06 Pa	0.8 m

A separate barometric altimeter is used to control the error growth in the INS vertical channel. GPS can be used for this operation, but in practice, shows significant bias and non-repeatability in height estimation. The barometer is located inside the aircraft’s fuselage and not connected to an external static port. The aircraft is well ventilated and testing has shown little to no effect on height estimation induced by either ram air or suction inside the fuselage. The verification of this assumption used duplicate sensors throughout the aircraft and compared the predicted altitude during flight. The resulting error in estimated height was dominated by the noise floor of the sensor. Given the resolution of the barometer, they were capable of coarse pitch and roll detection given their separation of approximately 2 m. Flight-testing revealed no significant bias on the altitude estimation using the barometric sensor attached to the INS circuit board.

3.4.5 GPS

A GPS receiver is used to provide localization information to control the growth of error in the inertial solution. Therefore, the total accuracy is implicitly related to the accuracy of the GPS receiver. GPS receiver characteristics can be broken down into performance classes in a similar manner to the other sensors used in this work. A figure detailing these classes is given in Figure 32.[74] For this work, a receiver in the mapping grade performance class is chosen as being representative of low-cost receivers typically used in sUAS systems. These receivers are typically single frequency devices that can use external aiding sources such as the wide-area augmentation system (WAAS). The specific GPS receiver used is a U-BLOX Neo-7M receiver with integrated antenna. This device provides position and velocity information in the ECEF coordinate frame at 4 Hz. The noise characteristics for the position and velocity data are given in Table 9. Dynamic vehicle motions that change the pitch/roll orientation of the antenna affect the accuracy of the GPS. These errors are typically small for shallow bank angles, but increase as the attitude excursions approach 90 degrees. Correcting the INS state estimates using pre-computed GPS location and velocity information makes this a loosely coupled system. The loosely coupled approach has been used in numerous GPS/INS fusion studies.[16, 30] Tightly coupled state-estimation in which GPS carrier phase information is included in the filter structure has proven effective in numerous other research endeavors.[16, 29, 75-77] The tightly coupled approach is typically much more computationally expensive than the loosely coupled approach.

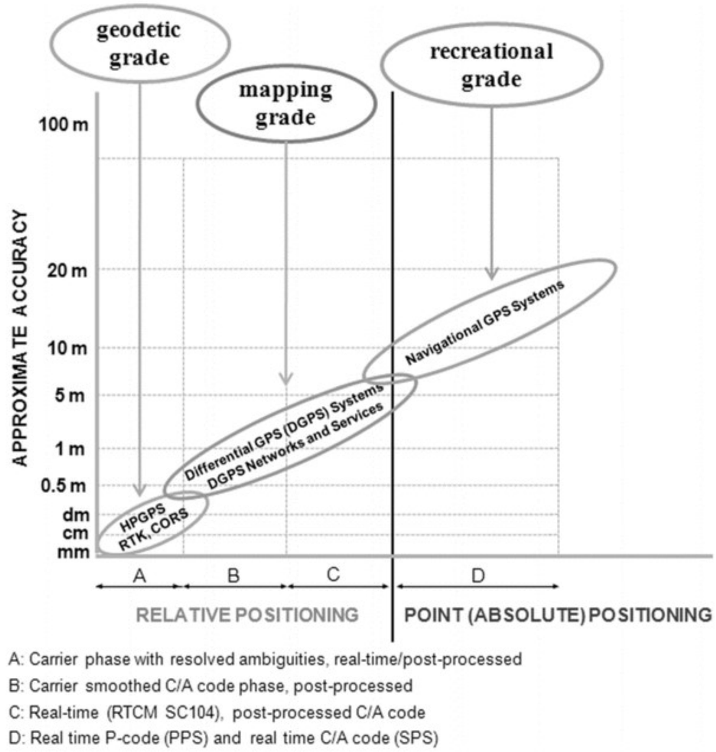


Figure 32: GPS receiver classification. Reprinted with permission from ASCE.org[74]

Table 9: GPS noise characteristics

Noise component	Noise estimate
Position	± 0.7 m static, x-y
	± 2.5 m dynamic, x-y
	± 2.5 m static, z
	± 5 m dynamic, z
Velocity	± 0.5 m/s static, x-y
	± 1.5 m/s dynamic, x-y
	± 2.5 m/s static, z
	± 5 m/s dynamic, z

The GPS estimate of the north and east position and velocity is typically more precise than the estimates provided for the vertical channel. This is due in part to the restricted elevation angle of the visible satellites. This reduction in vertical channel accuracy makes the use of the barometer for height and vertical speed estimation a more attractive option. Therefore, in practice, only the north and east components of the GPS position and velocity data are used.

3.5 Sensor Calibration

An optimization based calibration method is used to determine the scale factor, alignment, orthogonaltiy, and bias terms associated with the inertial sensors. Calibration is divided into two sections. The first occurs in the laboratory and addresses the calibration coefficients that are relatively constant during operation. This includes the scale factor, alignment, and orthogonality constraints. For MEMS devices, these coefficients are often strongly temperature dependent, which introduces another degree of freedom to the calibration matrices. The proposed calibration method relies on first aligning the gyroscopes to the body axes using a precision rate-table. After this, the accelerometers and magnetometer are corrected and aligned to the gyroscope axes. While this procedure can eliminate the largest sources of measurement error, an in-situ calibration for the bias terms still needs to be estimated by the sensor fusion algorithm. This algorithm is discussed in subsequent sections. The remainder of this section details the specific sensors used in the INS, and the laboratory calibration methods used.

3.5.1 Gyroscope

The gyroscopes are the fundamental sensor used for attitude estimation on the aircraft. As such, they are the primary sensor to which the others are aligned. Calibration of the gyroscopes is accomplished by individually exciting the three sensor axes with a precisely known rotation rate. From the general error model for the gyroscope presented in section 3.4.1, the calibrated output of the sensor is expressed as

$$\begin{bmatrix} \omega_{xB} \\ \omega_{yB} \\ \omega_{zB} \end{bmatrix} = \begin{bmatrix} S_x & A_{xy} & A_{xz} \\ A_{yx} & S_y & A_{yz} \\ A_{zx} & A_{zy} & S_z \end{bmatrix} \begin{bmatrix} \omega_{xB_{raw}} \\ \omega_{yB_{raw}} \\ \omega_{zB_{raw}} \end{bmatrix} + \begin{bmatrix} Bias_x \\ Bias_y \\ Bias_z \end{bmatrix} + \begin{bmatrix} C_x a_{xB} \\ C_y a_{yB} \\ C_z a_{zB} \end{bmatrix} + \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \varepsilon_z \end{bmatrix} . \tag{3.28}$$

The noise and acceleration terms are neglected by averaging the sensor data and minimizing lateral accelerations during table rotation tests. The bias terms are measured while the sensor is not moving before and after the calibration test. The resulting model is therefore reduced to determining the nine remaining scale factor and alignment components. The scaling and alignment matrix is split into scale factor, *S*, orthogonalization, *T*, and alignment, *M*, submatrices as follows

$$\begin{bmatrix} \omega_{xB} \\ \omega_{yB} \\ \omega_{zB} \end{bmatrix} = \begin{bmatrix} S_x & & \\ & S_y & \\ & & S_z \end{bmatrix} \begin{bmatrix} 1 & & \\ \cos(\alpha) & \sin(\alpha) & \\ \cos(\beta) & \cos(\gamma) & \sqrt{1-\cos^2(\beta)-\cos^2(\gamma)} \end{bmatrix} \begin{bmatrix} M_{11} & M_{12} & M_{13} \\ M_{21} & M_{22} & M_{23} \\ M_{31} & M_{32} & M_{33} \end{bmatrix} \begin{bmatrix} \omega_{xB_{raw}} \\ \omega_{yB_{raw}} \\ \omega_{zB_{raw}} \end{bmatrix} + \begin{bmatrix} Bias_x \\ Bias_y \\ Bias_z \end{bmatrix} \quad (3.29)$$

$$\vec{\omega}_B = S \cdot T \cdot M \vec{\omega}_{B_{raw}} + \overrightarrow{Bias} \quad (3.30)$$

The scale factor matrix adjusts the magnitude of the individual sensor axis in response to a known input. The orthogonalization matrix, *T*, is determined using a Gram-Schmidt orthogonalization algorithm. This algorithm uses a set of linearly independent vectors to generate a set of orthogonal vectors occupying the same subspace, resulting in the elimination of any cross coupling between sensor axes.[24] The process is guaranteed to generate an orthogonal vector set from a reference input. Therefore, the excitation of the sensor triad must be carried out in such a way to ensure the table rotation axes are orthogonal. This is the key parameter for accurate attitude estimation and is explained as follows. An error in the scale factor results in an over or underestimation of the rotational displacement. The attitude filter can correct this as it is observable by an individual external measurement such as any of the velocity or displacement components. In a similar manner, neglecting the alignment matrix results in reduced control effectiveness for the vehicle because the aircraft's coordinate frame is not aligned with the inertial frame. This can be roughly corrected by trim during flight. Orthogonalization however, does not have a physical analog. For example, an error in the x-y orthogonalization parameter means that a change in pitch would appear as a change in roll. This non-physical result can lead to divergence of the sensor fusion algorithm during periods of high rotation, i.e. fast turning motions. The orthogonalization axes are shown in Figure 33. From this figure, the assumed alignment angles are small, typically less than 5 degrees. This is guaranteed by precision construction of the sensor carrier board and published manufacturing tolerances. This allows for a simplification of the complete orthogonalization matrix using small angle approximations. The result is given as the matrix *T* in equation (3.30).

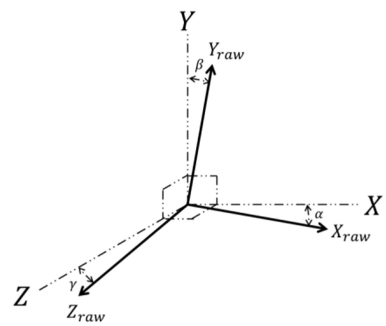
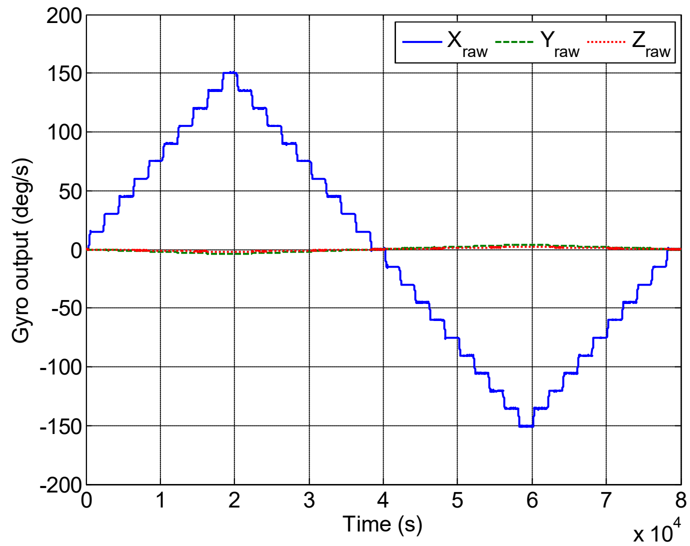


Figure 33: Orthogonalization correction axes

The gyro triad was excited using a precision rate table. This table uses a stepper motor capable of rotating the triad up to the maximum rate of the gyroscopes with an angular rate precision of ±0.001 deg/s. This was verified by monitoring the output of an optical encoder attached to the stepper motor output shaft. The triad was rotated in 15 deg/s increments up to the maximum supported rate for each sensor. An example time history of the excitation is shown in Figure 34 for a rotation about the triad's x-axis. The triad is excited in both the positive and negative rotation directions to highlight asymmetries in the sensor outputs. From Figure 34, the output on the Y and Z-axes is representative of combined alignment and orthogonality errors.



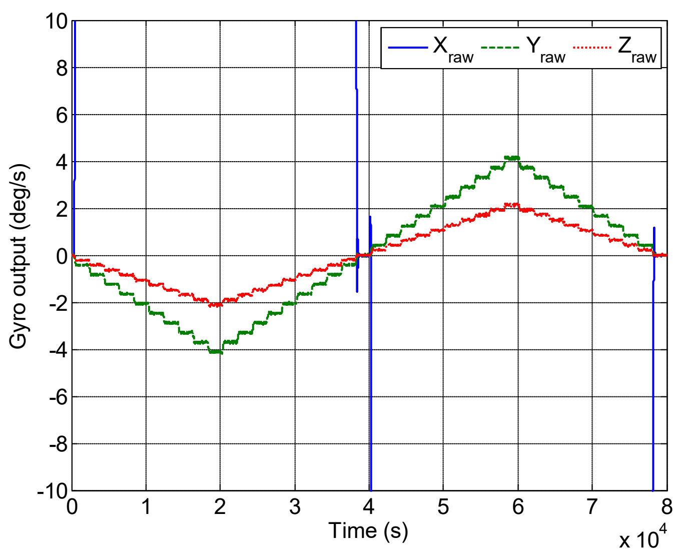


Figure 34: Raw X-axis gyro excitation (top) and zoomed in raw Y and Z axis values (bottom) for the CMS 390 gyroscopes.

The bias is removed from the measurement by subtracting the mean gyro signal at the beginning and end of the test. This results in the following alignment and scaling relation

$$\vec{\omega}_B = S \cdot T \cdot M \vec{\omega}_{B_{raw}} , \tag{3.31}$$

where $\vec{\omega}_B$ is the known input rate to the table. The raw measurement vector is expanded to incorporate off-axis excitation. This gives equation (3.32) where the first subscript is the excited axis and the second is the resulting off-axis measurement. Ω_B uses the known input rates, $\vec{\omega}_B$, and $\Omega_{B_{raw}}$ uses the measured raw gyroscope input rates, $\vec{\omega}_{B_{raw}}$.

$$\Omega_B = \begin{bmatrix} \omega_{x.x} & 0 & 0 \\ 0 & \omega_{y.y} & 0 \\ 0 & 0 & \omega_{z.z} \end{bmatrix}; \; \Omega_{B_{raw}} = \begin{bmatrix} \omega_{x.x} & \omega_{x.y} & \omega_{x.z} \\ \omega_{y.x} & \omega_{y.y} & \omega_{y.z} \\ \omega_{z.x} & \omega_{z.y} & \omega_{z.z} \end{bmatrix} \tag{3.32}$$

The individual scale, orthogonalization, and alignment matrices are estimated as follows. From Aydemir and Saranli, the known and measured angular rates are multiplied by their transpose[21]

$$\Omega_B \Omega_{B_{raw}}^{-1} \left(\Omega_B \Omega_{B_{raw}}^{-1} \right)^T = \left(S \cdot T \cdot M \right) \left(S \cdot T \cdot M \right)^T = \left(S \cdot T \right) \left(S \cdot T \right)^T . \tag{3.33}$$

The alignment matrix, *M,* is orthonormal by definition and when multiplied by its transpose results in the identity matrix. The scale, *S*, and orthogonality, *T,* matrices are found by Cholesky decomposition of the left-hand-side of equation (3.33)

$$S \bullet T = chol \left[(\Omega_B \Omega_{B_{raw}}^{-1} \left(\Omega_B \Omega_{B_{raw}}^{-1} \right)^T \right]^T . \tag{3.34}$$

Lower-Upper (LU) decomposition returns the scale factor matrix, *S,* as the upper triangular result, and the orthogonality matrix , *T*, as the lower triangular result

$$\left[T, S \right] = LU(S \cdot T) . \tag{3.35}$$

The alignment matrix *M* is found by solution of the original error equation (3.31)

$$M = T^{-1} S^{-1} \left(\Omega_B \Omega_{B_{raw}}^{-1} \right) . \tag{3.36}$$

From (3.36), the correction data for the gyroscope sensors used in this work are shown in Table 10.

Table 10: Gyroscope calibration parameters with 95% confidence bounds.

Manufacturer	Normalized scale factor	Orthogonalization angles α, β, γ (deg)	Alignment angles pitch, roll, yaw (deg)
Silicon Sensing CMS390	1.0037±0.004	-1.500±0.1	-1.571±0.02
	1.0049±0.004	-1.101±0.2	0.805±0.03
	1.0048±0.003	0.817±0.2	-0.988±0.02
VTI SCC-1300	0.975±0.004	0.190±0.3	-0.530±0.08
	0.960±0.006	1.503±0.2	-0.250±0.01
	0.965±0.007	1.240±0.2	0.272±0.12
InvenSense MPU-9150	0.985±0.004	0.094±0.4	-0.555±0.23
	0.990±0.002	0.001±0.2	-0.347±0.51
	0.970±0.006	0.483±0.2	0.682±0.44

From Table 10, the scale factors are all within the manufacturer's tolerance specifications. The orthogonalization and alignment angles for the InvenSense gyroscopes are better than the higher performance Silicon Sensing CMS390s. This is likely due to the co-location of the MPU gyroscopes in the same physical package. Additional testing has shown a wide variability in the alignment angles for the single-chip low performance gyroscope systems.[25, 26] After applying the corrections to the raw gyroscope signals (Figure 34), the resulting corrected data does not show significant cross coupling (Figure 35).

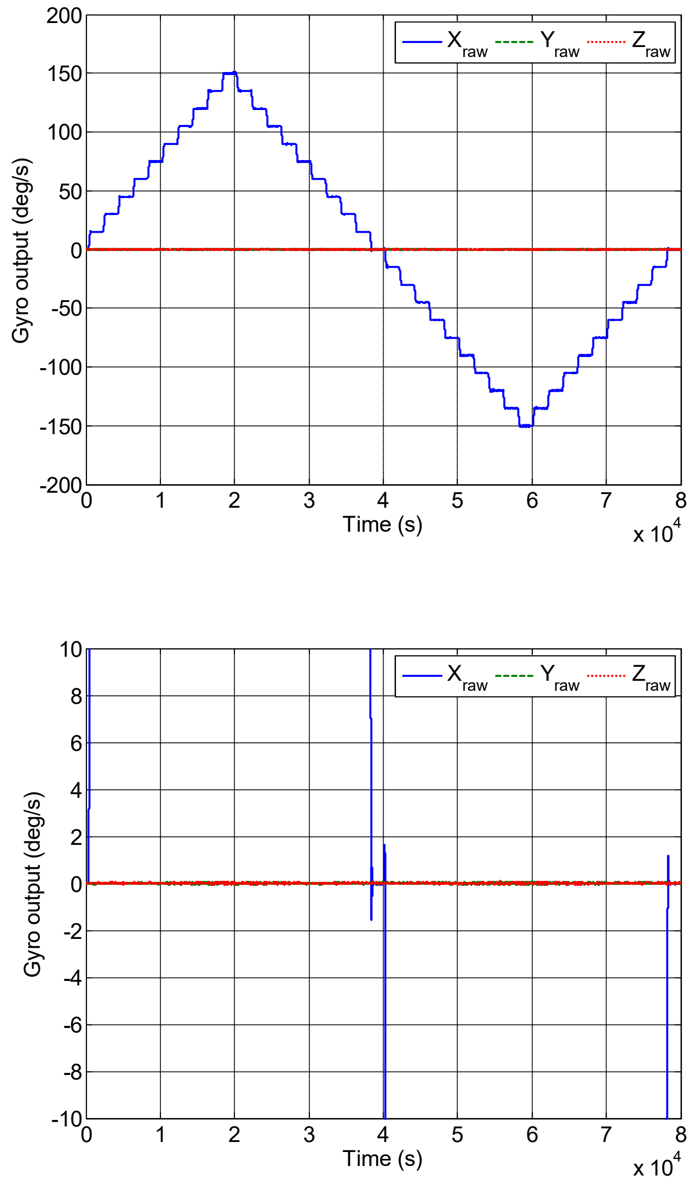


Figure 35: Corrected X-axis gyro excitation (top) and zoomed in raw Y and Z axis values (bottom) for the CMS390 gyroscopes.

To investigate the effects of temperature on the resulting calibration coefficients, the test stand was placed inside a commercial refrigerator and oven. This allowed the external temperature of the device to be varied from 0-40 C, which covers the expected range of operational temperatures the INS will likely be exposed to. The calibration routine was performed in 10 C increments from 0 to 40 C. The resulting changes in calibration coefficients are listed in Table 12 as a percent of full-scale change. The change in all coefficients was fit with a linear trend line and used to correct the in-situ sensor readings during operation.

Table 11: Gyroscope calibration temperature variation from 0 – 40 C.

Manufacturer	Normalized scale factor	Orthogonalization angles α, β, γ (deg)	Alignment angles pitch, roll, yaw (deg)
Silicon Sensing CMS390	±0.5%	±0.01%	±0.01%
	±0.4%	±0.02%	±0.02%
	±0.5%	±0.01%	±0.04%
VTI SCC-1300	±1.2%	±0.04%	±0.05%
	±1.2%	±0.04%	±0.02%
	±1.4%	±0.03%	±0.08%
InvenSense MPU-9150	±4.5%	±0.5%	±0.7%
	±3.2%	±0.7%	±0.8%
	±2.8%	±0.4%	±0.8%

3.5.2 Accelerometer

Calibration of the accelerometer triad is comprised of two serial operations. The first is to orthogonalize and scale the sensors. After an orthogonal and scaled basis is developed, the sensor outputs are aligned to the gyroscope axes. To develop the orthogonalization and scale matrices, as with the gyros, the sensors must be excited in the positive and negative directions. For accelerometer calibration, gravity is used as the known excitation force. An additional testing stand was developed to rotate the triad uniformly about all axes. The result of a rotation test is shown in Figure 36. Ideally, the sensor triad should measure a constant gravitational acceleration, independent of sensor orientation. However, due to the combined effect of scale and orthogonaltiy, the actual output varies as in Figure 37.

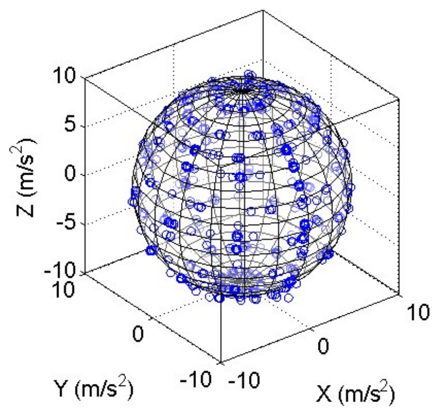


Figure 36: Raw acceleration signals for scaling and orthogonalization for the CMS390 accelerometers.

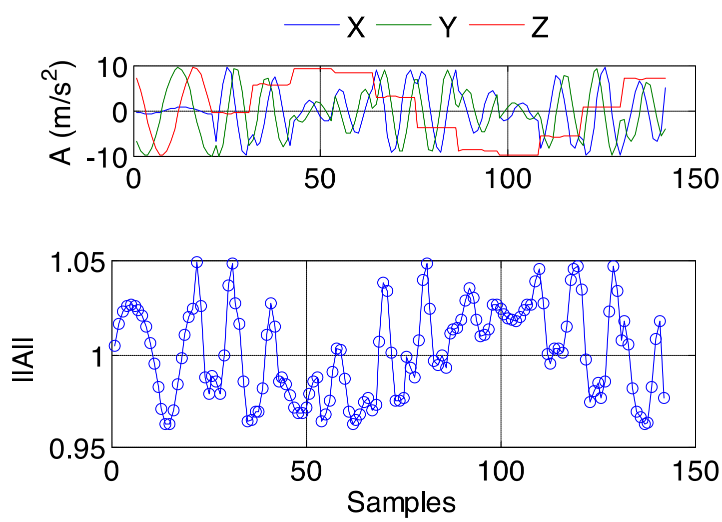


Figure 37: Raw acceleration signals for scaling and orthogonalization (top) and normalized acceleration magnitude (bottom) for the CMS390 accelerometers.

The scaling and orthogonalization of the triad is an extension of the work performed by Jurman *et. al*.[24] Instead of positioning the triad a known orientation, the unknown scale factor and orthogonality terms are estimated in an optimization routine where the objective function is the variance in sensor magnitude, given as

$$O\left(\bar{a}_B\right) = \frac{1}{N}\sum_{i=1}^{n}\left(g - \left\|\vec{a}_B\right\|\right)^2 \quad (3.37)$$

For the accelerometer triad, the output should be a constant acceleration of $1g$. Equation (3.27) can be reformulated to reflect only the orthogonality and scale factor errors, given as

$$\begin{bmatrix} a_{xB} \\ a_{yB} \\ a_{zB} \end{bmatrix} = \begin{bmatrix} S_x & & \\ & S_y & \\ & & S_z \end{bmatrix} \begin{bmatrix} 1 & & \\ \cos(\alpha) & \sin(\alpha) & \\ \cos(\beta) & \cos(\gamma) & \sqrt{1-\cos^2(\beta)-\cos^2(\gamma)} \end{bmatrix} \begin{bmatrix} a_{xB_{raw}} \\ a_{yB_{raw}} \\ a_{zB_{raw}} \end{bmatrix} + \begin{bmatrix} Bias_x \\ Bias_y \\ Bias_z \end{bmatrix} + \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \varepsilon_z \end{bmatrix} \quad (3.38)$$

In equation (3.38), the unknown quantities represent the scale factor, orthogonality, and bias terms. None of these terms depend explicitly on knowing the precise orientation of the device. Therefore, at least nine independent measurements are required to estimate the unknown parameters. In practice, the rotation table allows for several hundred data points to be collected in rapid succession. The optimization has nine degrees of freedom and uses Matlab's fmincon function estimate the alignment parameters subject to the global constraint listed in equation (3.37). The results of the optimization are shown in Figure 38. From this figure, the variance in predicted gravitational acceleration is reduced to the noise level of the sensors tested. At this stage, the gyroscopes have been corrected and aligned to the body frame, and the accelerometers corrected, but not aligned to the gyroscope axes. The final alignment of the accelerometers to the gyroscope axes is presented next.

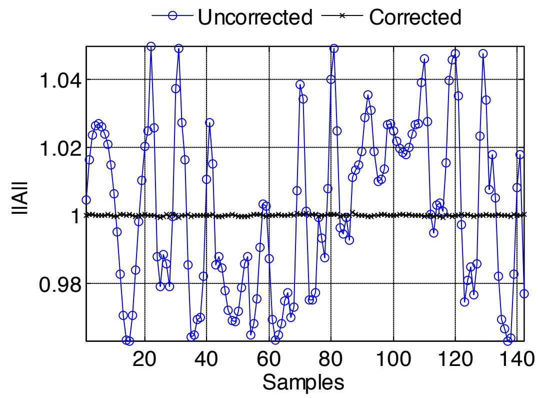


Figure 38: Acceleration magnitude before and after scaling and orthogonalization for the CMS390 accelerometers.

The final alignment of the accelerometers to the gyroscope's reference frame is achieved as follows. During positioning moves on the angular rate table, the gyroscope signals are recorded. From these signals, the change in orientation is estimated by integration between table rotations. The predicted accelerometer output at the end of the rotation is then determined by transformation of the acceleration signal before the rotation. The rotational change provided by the gyroscopes is given by

$$C_{gyro} = \int_{t=i}^{t=i+1} \overline{\omega}_B \, dt \quad . \tag{3.39}$$

The initial orientation estimated by the accelerometers is given by construction of a direction cosine matrix with an unknown azimuth angle. From this initial estimate, the accelerometer reading at the end of the rotation is estimated by

$$A_{end_{pred}} = C_{gyro} A_{start} \quad . \tag{3.40}$$

The difference between the measured and predicted acceleration values is an indication of the misalignment between axes, given as

$$C_{gyro} A_{start} = A_{end_{pred}} = M A_{end_{meas}} , \tag{3.41}$$

where C is the rotation between measurements estimated by the gyroscopes, A_{start} is the measured acceleration before rotation, A_{end} is the predicted and measured acceleration values after rotation, and M is the unknown alignment matrix. This matrix is parameterized by a yaw, pitch, and roll misalignment. The above relations hold if the gyroscope triad is at rest before and after the rotation. This allows the gyroscope bias to be removed and the relative change in orientation to be estimated with precision. Example integration windows are shown in Figure 39. From this figure, the gyroscope signals are integrated between the start and end points outlined in the plot. The alignment angles are then estimated by minimizing the difference between the predicted and measured acceleration signals. The resulting objective function is given mathematically as

$$O(\bar{a}) = \frac{1}{N} \sum_{i=1}^{n} \left(\frac{\cos^{-1}(\vec{a}_{B_{meas}} \cdot \vec{a}_{B_{pred}})}{\left\| \vec{a}_{B_{meas}} \vec{a}_{B_{pred}} \right\|} \right)^2 , \tag{3.42}$$

where $\bar{a}_{B_{meas}}$ is the measured accelerometer output after rotation, and $\bar{a}_{B_{pred}}$ is the predicted accelerometer output. The accelerometers are co-located with the gyroscopes, eliminating any lever arm effects. For the single axis accelerometers, the alignment angles are limited to the manufacturing tolerances of ±5 deg. The results for the scaling, orthogonalization, and alignment of the accelerometer triads are presented in Table 12. All errors are within the manufacturer's tolerance specifications. The advantage to the alignment method presented for the accelerometers is that the exact orientation of the device does not need to be known. This significantly reduces the complexity of laboratory calibration by requiring only a single data set for orthogonality, scale, bias, and alignment estimation.

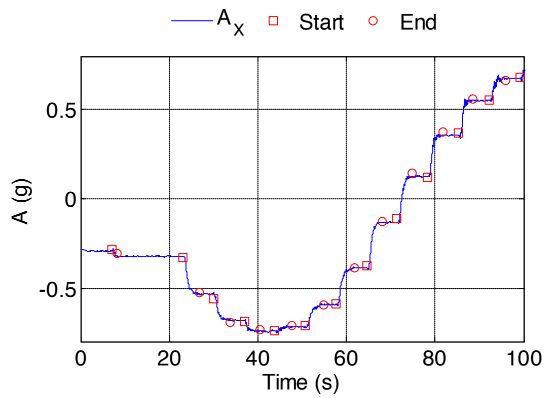


Figure 39: Integration windows for alignment estimation.

Table 12: Final calibration factors for accelerometers.

Manufacturer	Normalized scale factor	Orthogonalization angles α, β, γ (deg)	Alignment angles Pitch, roll, yaw (deg)	Bias (mg)
Silicon Sensing CMS390	0.997	91.918	-1.232	-2.17
	0.999	89.948	-0.462	-2.15
	1.001	89.162	0.431	-2.31
VTI SCC-1300	0.965	90.194	-0.545	-40.25
	1.105	89.750	-0.125	20.20
	0.950	88.655	0.250	15.10
InvenSense MPU-9150	1.003	90.037	1.248	8.00
	0.998	89.773	-1.057	8.20
	0.991	89.763	0.404	15.30

3.5.3 Magnetometer

The magnetometer is the primary instrument used for azimuth estimation. Three axes are necessary to recover a heading estimate in 3-D space due to the orientation of the Earth's magnetic field. For basic compassing operations in a level reference frame, the magnitude of the sensed magnetic field will be uniform. However, once the platform is tilted, the field strength will change based on the local magnetic field vertical component. This component varies in intensity over the surface of the Earth as documented and reported by Chulliat at the National Oceanic and Atmospheric Association (NOAA).[78]

To determine heading, the magnetometer data is derotated to the locally level frame using the attitude estimate provided by the gyroscopes. This is equivalent to gimbaling a standard two-axis needle type mechanical compass. The resulting locally-level data provides an azimuth estimate for the vehicle in the

NAV frame. This information is used in initial alignment of the inertial triad and in real-time estimation of gyroscope bias by the sensor fusion algorithm.

The calibration algorithm is similar to the method presented for the accelerometers. However, two additional bias terms require an *in-situ* estimation routine. The first bias component is related to the installation of the magnetometer in the vehicle. This bias is the result of localized distortions in the magnetic field termed hard-iron biases. These biases typically do not change during flight and can be accounted for during ground alignment. Azimuth-dependent deviations are termed soft-iron distortions and must be estimated by the sensor fusion algorithm. Typical causes of soft-iron distortions are the presence of ferrous objects such as other vehicles or intermittent use of high-powered transmitters. Once the aircraft has reached a sufficient altitude, these effects are typically small. Figure 40 shows the raw sensor data for the HMC-5883L magnetometer as installed in the aircraft. From this data, the hard-iron distortions are visible as the deviation from the ideal spherical representation of the magnetic field. The radius of the sphere will change depending on the local field strength as determined by the NOAA reference model.[78]

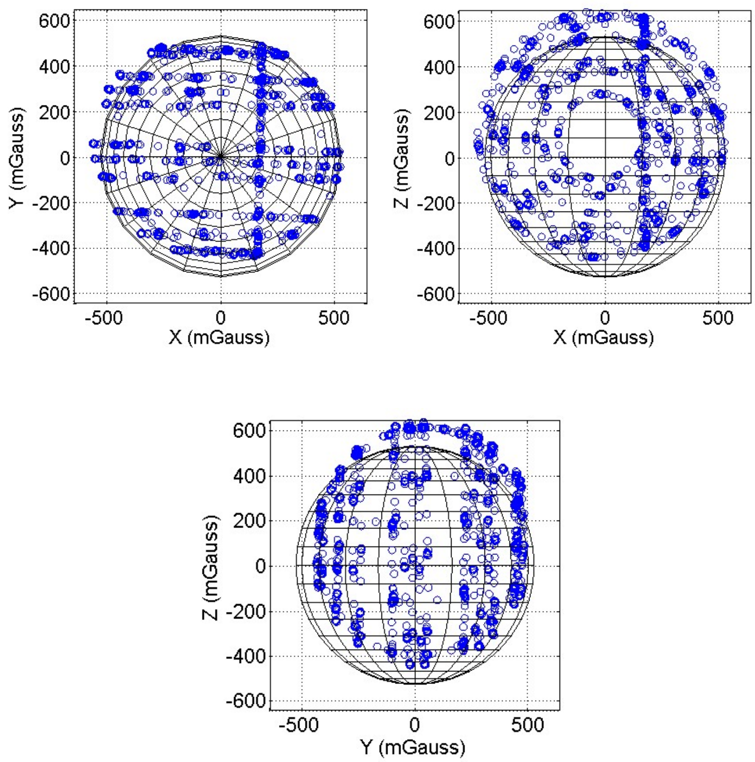


Figure 40: Raw magnetometer readings plotted on ideal spherical surface for the HMC-5883L.

The calibration of the magnetometer follows from the method presented for the accelerometers, and can be accomplished simultaneously. The major difference in sensor calibration is the reference vector in the scaling and orthogonalization optimization. For the accelerometers, the vector has a single component equal to the local gravitational acceleration. For the magnetometer, the alignment vector uses the three components of the local magnetic field. In practice, this does not complicate the optimization, but must be known *a priori* for proper estimation of the alignment matrix. In general, the direction of the magnetic field deviates significantly from the rotational axis of the Earth. In some locations, this can be up to 10 degrees. The offset introduced by the inclination and declination angles must be incorporated into the alignment matrix such that the estimated velocity and displacement match with the ECEF and NAV reference frames used by the INS and GPS systems. The orthogonalization, scaling, bias, and alignment matrices are formulated and computed in the same way as the accelerometer. The resulting calibration coefficients for the magnetometers are presented in Table 13. Figure 41 shows the corrected data and Figure 42 shows the normalized sensor output variation over a range of orientations.

Table 13: Magnetometer calibration coefficients

Manufacturer	Normalized scale factor	Orthogonalization angles α, β, γ (deg)	Alignment angles Pitch, roll, yaw (deg)	Bias (mGauss)
Honeywell HMC-2003	1.220	90.213	-0.262	-52.131
	1.114	89.102	-1.251	10.251
	1.224	88.225	0.015	15.795
HMC-5883L	0.973	87.867	-1.232	-16.198
	1.086	91.407	-0.462	35.620
	0.978	92.211	0.431	98.833
InvenSense MPU-9150	1.045	98.022	-6.956	-210.219
	0.982	82.475	-9.000	324.447
	0.961	101.807	-5.746	-87.433

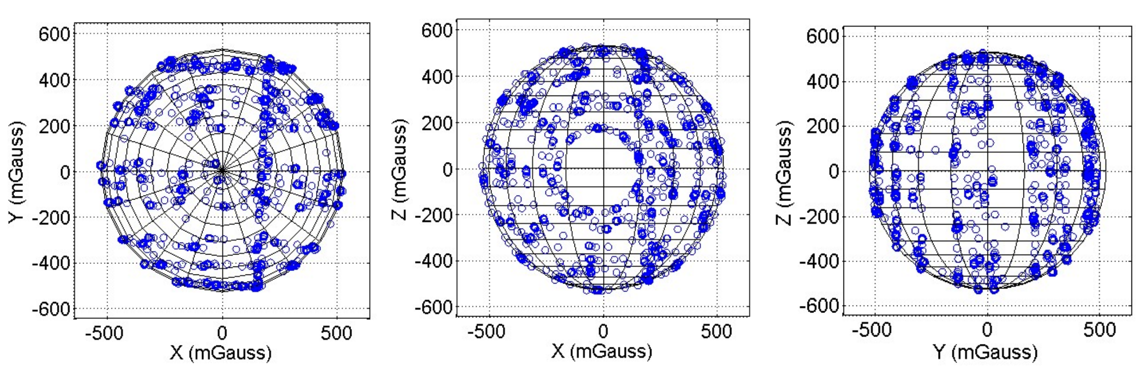


Figure 41: Calibrated output data for the HMC-5883L

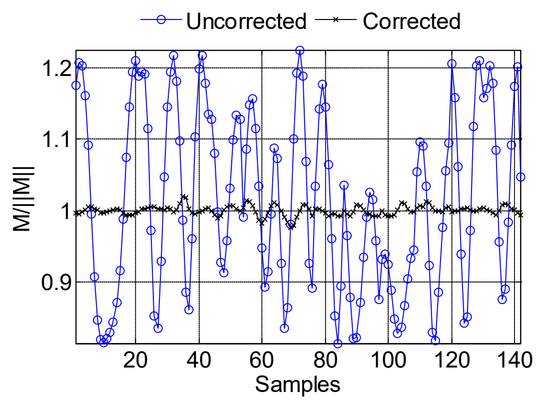


Figure 42: Normalized output from the HMC-5883L showing reduced variance over the sensors operating range.

3.5.3.1 Magnetometer In-Flight Bias Estimation

One substantial challenge associated with using magnetometer data in the onboard sensor-fusion filter is related to substantial noise present in the signal. This noise led to large uncertainties in azimuth estimation, which coupled directly to velocity and position errors as detailed in section 3.3. A typical example of the corrupted data is shown in Figure 43.

A Fourier analysis, shown in Figure 44, indicated that the source of the magnetometer noise was related to rapidly switching currents in wires and components (servos, motor, and telemetry systems) near the magnetometer. The FFTs shown in Figure 44 depict current switching noise, which is immediately traceable to events triggered by the onboard autopilot. The spikes centered around 40 Hz are servo switching noise. The two peaks near 75 Hz represent the motor commutation frequency. Telemetry transmission packets cause the low frequency spikes between 0-20 Hz. The onboard flight computer monitors the current signal at 10 kHz, which is significantly faster than the maximum commutation frequency of 1 kHz.

Fortunately, these systems are either monitored or controlled by the flight computer and their states are known in real time. By monitoring the instantaneous current usage, a signal mask can be created. During periods of high current surges, the magnetometer signal is not used. The current and magnetometer signals, shown in Figure 45, are normalized by one revolution of the motor. Therefore, an adaptive mask based on motor rotational velocity is introduced. The main advantages of this mask-based filter are the preservation of the sensor's bandwidth, and avoidance of the phase errors introduced by traditional filtering schemes. The HMC-2003s used in this guidance system has a maximum bandwidth of 1 kHz. To achieve a similar noise reduction, the magnetometer signal would need to be low pass filtered at approximately 5 Hz, which is insufficient relative to the correction filter requirements. A simple first-order digital low pass filter would result in a phase lag of up to 45 deg with a corresponding maximum group delay approaching 0.5 seconds. For an aircraft rotating at up to 45 deg/s, this correlates to a heading estimation error of up to 22.5 deg. This is error is large enough to cause divergence in the sensor-fusion algorithm used to correct the INS state estimates.

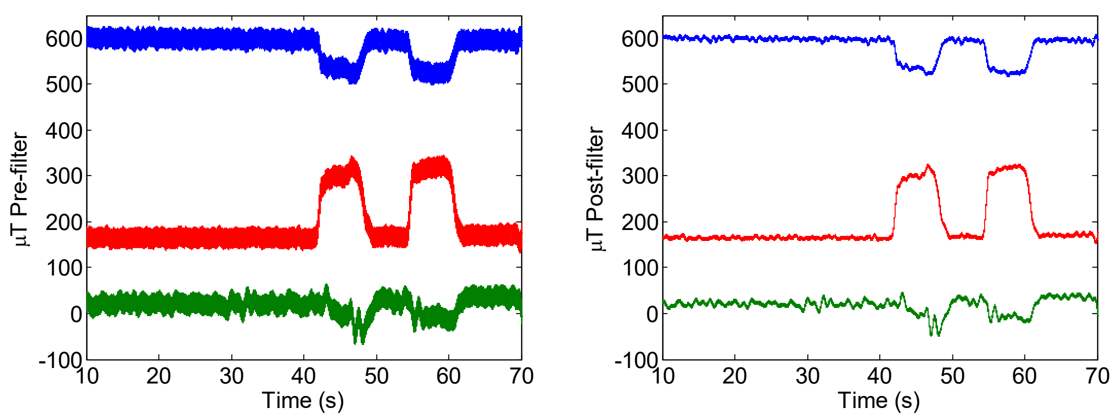


Figure 43: Raw magnetometer measurements (left) and mask filter corrected magnetometer measurements (right).

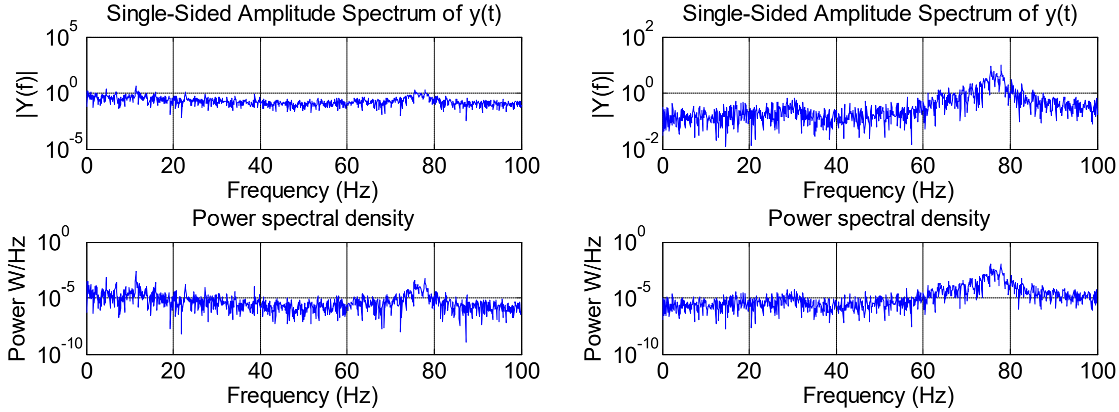


Figure 44: FFT of the magnetometer signal during a constant motor power phase of flight (left) and FFT of current signal (right).

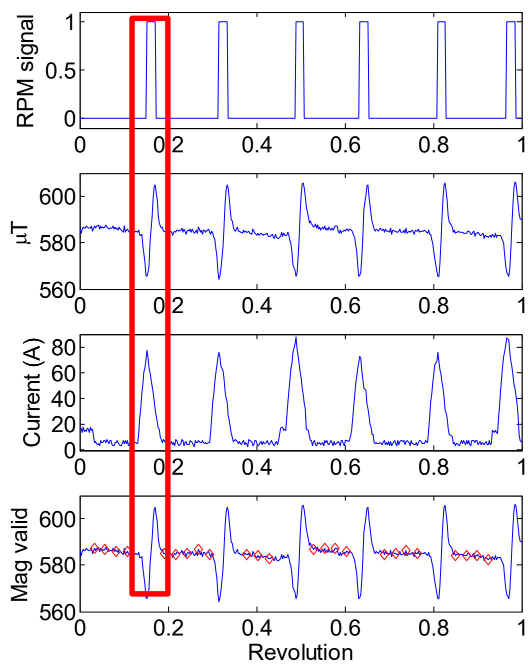


Figure 45: Magnetometer discrimination filter. The red block indicates a period where the magnetometer signal is invalid.

Noise in the INS sensors limits the ability of the flight computer to maintain a known altitude and airspeed. The INS used in this work employs an error-state filter capable of estimating the common sources of error in the inertial sensors[14]. These errors typically include contributions from bias, scale factor, and nonorthogonality. The scale factor and orthogonality errors are calibrated prior to flight using a rate table and typically do not change during the course of the flight. The remaining bias terms shift in flight and are correlated with EMI and vibration noise sources present in the airframe. This effect was demonstrated for the magnetometer, but similar errors occur in the gyroscopes, accelerometers, and barometer. As a practical example, the identification and elimination of the current based magnetometer noise allowed for an improvement in azimuth estimation, from ±9 deg to ±1 deg during periods of high rotation.

3.5.4 Air Data Calibration

This section details the calibration of the air data system. This includes the pitot-static system and barometric altimeter.

3.5.4.1 Barometric Pressure Calibration

The differential and barometric pressure sensors were calibrated against a known reference pressure transducer. A small PVC chamber was constructed which would house the pressure sensors and allow for the ambient pressure to be controlled via vacuum pump. Several barometric sensors were tested to determine the effects of temperature and ambient pressure on their output. In general, differential and barometric sensors are relatively insensitive to temperature variations as they are internally compensated. The resulting scale factor for each sensor was within the tolerances specified by the manufacturer. The offset bias terms varied greatly between sensors, yet were stable between tests. This fixed offset is likely the result of mechanical or thermal stress introduced on the component during assembly. The addition of mechanical stress via application of weights to the sensor significantly shifts the output. However, once the weight is removed, the output returns to its initial offset. The stability of the bias terms and the linearity of the sensors makes them amenable to a calibration. Calibration results for the BMP-085 barometric sensor are shown in Figure 46. From this figure, the linear curve fit has an R^2 value of 0.99 with a slope of 1.00.[79] While the scale factor and repeatability of the pressure sensors is evident, there is an initial bias in all sensors tested which needs to be removed. Pressure altitude is then calculated based on the relations presented in the 1976 standard atmosphere [79]

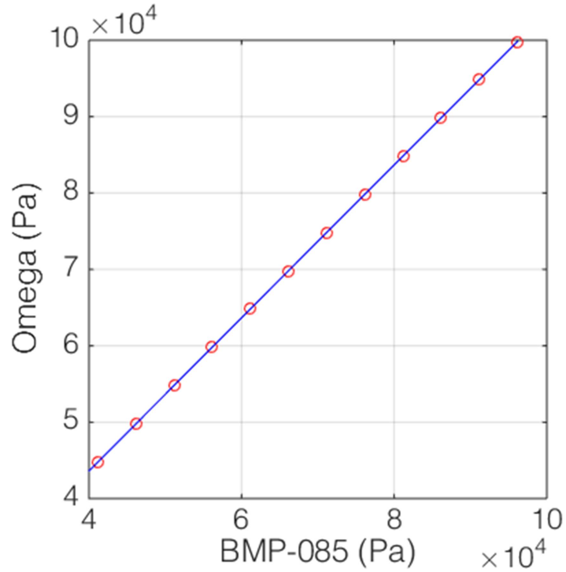


Figure 46: Pressure calibration data using an Omega PX409 reference signal.

3.5.4.2 Differential Pressure Calibration

The differential sensors used to determine airspeed were calibrated in a manner similar to the barometric sensors. To determine airspeed from the pitot probe, the incompressible Bernoulli relation is used

$$V_{airspeed} = \sqrt{\frac{2\mathrm{K}\Delta P}{\rho}}\ , \tag{3.43}$$

where K is a calibration factor which models the effects of mounting and directionality. This constant can be estimated as follows. The system model is a single scalar where K is modeled as a first-order Gauss-Markov process given as

$$\dot{K} = 0\ . \tag{3.44}$$

The wind estimate provided by the INS is subtracted from the ground and vertical speed provided by the GPS and barometer. The predicted airspeed is then given by

$$V_{pred} = C_{NAV}^{B} \left\| \vec{V}_{ground} - \vec{V}_{wind} \right\| \tag{3.45}$$

This model assumes that the airspeed vector is aligned with the fuselage of the aircraft. This assumption is valid over most of the flight range tested in this work. Error in the predicted airspeed will be increased if the vehicle is flown in an uncoordinated manner or at high angles of attack. These relations can be used in a standalone filter, or fused with other measurements from the INS error filter. The effectiveness of the calibration method was demonstrated using the wind tunnel outlined in chapter 2, and the aircraft pitot tube mounted on a sting. For this test the pitot and wind tunnel dynamic pressures were recorded. The estimated airspeed was computed based on comparing the tunnel speed with the measured dynamic pressure from the Pitot tube. In this way, the test corresponds to a flight in zero wind. A 100 mm x100 mm flat plate was placed directly behind the pitot tube at several downstream locations. As the plate is moved closer to the pitot tube, the ratio between the pitot and wind tunnel dynamic pressures decreases. The decreasing dynamic pressure ratio corresponds to mounting or other scale factor errors affecting the readings from the onboard dynamic pressure sensor. The proposed dynamic pressure calibration algorithm quickly converged on the ratio of the measured and actual dynamic pressures for all blockage locations as shown in Figure 47. Knowledge of this ratio allows the corrected airspeed to be computed onboard the aircraft. In practice, the flight calibration of the pitot sensor is carried out when low wind conditions exist. This limits any coupling between the INS wind and pitot scale factor estimation. For the aircraft considered in this work, the calibration factor K is 0.975. Once the calibration factor estimate stabilizes, it is locked in and treated as a constant. If the sensor or pitot tube is changed, the calibration routine is rerun during the next flight.

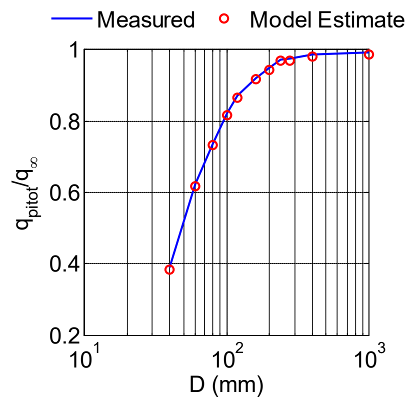


Figure 47: Measured and modeled dynamic pressure ratios for pitot tube interference testing.

3.6 Error-State Kalman Filter Formulation

The error-state equations introduced in section 3.3 are used with an extended Kalman filter to be introduced in this section. The basic Kalman filter algorithm will not be derived in this work given the prevalence in the available literature. Instead, the basic filter structure will be presented, with emphasis placed on the state transition and measurement matrices unique to the error-state INS application. Error quantification is presented for several test cases helpful in identifying the performance of the INS system.

3.6.1 Kalman Filter Equations

The Kalman filter was introduced by Rudolph Kalman in 1960.[80] Since then, the Kalman filter has gained widespread use for both linear and non-linear state estimation problems.[16, 30, 32, 75, 76, 81] Many applications of the filter use a common algorithm structure. In brief, the basic algorithm is summarized as follows.[23]

1) Propagate State:

   Given knowledge of the system states at time *t*, the states are predicted at time t+1 using a model of the system dynamics. For the filter presented in this work, the system model is given in section 3.3 and the details of the state transition matrix are presented in section 3.6.2. This process can be described as

$$\bar{x}_{t+1} = \Phi_t x_t \ , \tag{3.46}$$

where $\bar{x}_{t+1}$ is the predicted state at time t+1, Φ_t is the state transition matrix, and x_t is the current state at time t.

2) Propagate Covariance:

The covariance estimates are propagated to time t+1 using the state transition matrix and an additional Gaussian noise source, Q , termed the process noise,

$$\bar{P}_{t+1} = \Phi_{t+1} P_t \Phi_{t+1}^T + Q_{t+1} \ . \tag{3.47}$$

3) Measurement Update:

The first step in updating the predicted state with a measurement is to estimate the optimal Kalman gain, K_{t+1} . This reflects the trust in the estimated and predicted data. For linear filters, the steady-state gain is largely controlled by the noise covariance matrices, Q and R ,

$$K_{t+1} = \bar{P}_{t+1} H_{t+1}^T \left(H_{t+1} \bar{P}_{t+1} H_{t+1}^T + R_{t+1} \right)^{-1} \tag{3.48}$$

where P is the measurement covariance, and H is the measurement transition matrix. After the optimal gain is determined, the updated covariance is given as

$$P_{t+1} = \left[I - K_{t+1} H_{t+1} \right] \bar{P}_{t+1} \ . \tag{3.49}$$

Finally, the updated state is given as the weighed sum of the predicted and measured states

$$\hat{x}_{t+1} = \bar{x}_t + K_{t+1} \left[z_{t+1} - H_{t+1} \bar{x}_{t+1} \right] . \tag{3.50}$$

3.6.2 State Transition matrix

The basic Kalman filter structure is outlined in the previous section. The filter requires several unique components to reflect the intended application. The first component is the state transition matrix. This matrix is used to predict the states at time *t*+1 given the information available at time *t*. For the INS case considered in this work, the required state transition equations were derived in section 3.3. The fundamental equations are repeated here for clarity.

$$\dot{\psi} = \psi \times \left(\omega_{IE}^{NAV} + \omega_{EN}^{NAV} \right) + \varepsilon^{NAV} \tag{3.51}$$

$$\delta\dot{V}^{NAV} = -\left(\omega_{EN}^{NAV} + 2\omega_{IE}^{NAV} \right) \times \delta V^{NAV} + a^{NAV} \times \psi + \delta a^{NAV} \tag{3.52}$$

$$\delta\dot{S}^{NAV} = \delta V^{NAV} \tag{3.53}$$

For the basic filter, with no bias terms, equations (3.51), (3.52), and (3.53) can be combined as follows

$$\begin{bmatrix} \delta\dot{S}^{NAV_x} \\ \delta\dot{S}^{NAV_y} \\ \delta\dot{S}^{NAV_z} \\ \delta\dot{V}^{NAV_x} \\ \delta\dot{V}^{NAV_y} \\ \delta\dot{V}^{NAV_z} \\ \delta\dot{\psi}^{NAV_x} \\ \delta\dot{\psi}^{NAV_y} \\ \delta\dot{\psi}^{NAV_z} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \Omega_{2_z} & -\Omega_{2y} & 0 & -a_z & a_y \\ 0 & 0 & 0 & -\Omega_{2_z} & 0 & \Omega_{2_x} & a_z & 0 & -a_x \\ 0 & 0 & 0 & \Omega_{2_y} & -\Omega_{2_x} & 0 & -a_y & a_x & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \Omega_{1_z} & -\Omega_{1_y} \\ 0 & 0 & 0 & 0 & 0 & 0 & -\Omega_{1_z} & 0 & \Omega_{1_x} \\ 0 & 0 & 0 & 0 & 0 & 0 & \Omega_{1_y} & -\Omega_{1_x} & 0 \end{bmatrix} \begin{bmatrix} \delta S^{NAV_x} \\ \delta S^{NAV_y} \\ \delta S^{NAV_z} \\ \delta V^{NAV_x} \\ \delta V^{NAV_y} \\ \delta V^{NAV_z} \\ \delta\psi^{NAV_x} \\ \delta\psi^{NAV_y} \\ \delta\psi^{NAV_z} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \delta a_x \\ \delta a_y \\ \delta a_z \\ 0 \\ 0 \\ 0 \end{bmatrix}, \tag{3.54}$$

where $\Omega_1 = \left(\omega_{IE}^{NAV} + \omega_{EN}^{NAV} \right)$ and $\Omega_2 = \left(\omega_{EN}^{NAV} + 2\omega_{IE}^{NAV} \right)$. For the state transition matrix listed in equation (3.54), the estimated states include the position, velocity, and attitude errors. Equation (3.54) is of the form

$$\dot{x} = Fx + w\,. \tag{3.55}$$

To predict the states at time t+1, equation (3.55) must be numerically integrated. This is accomplished via first-order Euler integration, which can be described as

$$x_{t+1} = x_t + Fx_t dt = \left(I + F\right) x_t dt = \Phi_t x_t \quad . \tag{3.56}$$

3.6.2.1 Bias Estimation

Additional terms can be included in the above formulation for time-varying sensor bias estimation. The time-varying gyroscope bias is the only additional state considered in this work. The gyroscope bias is modeled as a first-order Markov process[29]

$$\Delta\dot{\omega}_{bias} = \frac{1}{\tau}\Delta\omega_{bias} + \varepsilon \ , \tag{3.57}$$

where τ is the time constant governing the evolution of the bias estimate and ε is an additive white noise component. For this work, the time constant is chosen as 10 seconds. This time constant is based on empirical testing of the INS sensors used in this work. The bias is numerically integrated based on the preceding equation as

$$\omega_{bias_t} = \omega_{bias_t} + \Delta\omega_{bias} \quad . \tag{3.58}$$

The bias terms can be incorporated into equation (3.55) as

$$\begin{bmatrix} \dot{x} \\ \Delta\omega_{bias} \end{bmatrix} = \left[\begin{array}{c|c} \boldsymbol{F}_{9x9} & \begin{matrix} \mathbf{0}_{6x3} \\ C_B^{NAV} \end{matrix} \\ \hline \mathbf{0}_{3x9} & I_{3x3}\left(1/\tau\right) \end{array}\right] \begin{bmatrix} x \\ \Delta\omega_{bias} \end{bmatrix} + \begin{bmatrix} \varepsilon_{INS} \\ \varepsilon_{bias} \end{bmatrix}, \tag{3.59}$$

where $\boldsymbol{F}$ represents the state transition matrix given in equation (3.54), and I is an identity matrix with the dimensions given in the subscript and ε is an additive noise term. This completes the outline of the relations necessary to predict the states at time t+1 according to step (1) listed in section 3.6.1. The next component of the Kalman filter is the measurement model, which relates the states to the outputs of the aiding sensors.

3.6.3 Measurement Model

The measurement model relates the calculated states to the measurements provided by other low-bias aiding sensors. These sensors were outlined in section 3.4 and include the magnetometer, barometer, and GPS. The error-state formulation ensures that the measurement equation is linear, greatly reducing the computational cost of the filter. The measurement equation is expressed as

$$z = \begin{bmatrix} \delta\overline{S} \\ \delta\overline{V} \\ \delta\psi_z \end{bmatrix} = \begin{bmatrix} \overline{S}_{INS} - \overline{S}_{GPS} \\ \overline{V}_{INS} - \overline{V}_{GPS} \\ Heading_{INS} - Heading_{meas} \end{bmatrix}. \tag{3.60}$$

The position error is evaluated in the NAV frame by transforming the ECEF position estimates provided by the GPS. The barometer, if available, provides the vertical position and velocity. If the barometer fails, the vertical state estimate provided by the GPS is used. ECEF velocity components are available from the GPS and are transformed to the NAV frame before processing. Heading information is provided by the corrected magnetometer components rotated into the NAV frame. Pitch and roll errors are estimated by the filter as no direct measurement of these states is available. The measurement transition matrix, H, is given simply as

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}. \tag{3.61}$$

This completes the measurement model description. The next section details the noise covariance estimates used in the filter, given by matrices Q and R in equation (3.47) and (3.48).

3.6.4 Noise Covariance Estimation

The noise covariance matrices represent the "tuning knobs" for Kalman filters. The process noise, Q is a diagonal matrix given by

$$Q = \begin{bmatrix} \sigma^2_{Position} & & & \\ & \sigma^2_{Velocity} & & \\ & & \sigma^2_{Angle} & \\ & & & \sigma^2_{Bias} \end{bmatrix}, \tag{3.62}$$

where the position, velocity, angle, and bias matrices are diagonal 3x3 matrices representing the noise in their respective states. For this work, the process noise is considered uncorrelated. Furthermore, the position and velocity error models are considered perfect, so zero additive noise is used. The angular noise is given by the angular random walk estimate (detailed in chapter 4), and the bias noise from the gyroscope bias stability, outlined in Table 4. Additional tuning is typically required based on operational conditions and external vibrations.

The measurement noise covariance matrix, R, is typically found by analysis of the aiding sensors (GPS, barometer, etc.). These noise statistics are approximated by analysis of the static and in-situ sensor noise. Based on the previously presented sensor descriptions, the final measurement uncertainty terms are shown in Table 14. In this table, the x-y components refer to GPS related uncertainties, and the vertical z components are provided by the barometer. Heading uncertainty is based on the corrected magnetometer data. The measurement noise covariance is given by

$$R = \begin{bmatrix} \sigma^2_{Position} & & \\ & \sigma^2_{Velocity} & \\ & & \sigma^2_{Heading} \end{bmatrix}, \tag{3.63}$$

where the position and velocity matrices are diagonal 3x3 matrices expressed in the NAV frame and the heading noise is a scalar.

Table 14: Measurement noise estimates

Noise component	Noise estimate
$\sigma_{Position}$	± 0.7 m static, x-y
	± 1.5 m dynamic, x-y
	± 0.5 m z
$\sigma_{Velocity}$	± 0.5 m/s static, x-y
	± 1.5 m/s dynamic, x-y
	± 1.0 m/s z
$\sigma_{Heading}$	±1 deg

In practice the noise covariance estimates are updated in flight using the circular error probability (CEP) output provided by the GPS. The CEP provides an estimate of the error in the GPS solution. The CEP is largely influenced by the number of visible satellites and whether an aiding source such as WAAS is available. For the flights conducted in this work, at least seven satellites were visible at all times with a maximum of 14. The filter is updated at the GPS output frequency of 4Hz.

3.7 Summary

This chapter presented the relationships and sensors necessary for characterization of an INS. This included the introduction of an error-state equation set used in subsequent chapters for assessing the accuracy of the inertial solution. Sensor error models were introduced and calibration methods proposed to both align and scale the sensor outputs to satisfy the small-error assumption implicit in the error-state formulation. While the accuracy of the sensors has been presented, the overall system performance still requires characterization. The system level performance estimation will assess the overall impact of measurement and estimation errors on the flight-test performance predictions presented in chapter 5. The next chapter will assess the performance of the attitude, velocity and position equations using the calibrated sensors introduced in this chapter. Static and in-situ INS error is detailed based on quasi-static and hardware-in-the-loop (HIL) testing. The result of the system testing will be the total state uncertainty present in the INS.

Chapter 4: Navigation System Testing and Validation

4.1 Background

This chapter details the testing necessary to quantify the uncertainty in INS state estimation. The testing used to validate the INS is a combination of quasi-static testing to assess sensor performance and calibration, and hardware-in-the-loop (HIL) testing to assess in-situ performance. The effects of sensor noise and bias are quantified as well as the effects of wind on overall state uncertainty.

4.2 Quasi-Stationary Testing

To verify the implementation accuracy of the attitude, velocity, and position algorithms and sensor calibrations presented in chapter 3, a quasi-stationary test of the INS was performed. There are several methods of assessing the accuracy of INS hardware and algorithms in the published literature.[16, 17, 29] These generally include a comparison to a known reference source, or other high-accuracy aiding solution such as differential GPS or Vicon type imaging system.[75, 77] Other methods of quasi-static, non-corrected INS testing rely on using synthetic data, which approximates the expected errors in the inertial sensors.[18, 30] While these tests provide valuable insight into the performance of the attitude, velocity, and position algorithms, they typically do not capture the in-situ performance of the physical INS. Additional complications can be due to a myriad of additional factors such as processor limitations, timing errors, quantization effects, and vehicle vibrations.

To test the in-situ accuracy of the INS system and the calibration methods presented in the previous chapter, a zero-displacement rotational test was performed with the completed INS. Such testing has been carried out with high-precision INS systems.[21, 26] These tests typically require highly accurate motion platforms that exceed the performance of the INS. For low-cost MEMS based systems, much insight can be gained using a testing apparatus with a positional accuracy in line with the predicted accuracy of the INS.

Others have demonstrated testing methods using rotation stands for low-cost INS systems.[24, 25] These systems are typically comprised of two or three degrees of freedom with coarse closed-loop control. Control of the test stand is typically accomplished using either servo or stepper motors in a direct-drive configuration. The testing presented in this section uses a testing stand matched to the performance of the INS to assess the in-situ stability and accuracy of the state estimation algorithms. First, an analysis of the factors governing the in-situ accuracy of the INS are presented.

For the INS presented in this work, the stability of the uncorrected attitude solution is limited by the gyroscope bias stability and output noise.[33] For the highest performance gyroscopes considered, the CMS390s, this equates to a bias drift 10 deg/hr, or 2.8×10^{-3} deg/s with a noise floor of 6.1×10^{-2} deg/s. For short integration intervals, the noise floor dominates the angular estimation error. Based on the noise statistics, the angular precision is 4.7×10^{-4} deg using the INS integration timestep of 1/128 s. This represents the theoretical resolution limit for the gyroscopes. However, continued integration of the noise corrupted signal results in a well characterized divergence of the angular position estimate. The drift is termed the angular random walk (ARW). For the CMS390 gyroscopes, the ARW is related to the noise floor and quoted as 6×10^{-2} deg/$\sqrt{s}$. Therefore, if the gyroscopes are integrated at 128 Hz for one second, the resulting ARW will give a total angular uncertainty of $\pm5.2\times10^{-3}$ deg. This statistic was verified by integrating the gyroscope signals for 1 second, and repeating 100 times. The resulting ARW traces are shown in Figure 48 for the CMS 390 gyroscopes. For this figure, the bias in each trace was removed before integration. The theoretical error growth based on the quoted noise statistics is shown for comparison (blue lines). The proposed quasi-stationary test is designed to last for 60 seconds, giving an angular uncertainty due to ARW of 0.04 deg.

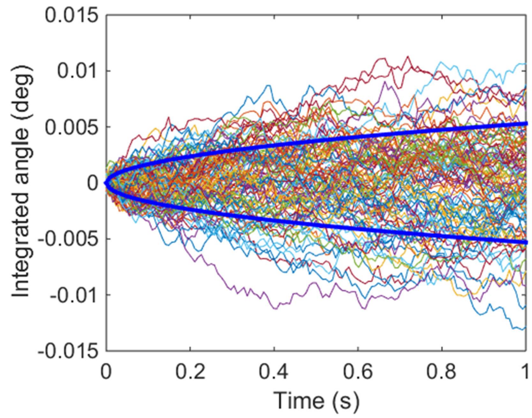


Figure 48: Angular random walk (ARW) for CMS 390 gyroscope for 100 samples. Blue lines indicate theoretical 1-σ error growth for 128 Hz integration time step.

The calibration test stand uses a closed-loop controller with servos (x-y axes) and a stepper motor (z-axis). This testing apparatus is similar to the unit used for accelerometer and magnetometer calibration in the previous chapter, with the addition of a feedback controller. The feedback controller uses a US Digital E3 optical encoder with a 10,000 counts/rev, (40,000 pulse/rev) quadrature output signal. The encoder is attached to the driving gear of the stage axis. With a 7:1 mechanical reduction provided by the motor-stage drive, the effective resolution is 280,000 pulses/rev, equating to 1.3×10^{-3} deg/pulse. The servo and stepper motors were controlled using a Parallax Propeller microcontroller operating at 80 MHz. The microcontroller logged the attitude, velocity, position, gyroscope, accelerometer, magnetometer, and encoder signals at 128Hz to an onboard SD card. This allows for a post-test reconstruction of the attitude, velocity, and position error.

Testing involved sinusoidally ramping the x, y, and z-axes of the stand to a maximum angular velocity of 30 deg/s. The path was then reversed, following the same path back to the start location and resulting in zero net displacement. The encoder track and INS estimated angular position are shown in Figure 49.

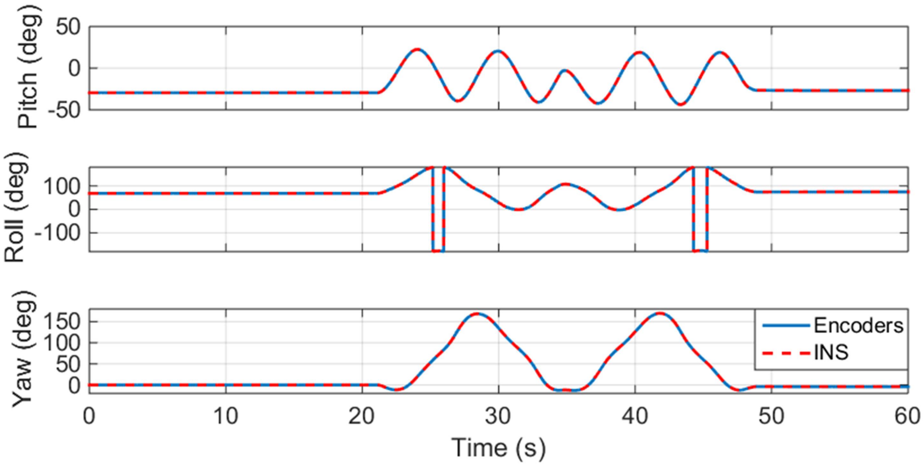


Figure 49: Motion table and INS angle estimate for zero-displacement test using the CMS390 sensors.

The angular position error estimates are shown Figure 50. From this figure, the relative error increases during the rotation period between 20 and 50 seconds. This is likely caused by resonant vibration and slight misalignments in the table axes. The rotation test outlined in this section was repeated fifty times. The resulting error statistics are presented in Table 15.

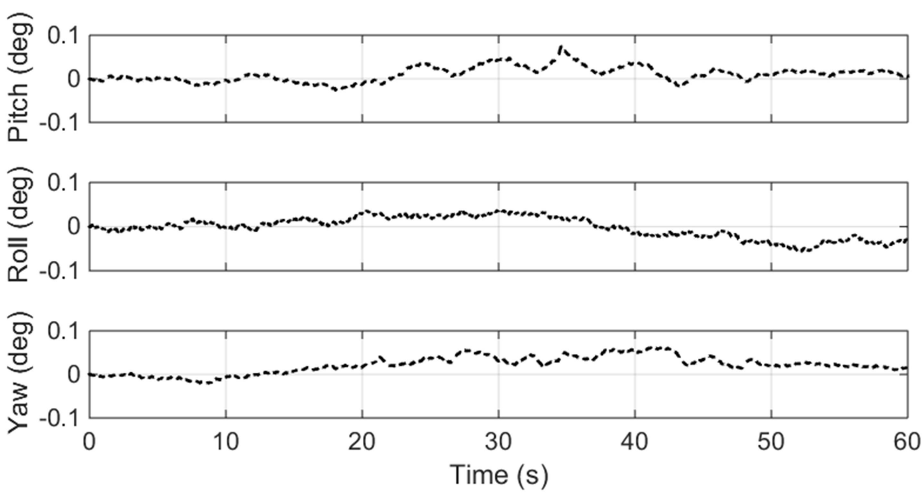


Figure 50: Attitude error for quasi-static rotation test using the CMS390 sensors.

Table 15: Angular error estimates from rotation table testing, 60 second integration window at 128 Hz.

Manufacturer	Noise Density (deg/√s)	RMS pitch error (±deg)	RMS roll error (±deg)	RMS yaw error (±deg)
Silicon Sensing CMS390	0.060	0.05	0.08	0.09
VTI SCC1300	0. 10	0.17	0.52	0.63
InvenSense MPU-9150	0. 52 (est)	0.92	0.61	1.24

The INS velocity and position follow from the relations presented in chapter 3. Velocity and position information from the encoder equipped rate table was compared to the estimates furnished by the INS. The results for the velocity in the NED frame are presented in Figure 51 and the NED position in Figure 53. From these figures, the angular error during platform movement resulted in an increase in velocity estimation error. The symmetric rotation of the table limits the velocity error buildup, shown in Figure 52. In this figure, the velocity drift is also clearly visible, due primarily to the accumulation of attitude errors. A constant attitude error results in a linearly increasing velocity error, and a parabolic increase in position error.[23] These effects are clearly visible in Figure 52 and Figure 54. Fifty tests were used to generate the statistics necessary for the tuning and evaluation of the error-state Kalman filter. The resulting uncertainty in velocity and position are given in Table 16 and Table 17. The difference in the north and east velocity and position errors is caused by an asymmetry in the rate table NED displacement and is not indicative of a unique error in the INS system.

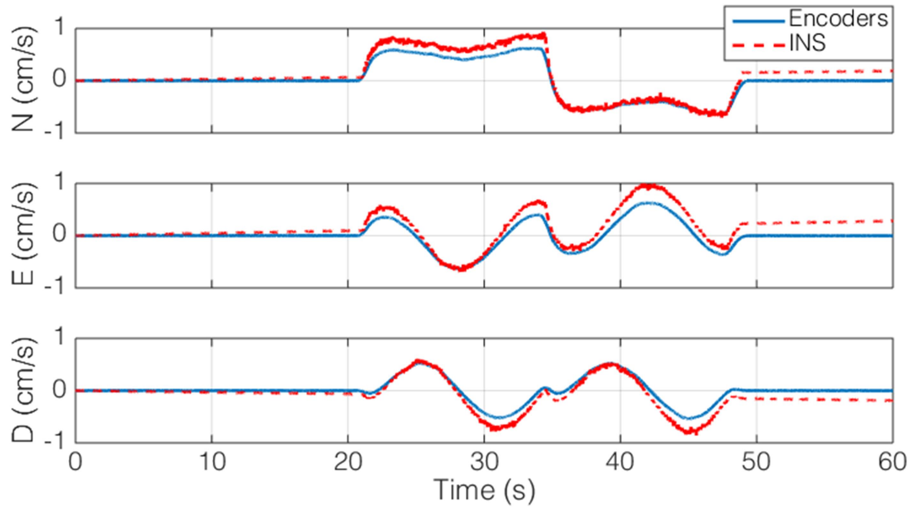


Figure 51: Velocity estimation for quasi-static rotation test using the CMS390 sensors.

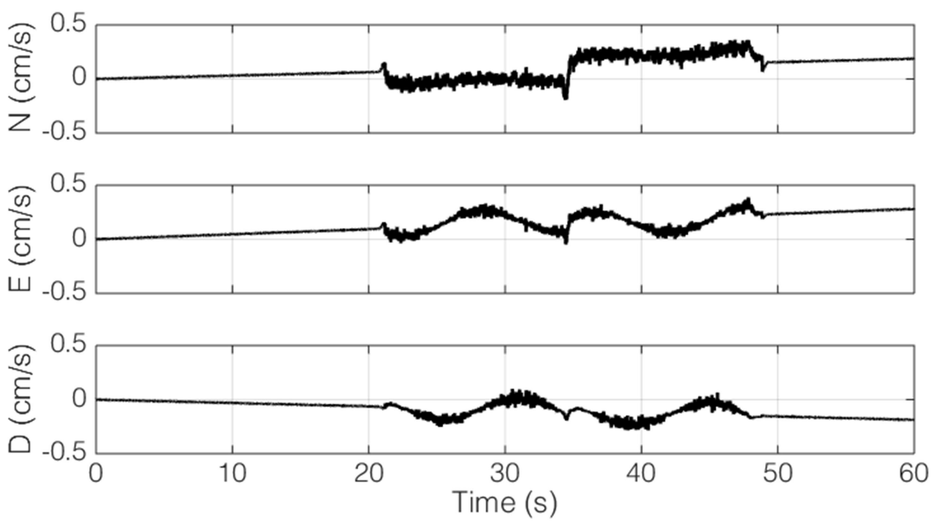


Figure 52: Velocity error for quasi-static rotation test using the CMS390 sensors.

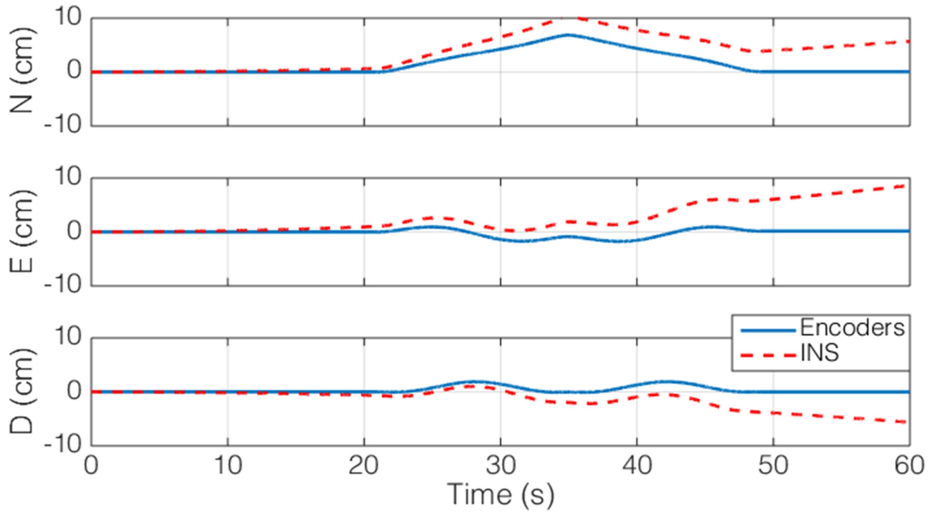


Figure 53: Position estimation for quasi-static rotation test using the CMS390 sensors.

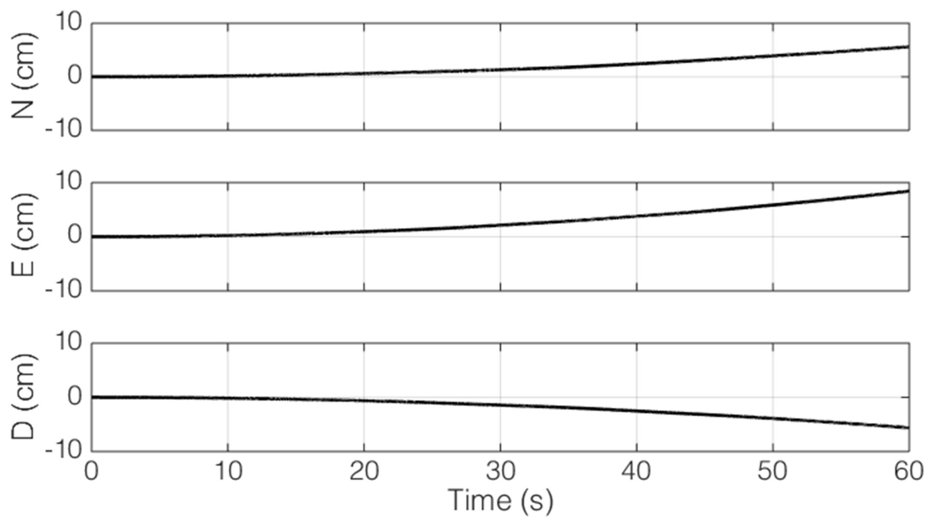


Figure 54: Position error for quasi-static rotation test using the CMS390 sensors.

Table 16: Velocity error estimates from rotation table testing, 60 second integration window at 128 Hz.

Manufacturer	RMS north error (±cm/s)	RMS east error (±cm/s)	RMS down error (±cm/s)
Silicon Sensing CMS390	0.43	0.68	0.77
VTI SCC1300	1.45	4.45	5.39
InvenSense MPU-9150	7.87	5.22	10.61

Table 17: Position error estimates from rotation table testing, 60 second integration window at 128 Hz.

Manufacturer	RMS north error (±cm)	RMS east error (±cm)	RMS down error (±cm)
Silicon Sensing CMS390	12.84	20.54	23.11
VTI SCC1300	43.65	133.51	161.75
InvenSense MPU-9150	236.20	156.61	318.36

The angular accuracy estimates formulated in this section are used as the process noise input to the error-state Kalman filter outlined in section 3.6.4. The effect of the measurement noise on the error-state filter still needs to be investigated to determine overall uncertainty in the estimated states. This is detailed in the next section using the INS hardware and the commercially available flight simulator X-Plane® for hardware-in-the-loop (HIL) simulations.

4.3 HIL Testing

HIL testing of INS systems has been recently proposed by others working on MEMS based INS systems.[35, 82] These studies focused primarily on flight vehicle control and not on state prediction. This section will demonstrate the applicability of HIL-based testing to the evaluation of INS performance. This testing provides the link between quantifying INS performance and the error statistics necessary for the flight-testing presented in the next chapter. In the quasi-steady state testing, the Kalman filter was not active. This was done primarily to test the accuracy of the INS navigation equations presented in section 3.2 and to generate an estimate of the process noise covariance necessary for using the Kalman filter. In this section, the Kalman filter is used to correct the state estimates computed by the INS. The measurement covariance matrix used the error statistics outlined in the previous section.

4.3.1 HIL Hardware and Setup

The HIL simulations used X-Plane® 10 to generate the required sensor data to test the INS system. X-Plane® is a physics based simulator with a relatively open architecture. This allows for near real-time input/output interfacing between the INS hardware and simulator.[83] The testing in this section utilized the physical INS hardware; therefore, a bridge between the sensor data generated by X-Plane® and the INS hardware is required. Custom software was written which established both a User Datagram Protocol (UDP) connection with X-plane® and a serial connection with the INS hardware. This software parses and scales the sensor outputs from X-Plane® and allows for the injection of sensor noise, biases, and quantization effects. The data rates are adjustable and designed to mimic the timing and output format of all the onboard sensors discussed in section 3.4. A schematic of the data flow is shown in Figure 55.

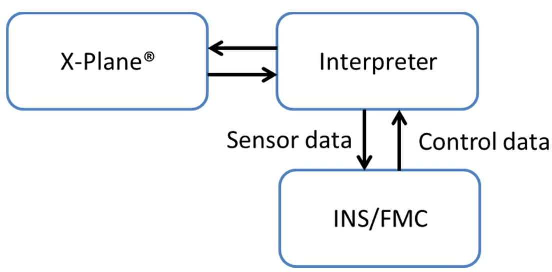


Figure 55: X-Plane® data flow diagram

X-Plane® is not natively designed for real-time applications. Therefore, an interpreter was developed to generate near real-time data for the physical INS. This software interfaces with X-Plane® via the available plugin software development kit. The interpreter interpolates the data provided by the simulator in order to provide a consistent time resolved data stream to the INS hardware. Interpolation is accomplished using the Spherical-Linear-intERPolation (SLERP) algorithm for incremental attitude and specific force estimation.[84] Interpolation necessitates buffering of the X-Plane® data, adding a delay between the sensor data output and INS processing. The X-Plane® data rate was matched to the INS at 128 Hz with a nominal variation of -20 Hz. To safeguard against latency and dropped packets, 100ms of data is buffered before being transmitted to the INS. The buffering ensures that the timing restrictions governing attitude, velocity, and position integration are satisfied. Buffering the output signals does not affect the INS state estimation as all of the simulated primary and aiding sensor data have an equal delay.

The interpreter repackages the X-Plane® data using the MAVLink protocol for serial transmission to the INS. MAVLink is a lightweight protocol which provides error checking and packet verification. When the INS receives the incoming data packet, a high-level interpreter injects the HIL data in place of the physical sensor data. The computed outputs from the INS and FMC are then passed back through the serial port to the interpreter. Vehicle control is delayed due to buffering of the output data. The result of this is a low frequency phase mismatch in the control system damping terms. To alleviate this effect, the damping terms (approximated as a scaled versions of the gyroscope signals) are neglected for the HIL tests. In practice, this makes the simulated aircraft more sensitive to gusts, but does not impact the state estimation algorithms.

4.3.2 HIL Flight-test Description

Control of the UAS is accomplished using a modified version of the open-source QGroundControl ground control station (GCS). This software establishes a command-and-control link with the UAS and allows for waypoint generation and system monitoring. For the purposes of INS evaluation, a simplified course was flown based on the flight-testing detailed in the following chapter. The salient features of the course are outlined in Figure 56. The flight is simulated at the Ohio State University's Don Scott airport (KOSU). The flight is initialized on runway 27L at zero velocity at the local field altitude. The first waypoint is located at the west end of runway 27L. After the simulation is initialized, the aircraft is accelerated to takeoff speed using the jet assisted takeoff system modeled in X-plane®. This simulates the accelerations experienced in a hand launch with a peak acceleration of 2g lasting for approximately 2 seconds. Once the aircraft reaches its minimum flight velocity of 12 m/s, it begins climbing to 130 m above ground level (agl) and turns towards waypoint 0. Once the aircraft reaches waypoint 0, it begins a right-hand turn towards waypoint 1. When the heading is within 5 degrees and position within 5 meters of the intended track, the motor power is cut, placing the aircraft in a glide. The aircraft then attempts to maintain constant airspeed by varying the glide angle. Once it has descended 50 m, full power is applied and the aircraft begins a constant airspeed climb back towards 130 m agl. After waypoint 1 is reached, the aircraft turns right again towards waypoint 2. Motor power is cut to slow the aircraft to its stall speed. Once stall is reached, full power is applied and the aircraft accelerates to its maximum velocity towards waypoint 2. After reaching waypoint 2, the aircraft turns right and enters a clockwise orbit around waypoint 3. After one complete orbit, the simulation is ended. This trajectory exposes the aircraft to the full range of velocities and attitudes encountered during flight-testing.

The HIL testing is used to quantify the INS error under a range of operating conditions. The first tests represent zero sensor noise, and zero wind. This is the nominal best-case scenario for state estimation. After this, a systematic error buildup is introduced in which the INS errors are computed after adding in sensor noise, random perturbations via wind, and gyroscope bias terms. Each of these error sources adds to the state uncertainty in the INS solution and must be quantified for the flight-test performance evaluation presented in the next chapter.

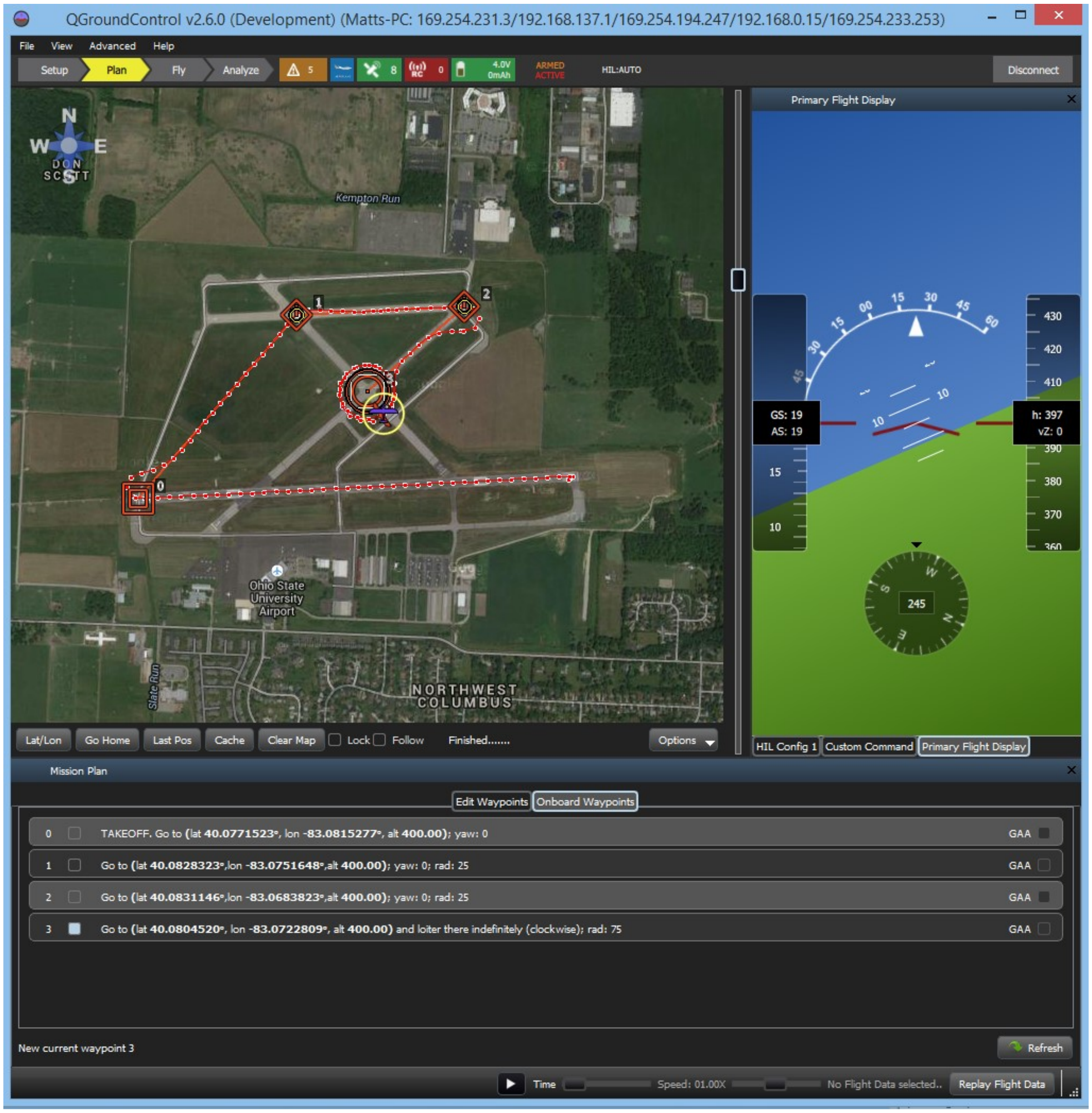


Figure 56: HIL ground station layout. The aircraft flight path is shown in left hand screen.

4.3.3 HIL Zero noise, zero wind

The first HIL-INS test uses zero additive sensor noise, and zero wind. This test therefore represents the best-case scenario. To test the performance of the error-state filter, the noise covariances were set based on the previously outlined data presented in section 3.6.4. No additional bias was added to the GPS data. Based on the data presented in Figure 57, the INS attitude error is equivalent to the uncorrected data presented in Figure 50. The estimated velocity and position are shown in Figure 58 and Figure 59, respectively. From these figures, the corrected velocity and position are within the uncorrected error bounds detailed in section 4.2. Mean error statistics are listed in Table 18.

Table 18: HIL state uncertainty estimates, 95% confidence.

Manufacturer	RMS angular error (±deg)	RMS velocity error (±m/s)	RMS position error (±m)
No-noise	1.738e-2	0.389	0.422

The flight trajectory in the HIL test covers the expected attitude and velocity envelope for the performance estimation flight-tests. This includes the impulse generated during hand launching of the UAS, glide tests, and full-power accelerations. The resulting errors in attitude, velocity, and position demonstrate the applicability of the HIL simulation in providing sensor data to assess the performance of the INS system. In the next section, sensor noise is added to the HIL simulation.

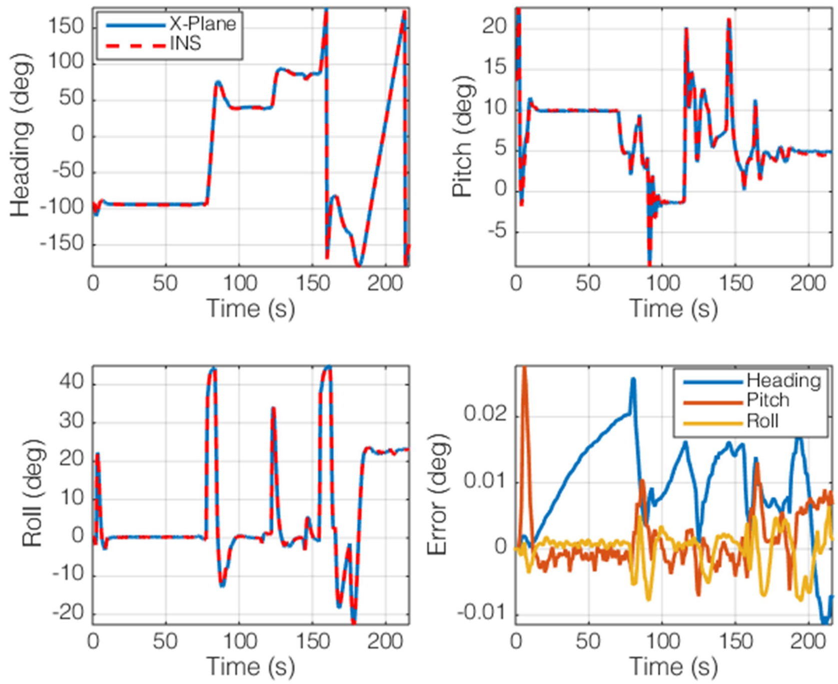


Figure 57: Attitude for HIL zero noise, zero wind.

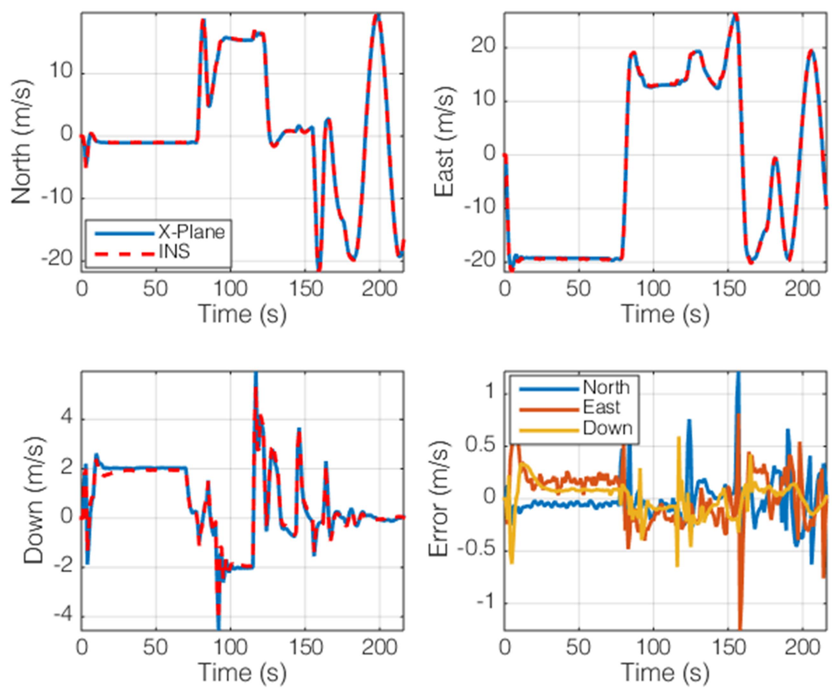


Figure 58: Velocity for HIL zero noise, zero wind.

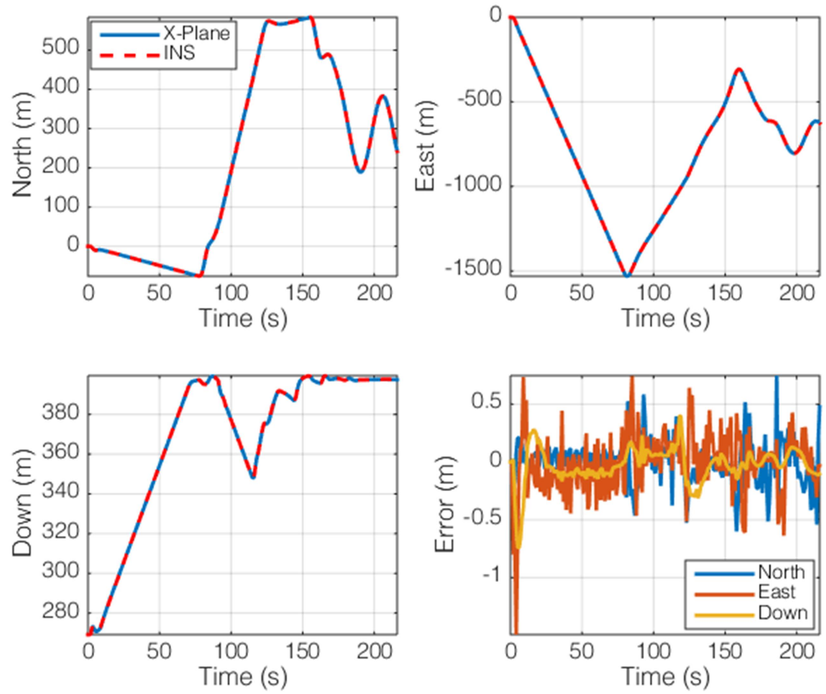


Figure 59: Position for HIL zero noise, zero wind.

4.3.4 Sensor noise, zero wind

Using the zero-noise, zero-wind case as a baseline, the effect of sensor noise is now considered. For this testing, noise is added to all of the primary and aiding sensors according to the statistics presented in section 3.4. The dataset is the same used in the zero-noise, zero-wind analysis detailed in the previous section. This allows for quantification of the effects of sensor noise characteristics on the INS states.

Zero-mean white noise was added to the primary INS sensors (gyroscopes and accelerometers) based on the noise statistics outlined in Table 4 and Table 6. The effect of bias on the INS solution is considered in the next sections. As discussed in section 4.2, integrating zero-mean sensor noise causes a well-characterized divergence of the INS solution. The divergence of the INS states from their true values is dictated by the noise floor of the sensors at the chosen sampling rate of 128 Hz. When the error-state filter (detailed in section 3.6) is used, the computed states are further influenced by errors in the aiding sensors.

While these sensors limit the long term divergence of the filter, they add to the error in the short-term state estimates. Therefore, a discussion of the effect of aiding sensor error on the INS solution is also presented.

The accuracy and noise characteristics of the GPS are related to the motion of the vehicle as outlined in section 3.4.5. For the purposes of this testing the GPS was considered static if the RMS velocity was below 2 m/s, and dynamic for any velocity exceeding this threshold. The velocity threshold is somewhat speculative, but reflects the best estimate for the changeover between GPS operational modes. A simple filter was applied to reflect the change in accuracy between static and dynamic conditions. Magnetometer noise is simulated based on the filter discussion presented in section 3.5.3.1. From this section, the noise in the magnetometer has frequency characteristics linked to the operation of high current devices on the aircraft. Therefore, noise was added to the magnetometer based on the engine RPM and the known 50 Hz servo update rate. The noise magnitude was approximated as ±25 μT using a sliding window of 1 second for calculation of the magnetometer noise statistics. The barometric pressure and airspeed noise is based on the discussion presented in section 3.4.4 and corresponds to ±0.4 Pa and ±0.5 Pa, respectively, at a sampling rate of 40 Hz.

The noise characteristics of the aiding sensors were assumed constant for the tests presented in this section. The Kalman filter tuning parameters were considered constant as outlined in section 3.6.4 and 4.2. Four tests were run to simulate the effect of aiding sensor noise only, and the combined aiding sensor – primary sensor noise associated with the different sensors used in the INS. The 95% confidence interval errors for the attitude, velocity, and position are detailed in Table 19. The position error in Table 19 is the precision error. An additional bias error exists and is equal to 2.2 m for the X-Y directions and 1.8 m for the vertical channel.

Table 19: HIL sensor noise state uncertainty estimates, 95% confidence.

Manufacturer	RMS angular error (±deg)	RMS velocity error (±m/s)	RMS position error (±m)
No-noise	1.738e-2	0.389	0.422
Aiding sensor noise only	0.265	0.413	0.491
CMS390 noise	0.315	0.413	0.496
VTI SCC1300 noise	0.542	0.546	0.634
InvenSense MPU-9150 noise	0.790	0.796	0.924

From Table 19, the no-noise error case corresponds to the results and discussion presented in the previous section. With the no-noise case acting as a baseline, the effect of adding in aiding sensor noise is readily apparent: the attitude error increases several orders of magnitude. When error is introduced into the position and velocity measurements, the filter commands a change in attitude to compensate for the perceived divergence. This leads to the increase in attitude uncertainty for the aiding sensor noise case.

Primary INS sensor noise further increases the state uncertainty. The effect is not as dramatic as the uncertainty introduced by the aiding sensors due to the relatively slow divergence caused by integration of the primary sensor noise components. Comparing the overall system performance based on the three different primary INS sensors, the effect of increasing noise is still visible. The gyroscope noise largely drives the attitude uncertainty. The increased attitude uncertainty couples with the accelerometer noise thereby increasing the velocity and position uncertainty. The magnitude of the INS state error is largely controlled by the filter based on the process and measurement noise statistics.

A discussion of the attitude, velocity, and position estimation errors for the noise cases involving the CMS390 primary INS sensors is now presented. Error tracks for the other primary sensors show similar characteristics with larger error magnitudes as outlined in Table 19. The attitude track for the HIL flight with additive noise in the primary and aiding sensors is shown in Figure 60. Of note in this figure is the relatively large azimuth uncertainty, which is driven primarily by noise in the magnetometer. When the magnetometer measurement noise term is increased, the resulting azimuth estimation exhibits a low frequency, large amplitude oscillation. Decreasing the measurement noise has the opposite effect and negatively impacts the gyroscope bias estimation. Extensive HIL testing has shown the magnetometer measurement noise given in section 3.4.3 and section 3.5.3.1 as the best compromise between long-term stability and short-term accuracy. The velocity error (Figure 61) shows relatively random oscillations corresponding to noise in the GPS velocity signal. The barometer noise is clearly visible in the vertical velocity estimate, though the resulting INS state estimate is in close agreement with the truth data provided by the simulator. The large velocity error excursion occurring near 1 s is the result of the impulsive jet assisted takeoff. The exact cause of this error is speculative, but post-flight reconstructions of the INS state estimates point towards the discrete nature of the sensor output the likely culprit. The impulsive launch is applied "instantaneously" thereby leading to an attitude error (see Figure 60) which manifests as a bias error in the velocity and position estimation. This error grows quickly before being corrected by the filter. Practically speaking, this sort of impulse is not commonly encountered in real-world testing and should not be considered a limitation of the INS algorithm or filter.

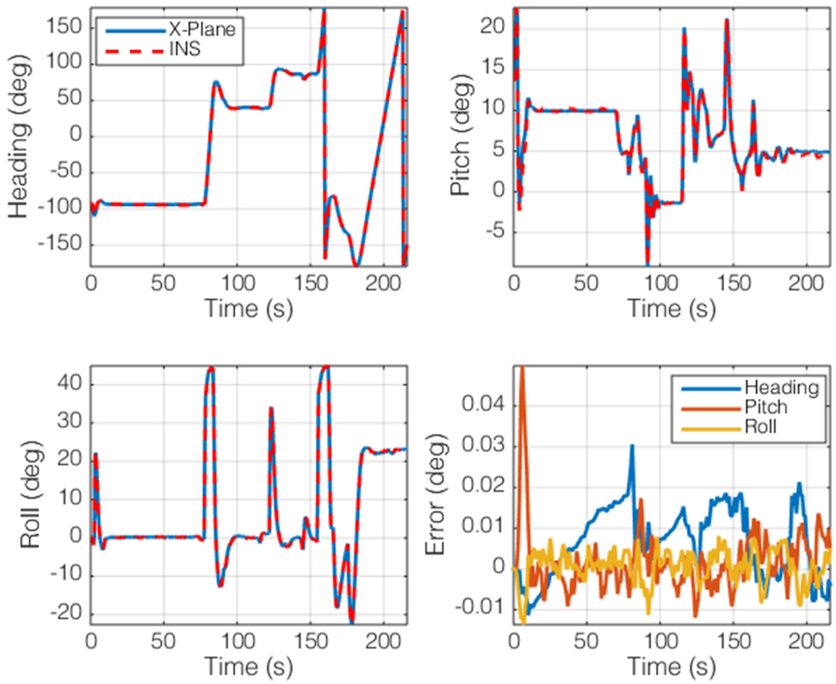


Figure 60: Attitude for HIL with noise, zero wind.

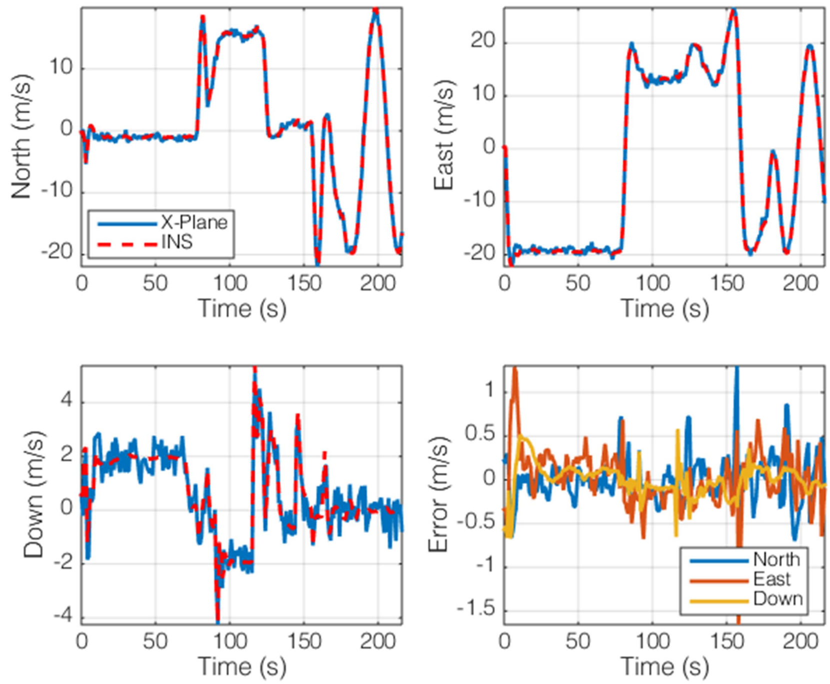


Figure 61: Velocity for HIL with noise, zero wind.

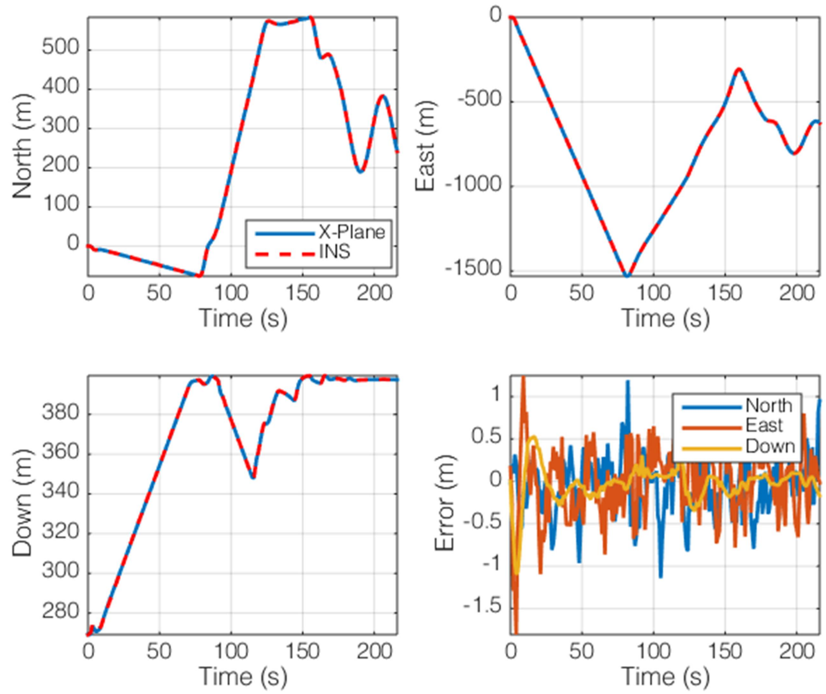


Figure 62: Position for HIL with noise, zero wind.

4.3.5 Sensor noise with wind

The effect of wind on the state estimation is another potential source of error. Atmospheric turbulence can further act as a noise source in the primary INS computations. The body-sensed angular rates and accelerations are rotated and integrated in the NAV frame. The sensed rates and accelerations are measured with respect to an inertial reference frame and are therefore independent of the aerodynamic and atmospheric forces acting on the vehicle. This implies that wind should have a minimal effect on the accuracy of the computed solution. However, turbulence causes random vibration like motions which may add to the state uncertainty in a manner similar to that detailed for the primary and aiding sensor noise contributions. To explore this effect, the wind model in X-Plane® is used to simulate both constant and varying wind conditions.

A constant wind component was added to the X-Plane model and the same flight path detailed in section 4.3.2 was flown. Three tests with a constant wind magnitude and direction were flown. The wind magnitude was chosen as 5, 10, and 15 m/s, covering the range of flight conditions commonly encountered in actual flight testing. The wind direction was chosen as 45 degrees to excite any cross-coupling errors in the NED reference frame. A turbulence case was also flown with random gusts up to 10 m/s from random directions. The gust model was severe in that the temporal variation in both wind magnitude and direction exceeded that measured in any flight-test cases. The baseline accuracy model uses both primary and aiding sensor noise. The CMS390 sensors' noise models were chosen as the reference case presented here, but results from the other primary INS sensors were also measured and showed similar trends. Results from the constant wind and turbulence test cases are presented in Table 20. From this data, the state uncertainty does not show a significant dependence on the wind magnitude. The gusting case considered does show a slight increase in the estimated uncertainty, but the effect is still small when compared to the baseline case.

This section has shown the effects of unbiased sensor noise and wind on the resulting state uncertainties. The filter was shown to be capable of bounding the growth of error in the state estimates in a manner proportional to the noise characteristics of the individual sensor components. The final source of error that needs to be considered is sensor bias. As mentioned previously, attitude errors lead to a bias in the velocity and position estimates. Integration of the bias leads to rapid divergence of the state estimates in a manner proportional to the noise in the primary INS sensors. The effect of additional bias terms on the state uncertainty is considered in the next section.

Table 20: HIL sensor noise state uncertainty estimates for wind and turbulence cases.

Manufacturer	RMS angular error (±deg)	RMS velocity error (±m/s)	RMS position error (±m)
CMS390 noise	0.315	0.413	0.496
5 m/s wind 45°	0.258	0.462	0.519
10 m/s wind 45°	0.343	0.515	0.582
15 m/s wind 45°	0.303	0.494	0.582
10 m/s random gust	0.335	0.505	0.537

4.3.6 Bias estimation

Bias in the attitude and velocity estimates can quickly lead to the divergence of the uncorrected INS state estimates. While some form of bias is present in all of the sensors used in this work, the time-varying gyroscope bias is one of the more important, yet difficult to capture components. The influence of the gyroscope bias was demonstrated in section 4.2. In the uncorrected INS state estimation routine, ARW lead to an eventual divergence of the attitude solution. The divergence of the attitude solution caused an apparent bias in the measured specific force viewed in the NAV frame. Integration of this perceived bias lead to a rapid divergence of the INS solution. The two straightforward methods for correcting this error are to increase the reliance on the measurement data, or to artificially increase the filter's process noise. Increased reliance on the measurement data drives up the overall uncertainty in the state estimates, bounded by the errors in the aiding sensors. Artificially increasing the process noise leads to a similar effect with a pronounced temporal oscillation of the solution. Therefore, both solutions are unacceptable from the perspective of driving down the overall INS uncertainty.

The MEMS gyroscopes used in this work have an additional bias instability component that increases the state uncertainty in conjunction with the white noise. The bias instability discussed in section 3.4.1 is a temporally varying term that can outweigh the effects of ARW in dynamic environments. This is primarily due to changes in the gyroscope temperature and exposure to shock and/or jerk during takeoff and landing.

These same factors lead to shifts in the accelerometers offset bias, but the net effect is often small in comparison to the gyroscope bias.

From section 3.6.2.1, a time-dependent bias estimation state was added to the Kalman filter. The gyroscope bias is considered to be randomly varying in a manner similar to the ARW model. Numerous INS Kalman filter implementations have chosen to model the gyroscope bias in this way.[18, 23, 65, 76] Estimation of the bias depends on the process noise used in the filter and the time constant associated with the Gauss-Markov bias model. For this work, the noise characteristics were modeled using the bias instability parameter listed for each gyroscope in section 3.4.1. The time constant was chosen to be 10 seconds based on both ground and flight-testing. This value best limited the oscillations and growth rate of the bias estimate. The time constant is a variable tuning parameter and is largely sensor and implementation specific.

Validation of the bias estimation method was accomplished through two HIL tests. The first used a constant bias in the gyroscope signal. The second used a linearly increasing bias. For each test, sensor noise was simulated in the primary and secondary INS sensors as outlined in the previous section. The flight data from the no-wind case and CMS390 noise statistics were used for comparative purposes. Figure 63 shows the results from the constant and ramping bias cases. In this figure, the dashed lines represent the bias applied in the body frame to the indicated gyroscope axis. The first test used 0.1 deg/s on the x-axis, 0.2 deg/s on the y-axis, and 0.3 deg/s in the z-axis. The second test used a linearly increasing bias from 0 deg/s in all axes to a maximum equal to the constant bias used in the first test. In both tests, the filter quickly converged to the applied bias. Accurate estimation of the bias serves two purposes. Firstly, it limits the state uncertainty when the filter is running. Secondly, it limits the growth of error when the aiding sensor data is invalid or unavailable. In the second scenario, during aiding sensor outages, the bias is considered constant and equal to the magnitude immediately preceding the outage. In both tests, the resulting error in state estimation after bias convergence is equivalent to the zero bias case.

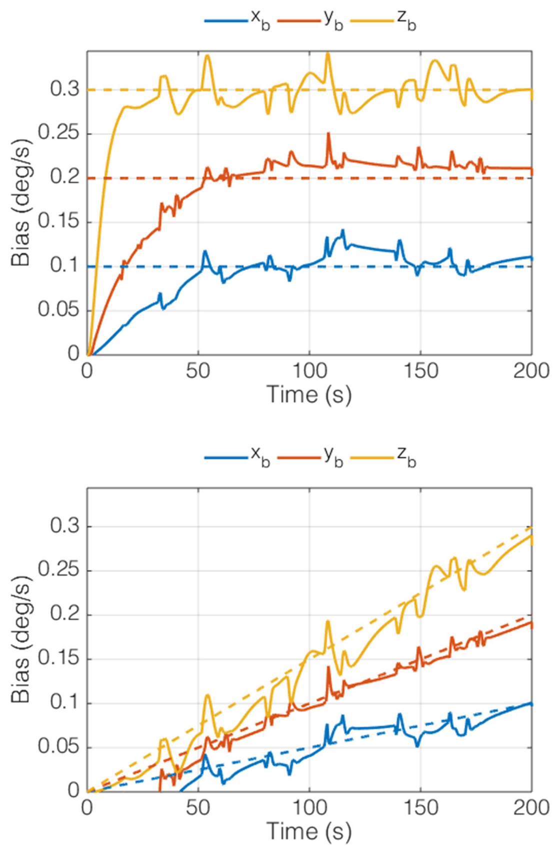


Figure 63: Bias estimation for constant gyroscope bias (left), and ramping bias (right).

4.4 Summary

This chapter demonstrated the testing and characterization of the proposed INS. This system is designed to provide state estimates which can be used for flight-test performance evaluation. To assess the accuracy of this system, several tests were performed. The first was a quasi-static test which assessed the accuracy of the navigation equations and sensor calibrations presented in chapter 3. Based on this testing, the noise characteristics of the sensors were shown to drive the resulting state uncertainty in the INS. The sensor noise characteristics were then used with the proposed error-state Kalman filter in a HIL test to demonstrate the resulting state accuracy. This testing allowed for quantification of the contribution of individual noise sources to the resulting state error. In addition to noise estimates, the efficacy of the bias estimation was demonstrated. The resulting state estimation uncertainty is used in the next chapter to estimate the flight performance of a fixed-wing sUAS. Performance estimation relies heavily on the previously presented propulsion model (chapter 2), and INS state accuracy estimates (chapter 4).

Chapter 5: Flight-testing

5.1 Background

This chapter provides an overview of the fixed-wing sUAS flight-testing campaign. The flight-testing performed in this work is aimed at validating the design estimates for an electrically powered sUAS. Therefore, a brief design overview is provided to provide the framework for flight-test evaluation. The specific flight-tests considered are aimed at determining the lift to drag (*L/D*) ratio, zero lift drag coefficient (C_{D_O}), and Oswald efficiency factor, e. Determination of these parameters uses the propulsion system model presented in chapter 2, and the INS error statistics listed in chapters 3 and 4. The flight-testing data and propulsion system model are then used to predict the range and endurance of the aircraft. Specific focus is placed on the impact of estimation uncertainty and Reynolds dependent power available models on the resulting range and endurance estimates.

5.2 Aircraft Model

The fixed-wing sUAS used in this work was designed in-house and named the Peregrine. The initial design of the Peregrine was done using statistical parameters from past UAS projects of similar size and configurations. For the intended mission profile, the configuration shown in Figure 64 was chosen as the best candidate for detailed design and analysis. This aircraft features a relatively high aspect ratio wing with V-tail stabilizer control surfaces. The center section of the aircraft comprises the payload section which houses various cameras used for precision agriculture applications. The nose compartment contains the aircraft’s power systems including the motor controller and propulsion battery. The flight computer and INS are located in the fuselage behind the main wing. Radio and telemetry transceivers are located in the wing tips to minimize the EMI impact on the INS sensors. The transceivers provide in-flight data monitoring and control capabilities. The airframe was constructed in-house using foam molds machined on a small-scale three-axis CNC machine. The majority of the airframe is a carbon fiber/Divinycell/carbon fiber sandwich construction. This results in a rigid and lightweight airframe with a high degree of dimensional stability. This construction method prevents airframe flexing and deformations that can lead to deviations from the baseline performance predictions.

The aircraft is designed to cruise at 14 m/s with an endurance of one hour. This speed represents the velocity necessary to traverse 3 km^2 while operating at 120 m above ground level (agl) with a visual swath area of 30 m. Initial weight estimates gave an empty weight of 2 kg. The maximum gross weight of the aircraft is 5.4 kg, for a useful payload weight of 3.4 kg. The stall speed is 10 m/s, which corresponds to a typical hand launch velocity for aircraft of similar weight. Using this data, an iterative wing optimization routine was developed using a Matlab interface to XFOIL.[56] The constraints bounding the design include limits on chord Reynolds number, wing area, taper ratio, aspect ratio, and section washout (see Table 21). The objective function is the total drag coefficient of the wing operating in the cruise configuration. The wing model was discretized into 10 spanwise stations distributed evenly along the half span. A simple panel model was created to capture the three-dimensional geometric effects associated with finite wings.[11, 85, 86] This model is summarized as follows. A complete discussion of the above method can be found in Phillips and is repeated here for clarity.[71] For a twisted, finite wing, the local circulation can be determined from a Fourier series expansion given by:

$$\begin{aligned}
&\Gamma(\gamma) = 2bV_\infty \sum_{n=1}^{N} A_N \sin(n\gamma) \\
&A_N \equiv a_n \left(\alpha - \alpha_{L=0}\right)_{root} - b_n \omega \\
&\sum_{n=1}^{N} a_n \left[\frac{4b}{C_{L,\alpha} c(\gamma)} + \frac{n}{\sin(\gamma)}\right] \sin(n\gamma) = 1 \\
&\sum_{n=1}^{N} b_n \left[\frac{4b}{C_{L,\alpha} c(\gamma)} + \frac{n}{\sin(\gamma)}\right] \sin(n\gamma) = \omega(\gamma)
\end{aligned} , \quad (5.1)$$

where $\gamma = \cos^{-1}\left(-\frac{2c}{b}\right)$. For this model, Γ is the local circulation, b is the wing span, a_n and b_n are the series Fourier coefficients, c is the local wing chord, and $C_{l,\alpha}$ is the airfoil lift curve slope (determined from XFOIL).

An intermediate output associated with the optimization is an approximation for the wing efficiency factor, e (shown in (5.2)). This is an important design constraint governing the overall performance of the wing. Validation of the wing efficiency parameter is discussed further in the flight-testing section. The initial and optimized wing shape is shown in Figure 65.

Figure 64: Peregrine UAS. Structure is carbon monocoque sandwich with selective fiberglass paneling for RF transmission.

Table 21: Constraints used in aircraft wing optimization.

Constraint	Minimum Value	Maximum value	Final value
Mean Reynolds number	100,000	n/a	432,885
Wing area	0.25 m^2	2 m^2	0.36 m^2
Taper ratio	0.4	3.0	2.67
Aspect ratio	7	20	11.04
Tip washout	-5 deg	5 deg	-1.24 deg

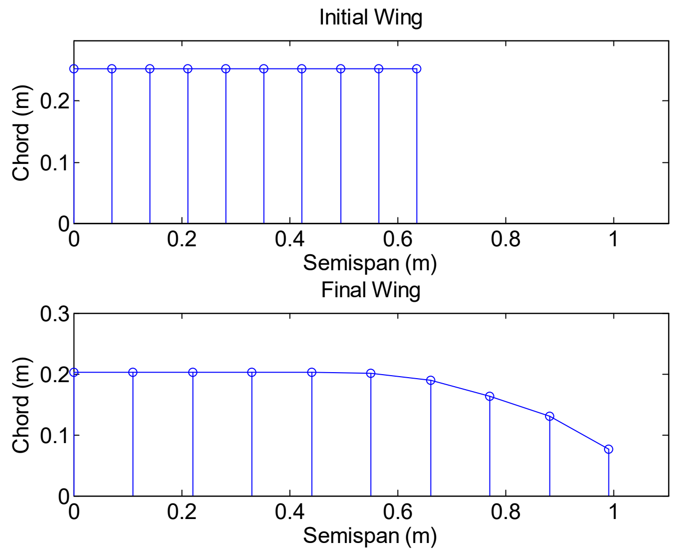


Figure 65: Initial wing (top) and optimized final wing (bottom). Final wing is the result of applying the optimization constraints listed in Table 21.

Based on the optimal solutions to (5.1), the lift, drag, and efficiency terms for the wing can be determined from

$$\begin{aligned} C_L &= A_1 \pi AR \\ C_{D_i} &= \pi AR \sum_{n=1}^{N} n A_n^2 \\ e &\equiv \left(1 + \sum_{n=1}^{N} n \left(\frac{A_n}{A_1}\right)^2\right)^{-1} \end{aligned} \quad . \tag{5.2}$$

Extensive background investigations into low Reynolds number airfoils yielded the SD7043 as the best candidate for this aircraft.[11, 87-89] The minimum chord was sized to ensure the Reynolds number remained above $1{\times}10^5$, giving a maximum lift coefficient of 1.4. A 20-degree plain flap occupying 30% of the chord and 60% of the semispan locally increases the maximum lift coefficient to 1.8. The maximum aspect ratio is bounded by structural considerations. The V-tail arrangement uses symmetrical sections, making the left and right stabilizers symmetric. This simplifies the mold construction as only one master mold is necessary. The aircraft is designed to belly land; therefore, the tail configuration also minimizes the chance of ground impact damage. The next section details the vehicle's predicted performance that will be validated through flight-testing in later sections.

5.3 Predicted Aircraft Performance

The total aircraft lift-to-drag ratio (*L/D*) was approximated using semi-empirical drag estimation techniques and the results from the Matlab and XFOIL based wing optimization. The zero-lift drag coefficient was estimated using a drag buildup of all major components and historical data from previous aircraft.[90, 91] The estimated C_{D_O} is 0.0261 for the Peregrine aircraft. The details of the drag buildup are presented in Appendix 1. The power required follows from the definition given in equation (5.3), which depends predominantly on the aircraft zero-lift drag coefficient, geometry, and operational conditions.[40] The power required is tabulated based on the aircraft's geometry outlined in Table 21 and operation at 365 m msl at standard temperature and pressure. The wing efficiency factor, *e*, was determined iteratively by summation of the Fourier coefficients given in equation (5.2).[71] For the Peregrine, e is predicted to be 0.90. The predicted power required and *L/D* are shown in Figure 66 and Figure 67, respectively. These represent key performance parameters estimated in the initial design phase and are experimentally verified through a series of flight-tests.

$$P_R = \frac{1}{2}\rho V^3 A C_{D_0} + \frac{W^2}{\frac{1}{2}\rho V A}\left(\frac{1}{\pi e AR}\right) \tag{5.3}$$

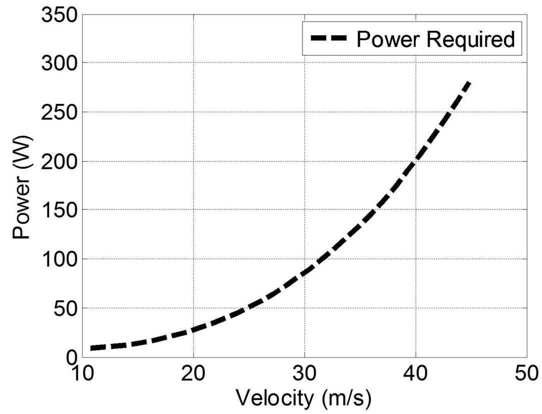


Figure 66: Predicted power required for the Peregrine UAS in steady, level flight.

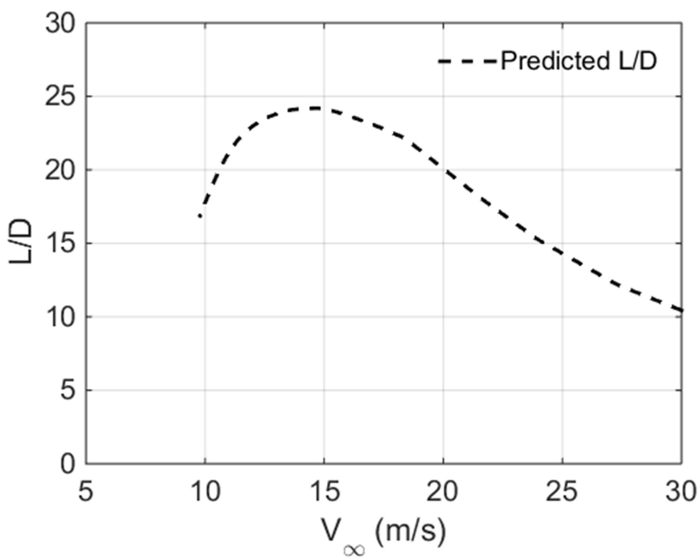


Figure 67: Predicted *L/D* for the Peregrine UAS.

The power required and *L/D* curves shown in Figure 66 and Figure 67 use Reynolds dependent aerodynamic parameters for lift and drag estimation. The span efficiency, *e,* is implicitly a function of the total circulation and therefore Reynolds number. The magnitude of the dependence is relatively weak, with a total variation on the order of 1% over the flight envelope. The limited speed and altitude capabilities of the vehicle preclude significantly changing the flight Reynolds number during testing. Based on this reasoning, the effect of Reynolds number on the performance predictions outlined in Figure 66 and Figure 67 is minimal. In contrast, the propeller used for propulsion exhibits a relatively strong dependence on localized Reynolds effects. These effects will be addressed in the power required flight-testing section, 5.5. First, the *L/D* testing methods and results are presented.

5.4 Lift-to-Drag Ratio

This section details the three methods used to reconstruct the aircraft's *L/D*. These methods rely on the accuracy of the FMC in maintaining airspeed and the onboard air data sensors for determining the descent rate and distance. First, three methods for estimating the *L/D* are outlined. The sensors and control techniques are then described for implementing the three *L/D* estimation techniques. Finally the results for each method are presented. For each method, the effect of sensor uncertainty on the final *L/D* estimates is addressed based on the analysis presented in chapter 3.

5.4.1 Relations

Estimating the aircraft's *L/D* polar requires gliding the aircraft over a range of fixed airspeeds with zero net thrust. The assumption of zero thrust is uniquely met by turning off the motor and allowing the propeller to fold back. For this work, three methods are compared. The first is the standard rate method outlined by Kimberlin.[40] The rate-based calculation for *L/D* involves comparing the vertical descent rate to the average airspeed. The descent angle and corresponding *L/D* is given by[40]

$$\begin{aligned} \theta &= \sin\left(\frac{dh/dt}{V_\infty}\right) \\ L/D &= \frac{1}{\tan(\theta)} \end{aligned}, \tag{5.4}$$

where dh/dt is the change in altitude during the test divided by the total time taken to descend, V_∞ is the airspeed, and θ is the descent angle. For this method, the airspeed is averaged over the testing window. For the second *L/D* estimation method, the rates in equation (5.4) are replaced by the integrated displacements during the unpowered descent. The total vertical distance is represented by the difference in altitude over a single testing window. The major difference between the methods arises from replacing the average airspeed with the total displacement along the flight path, detailed in equation (5.5). Initial results presented in 5.4.4.2 indicate that the total displacement method better accounts for airspeed variations induced by wind and gusts. Finally, a method for estimating the *L/D* for vehicles without an airspeed sensor is demonstrated. For vehicles of the size considered in this work, the wind speed can be a large fraction of the vehicle's airspeed, complicating the determination of horizontal distance traveled. Incorporation of INS based wind speed estimates are used to correct for this error and are detailed in 5.4.4.3.

$$
\begin{aligned}
S_{hypotenuse} &= \int_{t=0}^{t=i} V_{\infty} dt \\
S_{vertical} &= h_i - h_0 \\
\theta &= \sin\left(\frac{S_{vertical}}{S_{hypotenuse}}\right) \\
L/D &= \frac{1}{\tan(\theta)}
\end{aligned}
\tag{5.5}
$$

5.4.2 Measurement methods

L/D testing relies on the onboard sensors and fusion algorithms inside the INS to provide a real-time estimate of the aircraft's state. Of critical importance are the estimates of the airspeed and both the vertical displacement and velocity. Velocity and displacement information is provided by integration of the Earth frame acceleration and is corrected by barometric altimeter. The sensors and fusion techniques are discussed in detail in chapter 3. For comparative purposes, the *L/D* estimates calculated using the INS are compared to estimates based on unaided GPS information. The GPS sensor used in this work is outlined in section 3.4.5.

L/D errors traceable to sensor performance are addressed in section 5.4.4. The noise characteristics used in developing the error estimates are summarized in Appendix 1. Error bounds for the *L/D* data were generated using the Monte-Carlo methodology in a similar fashion to the methods outlined in the propulsion testing.[62]

5.4.3 Flight-test setup

Data generation for *L/D* flight-testing requires holding a constant airspeed and measuring the corresponding descent rate or distance. To automate this procedure, the FMC was altered to fly a racetrack pattern at various airspeeds. The expected operational airspeed range of the Peregrine is between 12 and 24 m/s. This speed range was discretized into 2 m/s increments to allow for reconstruction of the *L/D* curve. Therefore, six laps were required to fully capture the glide performance of the aircraft. The four phases of flight outlined in Figure 68 are explained as follows.

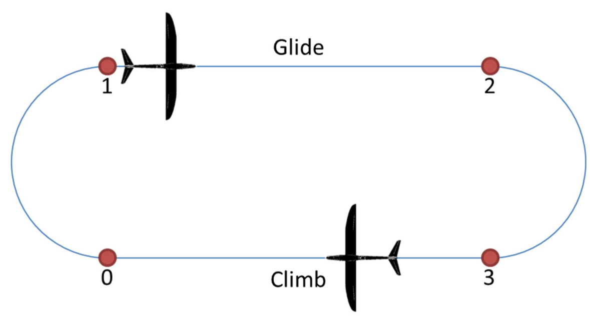


Figure 68: *L/D* estimation flight profile

The first phase is a full power climb to altitude. During this phase, the aircraft is also navigating back to the initial waypoint 0 for the impending descent. The Peregrine has a thrust to weight ratio approaching unity, thereby allowing the climb to be completed well before the aircraft reaches the initial waypoint. The second phase is the turn towards the descent, waypoint 1. The aircraft is bank limited to 30 degrees to prevent tip stalling at low airspeeds. As the aircraft approaches waypoint 0, the throttle is reduced to idle and the bank initiated. During the turn, the aircraft attempts to hold altitude, resulting in a reduction in airspeed. The throttle is then used as required to maintain the set point airspeed and altitude until the turn is complete. The third phase is an unpowered, constant airspeed descent towards waypoint 2. During this flight phase, the motor is shut down to limit the additional drag imposed by a windmilling propeller. During this phase of flight, the aircraft adjusts its pitch attitude to maintain the required airspeed. Once the aircraft has descended to the altitude floor imposed by the operational environment (nominally 20 m agl), the motor is turned back on and the climb to the initial waypoint altitude is started. Waypoint 3 represents the start of the full-power climb. For the highest descent rates, the descent leg is extended horizontally after the motor is switched back on. This ensures adequate time for the ascent to the initial waypoint. Flight data is logged during the entirety of the flight for post flight reconstruction. An internal flag is set to mark the beginning and end of the unpowered descent legs. This allows for post processing of the descent data to estimate the aircraft's *L/D*.

During flight, the racetrack orientation does not change. Therefore, before flight, the course is aligned such that the descent leg is aligned with the prevailing winds. This alignment is only important for the INS based estimation techniques presented in 5.4.4.3. Additional testing using a random azimuth for each circuit of the course did not yield significantly different results to justify the increase in required flight area.

5.4.4 Comparison and discussion

Based on the testing methods outlined in section 5.4.2, results for each of the three *L/D* estimation methods are now presented. The first method is the rate method in which the vehicle's average airspeed and total descent distance are compared. The second method replaces the average airspeed with the total descent distance obtained by integration of the airspeed. The final method demonstrates performance estimation using an INS without an air data system using the relations for total descent distance and flight path distance traveled.

5.4.4.1 Rate method

This section details the *L/D* estimation using the three previously identified methods. The first method uses airspeed and altitude information provided by the onboard air data system. The specific instrumentation is outlined in chapter 3. Based on the relations presented in equation (5.4), the average airspeed and total descent distance are required to estimate the descent angle and *L/D*. Twenty-four flight-tests were conducted in calm wind conditions in order to gather the data required to estimate the aircraft *L/D* across the operating airspeed range. An example airspeed plot for four set points is shown in Figure 69, with error bounds corresponding to the estimated airspeed sensor error outlined in Table 8. This figure indicates the effectiveness of the aircraft's FMC in holding a specified airspeed in unpowered flight. This capability is critical as will be demonstrated for flights conducted with wind in section 5.4.4.2. The change in altitude was computed based on the barometric altimeter (Figure 70). The difference in starting altitudes in Figure 70 is related to the time necessary for the airspeed to settle to within the required set point for the test.

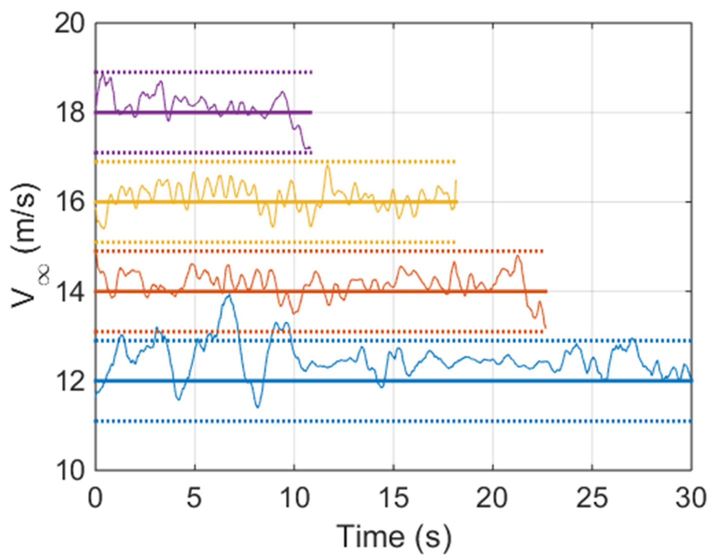


Figure 69: Airspeed hold during gliding descent

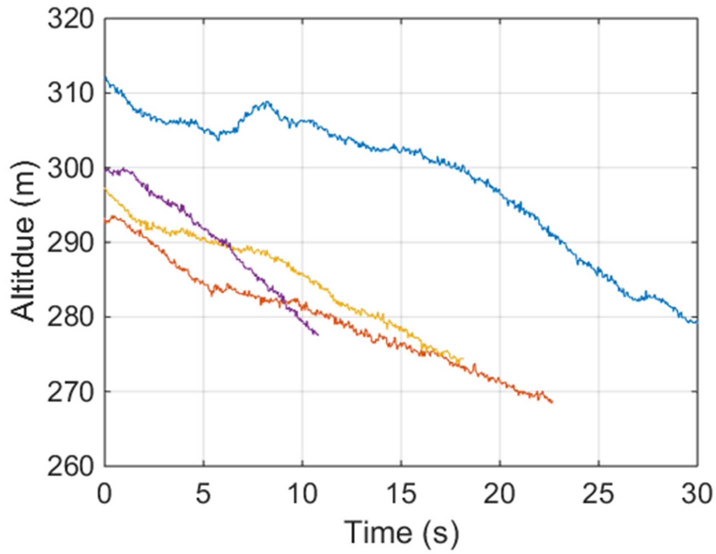


Figure 70: Altitude plots for airspeeds corresponding to Figure 69

Based on the data presented in Figure 69 and Figure 70, the descent angle can be computed (Figure 71). The resulting *L/D* is shown in Figure 72. Error estimates are based on the sensor characteristics outlined in Table 8 for the airspeed and barometric altitude sensors. The prime drivers for the resulting uncertainty are primarily related to the noise characteristics of the airspeed sensor. This result is echoed by Ostler's flight-testing results.[38] Figure 72 shows good agreement between the predicted and measured *L/D* for the aircraft for flights with little to no wind. Flights at higher airspeeds were attempted, but the limited operating ceiling did not allow the aircraft to reach a steady state descent rate and airspeed. Low speed testing was limited by loss of aircraft control occurring at speeds below 12 m/s. The next section will discuss the rate integration method for determining the *L/D*.

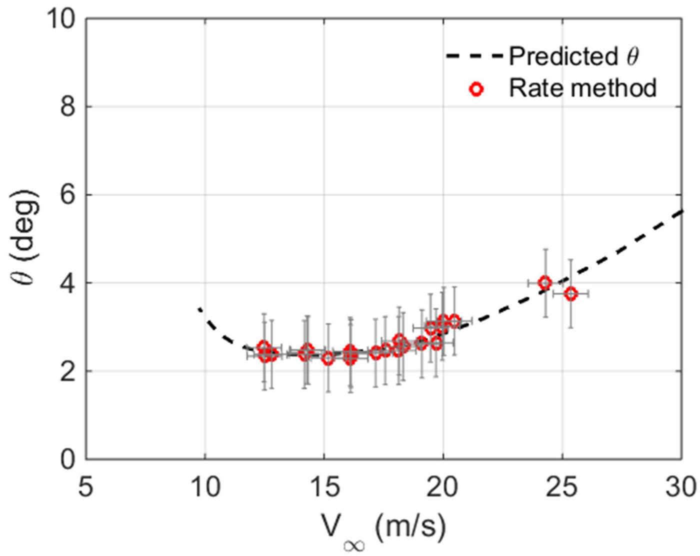


Figure 71: Descent angle versus airspeed for rate-based *L/D* estimation

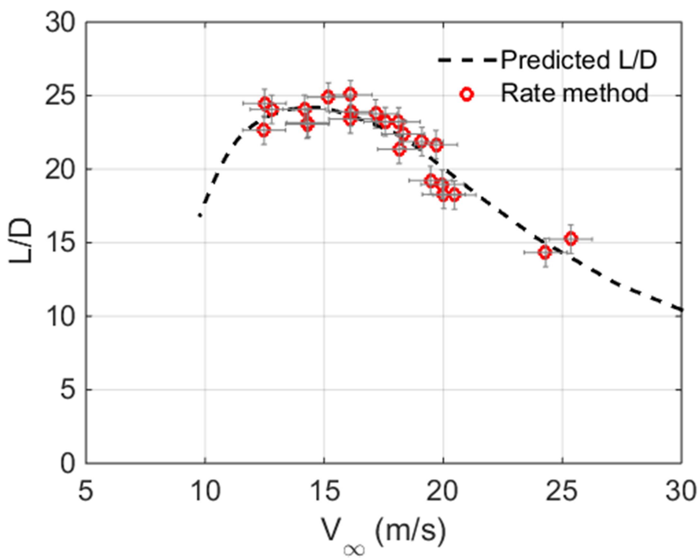


Figure 72: *L/D* estimates using rate method

5.4.4.2 Rate Integration method

The *L/D* estimates outlined in the previous section agree with the predicted performance within the uncertainty limits. This demonstrates the efficacy of this method in predicting the *L/D* when wind conditions are either minimal or steady. At the low flight speeds and altitudes the vehicle operates, this requirement places strict limitations on the wind conditions in which testing can be completed. To address this limitation, the rate integration method is used. Results for calm conditions are shown in Figure 73. From this figure, the estimated *L/D* for the rate based and integration based methods are nearly identical.

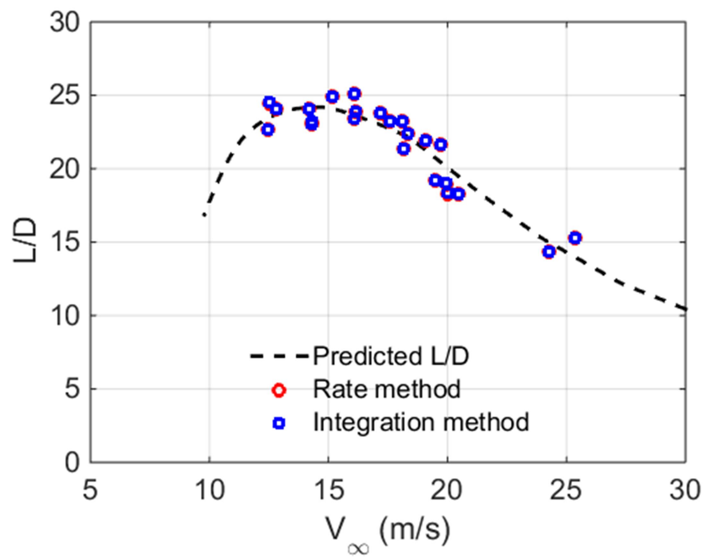


Figure 73: Rate and integration based *L/D* estimation methods for low wind case.

When the same test is carried out in moderately gusty conditions (max ~5 m/s measured at ground level), the resulting *L/D* estimation for the rate and integration methods are shown in Figure 74. Clearly, the standard rate based estimates shown in this figure show a larger scatter compared to the integrated method, and the low-wind case previously discussed. The trending for increasing wind speed gust velocity is shown in Figure 75. The data presented in this figure was taken with a nominal ground level wind velocity on the order of 10 m/s.

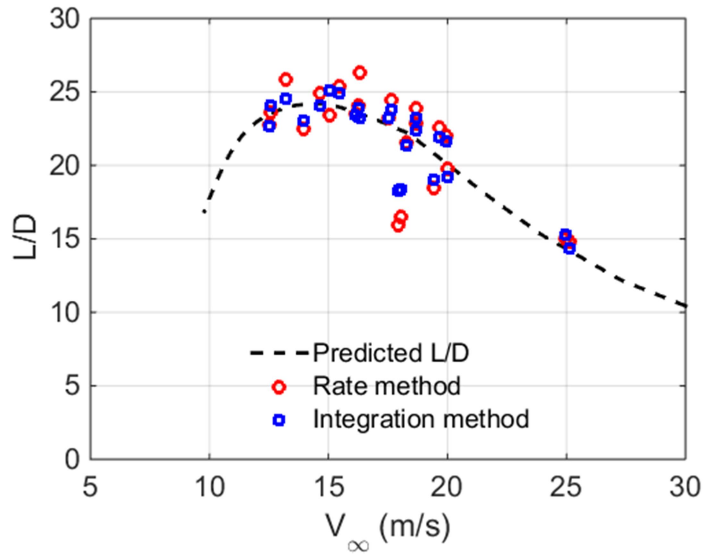


Figure 74: Rate and integration *L/D* estimation in gusty conditions (~5 m/s wind at ground level).

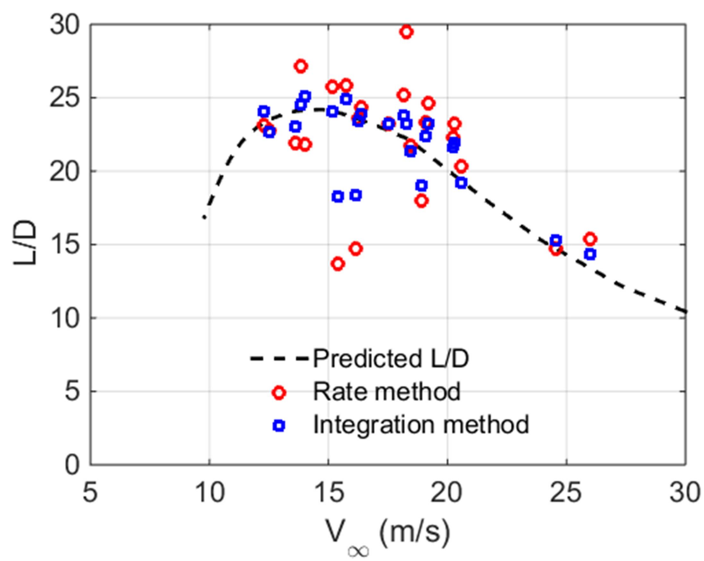


Figure 75: Rate and integration *L/D* estimation in high wind and gust conditions (~10 m/s wind at ground level).

The influence of wind on the overall estimation uncertainty is clearly visible. However, in all cases, the integration based method is either equivalent, or exceeds the performance of the rate based method. It should be noted that the efficacy of the rate method is related to the fidelity and update rate of the airspeed sensors on board the aircraft. If the data rate of the onboard differential pressure sensor is decreased from

100 Hz to 10 Hz, the noise in the *L/D* estimate increases (Figure 76). As the data rate of the sensor is increased, the resulting *L/D* quickly approaches the estimate shown previously in Figure 75. The frequency dependence of the integration based method points towards aliasing as being the likely cause of the increased scatter at lower sampling rates. The next section discusses *L/D* estimates computed without onboard air data sensors.

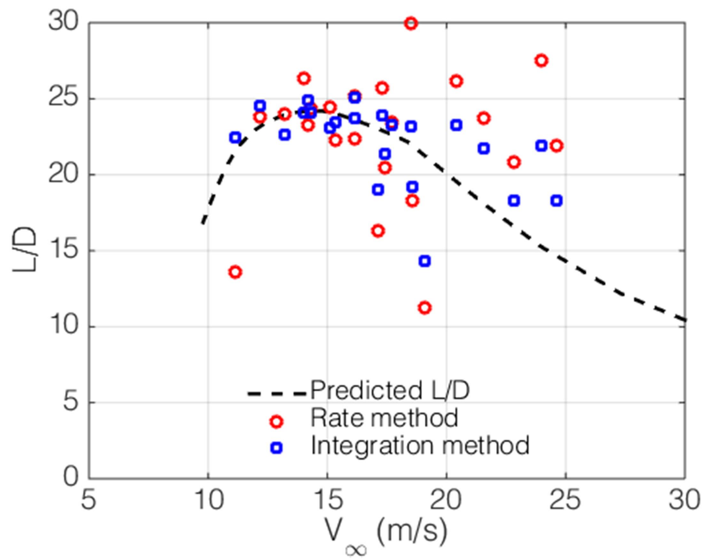


Figure 76: Rate and integration *L/D* estimation in high wind and gust conditions at reduced airspeed data rate of 10 Hz (~10 m/s wind at ground level).

5.4.4.3 INS based *L/D* estimation

While numerous INS systems have air data measuring capabilities, due to cost/space/size constraints, many forgo these sensors in favor of a strictly INS based attitude and control system. Therefore, it is necessary to evaluate the performance of a strictly INS based approach to estimating the aircraft's *L/D*. There are numerous advantages to this method, the first of which is the simplicity of integration into an existing vehicle. Tying into an existing air data computer or installing and calibrating a separate unit can be time consuming and costly. However, if the GPS or INS data is used directly, the resulting *L/D* estimates are likely to be highly erroneous. Figure 77 shows the low wind estimate of the aircraft's *L/D* and the corresponding data generated using the rate method.

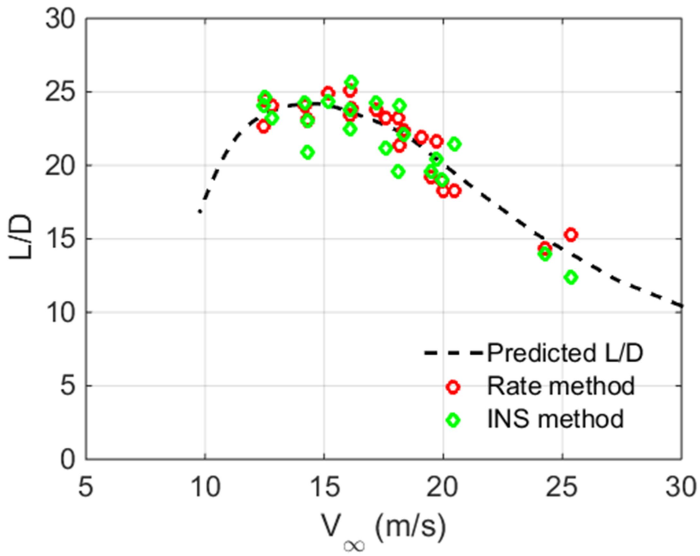


Figure 77: INS based *L/D* estimates for low wind flights

The INS *L/D* estimates in Figure 77 were generated using the Earth frame velocity as outlined in chapter 3. For the case of low wind, the resulting *L/D* estimates are in reasonable agreement with the rate method previously outlined. However, when the wind/gust speed increases, the accuracy of the *L/D* estimate rapidly diverges (Figure 78). Based on equations (5.4), this is directly a result of using the ground speed in place of the airspeed. If however the wind speed estimates furnished by the INS are used to correct the ground speed estimates, a pseudo airspeed estimate can be established. An example of the estimated wind speed furnished by the INS is shown in Figure 79. During a power off glide test, the vehicle state is relatively constant, resulting in a wind estimate not able to capture high frequency variations induced by gusts. Given this limitation, incorporation of the estimated wind speed can dramatically improve the resulting *L/D* estimates for systems not equipped with air data systems (Figure 80).

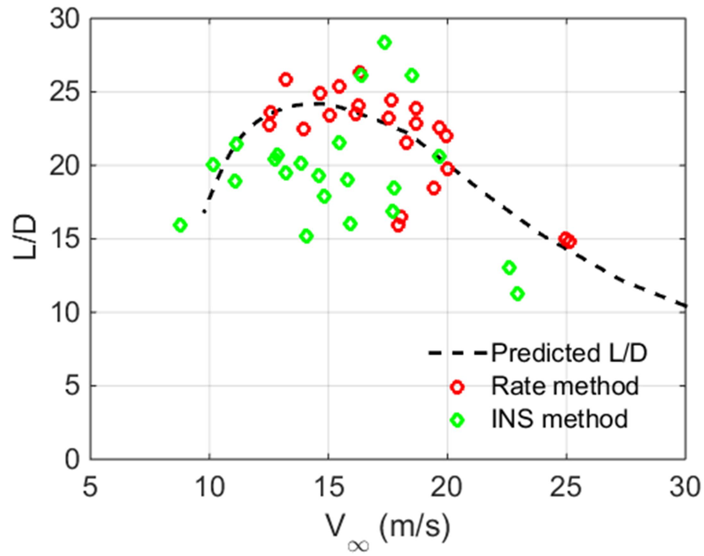


Figure 78: ***L/D*** **estimates for INS based method with moderate wind (~5 m/s measured at ground level).**

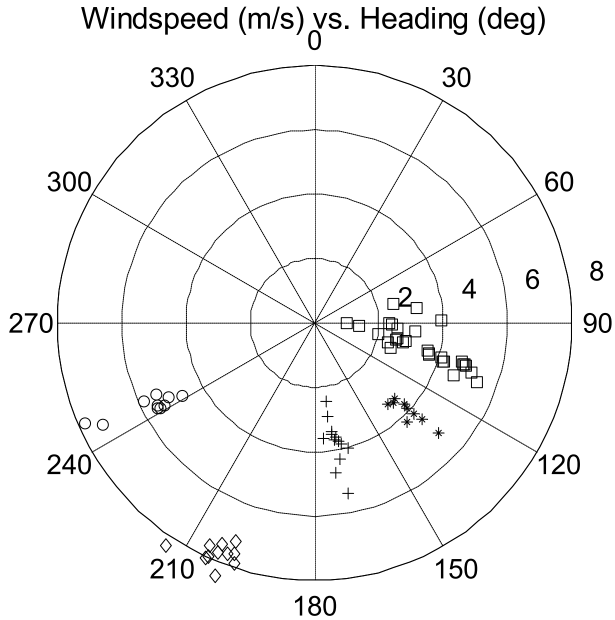


Figure 79: Wind speed estimation for GPS ground displacement correction. Each group of symbols is the wind over one test run.

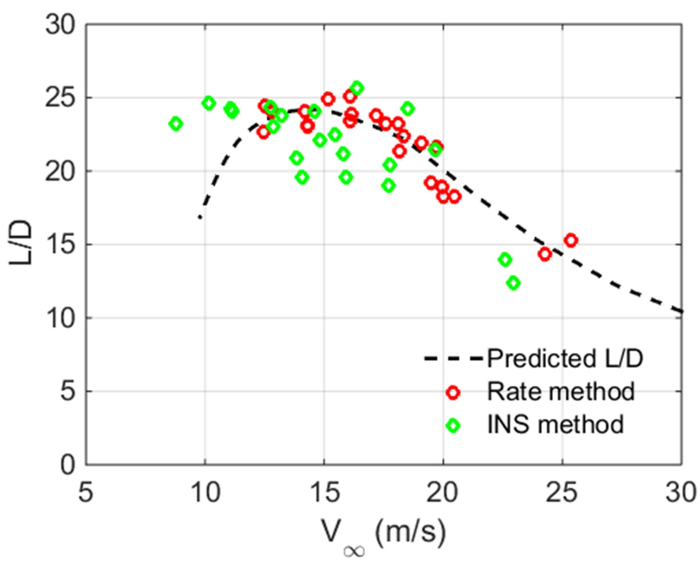


Figure 80: ***L/D*** **estimates for INS based method with wind correction and moderate wind (~5 m/s measured at ground level).**

5.5 Power Required

This section details the validation of the power required outlined in section 5.3. Two methods are used to establish the power required. The first is a full power level acceleration, and the second is a full power, constant airspeed climb.

5.5.1 Relations

This section details the flight-testing performance estimation for the power required. Based on the relations presented in the aircraft overview section 5.2, the excess power can be expressed as the difference between the available and required power. Given as

$$P_E = P_A - P_R = W\left(\frac{dh}{dt} + \frac{V}{G}\left(\frac{dV}{dt}\right)\right), \tag{5.6}$$

where P_E is the excess power, P_A is the available power, P_R is the required power, dictated by the aircraft design, dh/dt is the climb rate, and $\frac{V}{G}\left(\frac{dV}{dt}\right)$ is a velocity-dependent acceleration term. The power available estimation is covered in detail in chapter 2. In brief, the power available can be estimated based on the measured electrical power, propeller RPM, and airspeed. Therefore, to validate the power required estimates outlined in section 5.2, two independent tests can be performed. The first is a level acceleration, in which the rate-of-climb (dh/dt) is held constant. This reduces equation (5.6) to

$$P_E = P_A - P_R = W\left(\frac{V}{G}\left(\frac{dV}{dt}\right)\right). \tag{5.7}$$

In this expression, the available power can be inferred based on propulsion system measurements, the weight is fixed by the design, and the airspeed and acceleration can be measured by the INS. This test requires that the height be kept approximately constant during the test run.

The second test method is the constant airspeed rate-of-climb. For this test, the acceleration term in equation (5.6) is zero, reducing the expression to

$$P_E = P_A - P_R = W\left(\frac{dh}{dt}\right) \tag{5.8}$$

In a manner similar to the level accelerations, the power available is inferred through inflight measurements and the propulsion model developed in chapter 2. In this test, the ascent rate is measured in a manner similar to the descent rate presented in the *L/D* estimation.

The level acceleration and constant airspeed rate-of-climb produce estimates for the excess power. Using the available power estimates provided by the propulsion system model, the power required can be estimated. This information can be linked back to the initial design of the aircraft based on equation (5.3) (repeated here for convenience). Using this expression, C_{D_O} and e can be estimated using the geometric and operational parameters for a given flight.

$$P_R = \frac{1}{2}\rho V^3 A C_{D_0} + \frac{W^2}{\frac{1}{2}\rho V A}\left(\frac{1}{\pi e AR}\right) \tag{5.9}$$

The next section details the measurements necessary for estimating the power required.

5.5.2 Measurement methods

The measurements required for level acceleration estimation and full powered climbs are now presented. For these tests, the motor RPM and electrical power must be monitored. For the results presented in this chapter, the APC 10x6E propeller was used. This propeller does not fold back during unpowered flight which prevents power required and *L/D* flights from occurring simultaneously.

5.5.2.1 RPM

The inflight RPM signal is monitored and incorporated into the flight computer control scheme in a manner similar to that outlined in chapter 2 for ground based propulsion testing. The RPM is sensed using a back-EMF sensing technique in which the voltage spike resulting from the commutation of the motor is fed through a comparator circuit. The output of the comparator is a square wave representation of the physical rotation of the motor and propeller. This method has proven effective in high frequency monitoring of the motor RPM as there are multiple pulses per revolution (12 for this testing), opposed to 2 if the propeller were monitored optically. As an additional benefit, vibration and environmental effects have little to no effect on the accuracy of the RPM estimate. The maximum propeller RPM is approximately 15,000 RPM, or 250 Hz. This is well below the maximum operating frequency of the comparator at 15 MHz. This allows for additional low pass filters to be used in line to reduce the effects of EMI and feedback produced by a high power electrical motor. The accuracy of this sensor is detailed in appendix 1.

5.5.2.2 Electrical power

The onboard electrical power is recorded using similar hardware to the ground based propulsion system characterization presented in chapter 2. The electrical power is the product of the instantaneous voltage and current supplied to the speed controller. The same components used for ground based estimation are used here for monitoring the voltage and current. The electrical power is measured in flight using a combination current/voltage circuit and a Coulomb counter. The current sensor is the Allegro ACS758LCB-100B 100 A. The current output voltage and battery voltage is monitored through a precision resistor divider and onboard high speed 16 bit ADC. The Coulomb counter is an ST STC3100 which provides a check on the total energy consumed during a given maneuver. These chips are ground calibrated before flight using a precision power supply and motor setup. The sensors are sampled at 100 Hz during the flight-test maneuvers to allow the flight computer to generate real-time control updates.

The current and voltage sensors are placed upstream of the battery and immediately downstream of the speed controller/motor. This allows the battery efficiency to be decoupled from the power available propulsion model. The accuracy of the voltage and current measurements are identical to the ground based testing and are summarized in appendix 1.

5.5.3 Flight-test setup

Two methods for estimating the power required are presented in this section. The first is a level acceleration test, and the second is a constant airspeed, full power climb. These two methods produce equivalent results but require significantly different controller operation. In order to facilitate the most efficient use of flight-testing time, the power required methods were integrated into the general pattern flown for the *L/D* estimation presented in section 5.4.

For the level acceleration test, the climb leg of the *L/D* test is replaced with a full power level acceleration followed by a zoom climb to the altitude required for the constant airspeed glide. The climb is initiated after the aircraft reaches its maximum level airspeed of approximately 35 m/s. The relatively high power to weight ratio for the aircraft results in a total test time of approximately 10 seconds after full power is applied. The low inertia of the motor and propeller limit the spool-up time to approximately 100 ms after full throttle is commanded. After the test run, the motor is left at full power and a 2g pull up is initiated to a pitch attitude of 20 degrees. The maximum programmed altitude rarely exceeds 100 m, thereby requiring several seconds of climb time to the target altitude. The second power required method relies on a full power, constant airspeed climb. This test is similar to the constant airspeed glide phase of flight. For this testing, only slight modification to the flight computer is necessary. For these tests, the flight computer gates the saved data such that markers are present when the airspeed has settled to a relatively constant value.

Given the limited space in which the flights typically occur, an adaptive controller was designed to roughly predict and set the angle of attack for a required airspeed. This is especially critical for the level acceleration tests in which the high thrust-to-weight of the aircraft makes maintaining a constant altitude difficult. The angle of attack is determined by a first order approximation for the lift curve slope of the aircraft. The required angle of attack is given by

$$C_L = \frac{\partial C_L}{\partial \alpha}\alpha + \alpha_{L=0} = \frac{L}{\frac{1}{2}\rho V^2 A} \tag{5.10}$$

From this expression, a rough estimate of the required angle of attack, α, can be determined. In practice, this rough estimate of α is sufficient to prevent the majority of oscillations occurring after full power is added. If the angle of attack is approximately maintained during the first seconds after full throttle is applied, the airspeed and climb rate stabilize quickly. If no attempt is made to initially match the angle of attack, airspeed and climb rate oscillations persist longer as energy is transferred between the two states. This method of predictive control is loosely based on feed-forward techniques successfully applied to numerous dynamic systems.[38, 92, 93] Additionally, the autopilot pitch gains must be re-tuned for full power testing. This is caused by the additional control effectiveness afforded by the increased airflow over the stabilizers. In practice, the pitch damping is reduced to prevent the autopilot from overcompensating based on the pitch rate input.

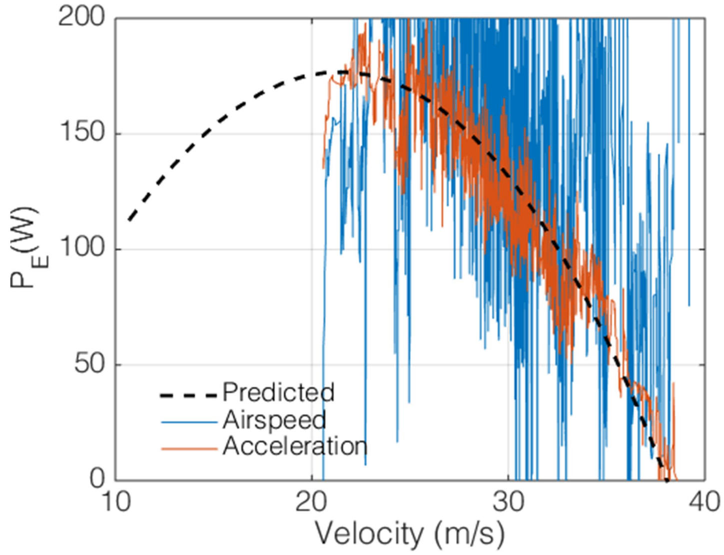
After the full power level acceleration or constant airspeed climb, the aircraft proceeds to the first waypoint (1 in Figure 68) and is setup for an unpowered glide as detailed in the *L/D* estimation section. As with the unpowered flight-testing, flight data is logged during the entirety of the flight for post flight reconstruction. An internal flag is set to mark the beginning and end of the level acceleration or constant airspeed climb legs.

5.5.4 Results

The power required estimation for the level acceleration and rate-of-climb tests are discussed in this section.

5.5.4.1 Level acceleration

The excess power is estimated during the level acceleration flights based on equation (5.7). For this test, the weight of the vehicle is constant during the acceleration run. The velocity information is provided by the air data system previously discussed in section 5.4. The derivative term can be approximated using the differentiated airspeed data, or the Earth frame acceleration data. The excess power estimated using both methods is shown in Figure 81 (top). From this figure, the differentiated airspeed shows much greater noise than the Earth frame acceleration. This is primarily the result of differentiating a high speed digital signal. This noise can be reduced by using a low pass filter (Figure 81 bottom) or by reducing the bandwidth of the sensor (Figure 81 bottom). The predicted excess power is given by the difference between the estimated power required (Figure 66) and the power available found using the system efficiency surface fit outlined in section 2.5.4.



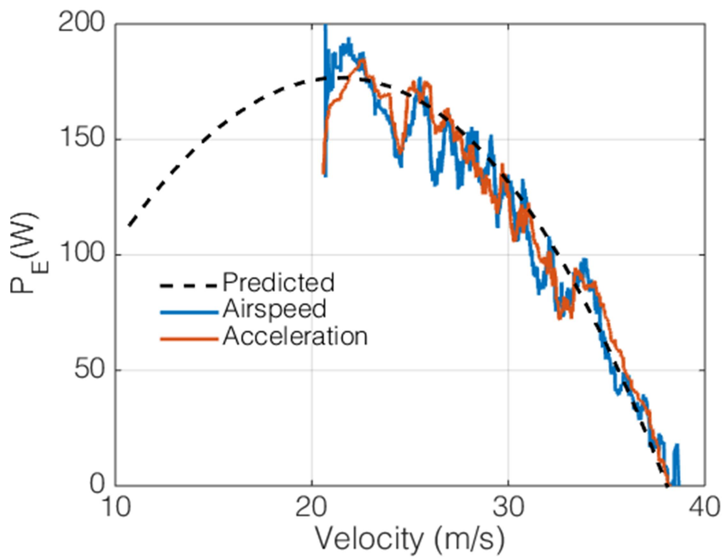


Figure 81: Excess power for level acceleration using differentiated airspeed and accelerometer signals. Unfiltered (top) and filtered (bottom).

Figure 82 shows the average excess power and 95% confidence interval generated using the Earth frame acceleration for three runs averaged together. From this figure, it is clear that the level acceleration captures the estimated excess power within the confidence bounds based on the INS sensor performance outlined in section 3.4.2. The errant data point at the lowest airspeed in Figure 82 is linked to difficulties in maintaining control when the vehicle is near stall. Extensive testing shows large variations in the excess power measured at low airspeeds due to excessive out of plane vehicle accelerations.

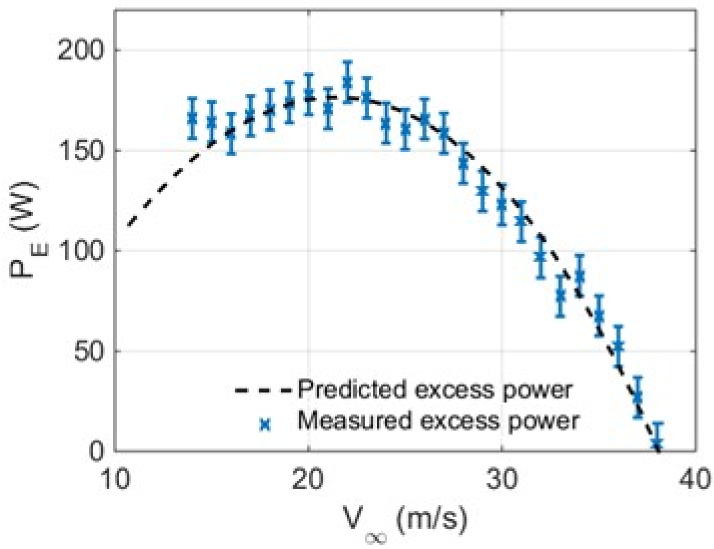


Figure 82: Excess power for level acceleration, mean and 1σ variation.

The Earth frame acceleration used to generate the excess power plots is fundamentally tied to the attitude estimates provided by the INS. If the body frame accelerations are used instead, the excess power follows the trend in Figure 83. In this figure, the divergence from the predicted excess power at low airspeeds is clearly visible. This is related to misalignment between the direction of the vehicles acceleration and the body's longitudinal axis (angle of attack).

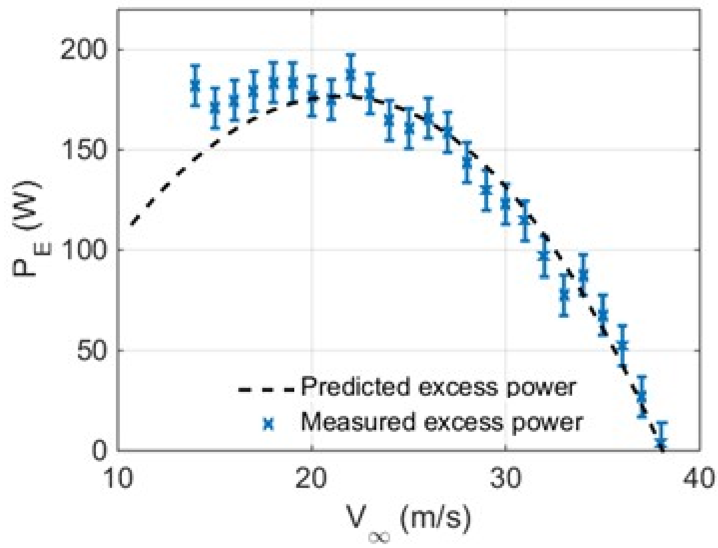


Figure 83: Excess power for level acceleration calculated with body referenced acceleration.

Based on the excess power data and the power available estimates provided by the propulsion model and measured electrical power, the power required can be determined from equation (5.7). The power required estimates, combined with its definition in equation (5.9) provides a means for estimating C_{D_O} and e. This is accomplished by multiplying both sides of the equation by the freestream velocity (shown in (5.11)). When the resulting power required times velocity is plotted against V^4, the relation for C_{D_O} and e is linear. An example of this plot is shown in Figure 84. From this plot, $1/e$ is the y-intercept of the best fit line (R^2 = 0.92), and C_{D_O} is the slope. The numerical results of this analysis are tabulated in Table 22. From this data, C_{D_O} and e are well captured by the level acceleration flight-test. Figure 84 also shows the uncertainty estimates for the level acceleration data. The airspeed sensor drives the majority of the uncertainty as the velocity is raised to the fourth power.

$$P_R = \frac{1}{2}\rho V^3 A C_{D_0} + \frac{W^2}{\frac{1}{2}\rho V A}\left(\frac{1}{\pi e A R}\right) \tag{5.11}$$

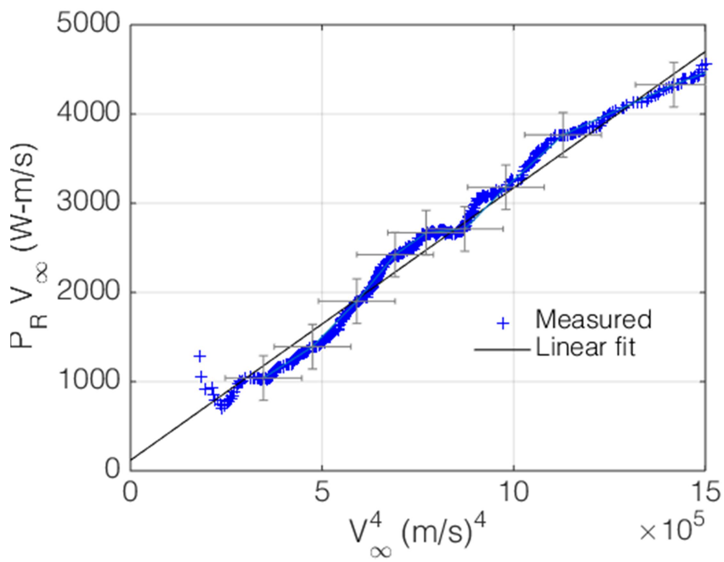


Figure 84: Power required estimate for C_{D_o} and e

Table 22: Analytic and experimental drag estimation.

Parameter	Analytic	Experimental Level acceleration	% Difference
C_{D_o}	0.0261	0.0220	15.7%
e	0.90	0.85	5.5%

5.5.4.2 Rate-of-climb

The rate-of-climb tests were carried out using the methods outlined in section 5.5.3. As in the *L/D* testing, plots of the airspeed and altitude for several test runs are shown in Figure 85 and Figure 86. In these figures, the airspeed is provided by the onboard pitot probe, and the altitude is the fused estimate provided by the INS. It should be noted that there is no significant difference between the altitude plots provided by the barometer and vertical INS channel. This is due in part to the high resolution and excellent repeatability of the barometer. The INS does provide enhanced control capabilities by smoothing out the high frequency oscillations induced by propulsion system vibrations. Remnants of the vibrations and pitch oscillations are still visible in the airspeed plot shown in Figure 85. As noted by Ostler, vibrations play a major role in the set point following performance of a low wing-loading sUAS.[38]

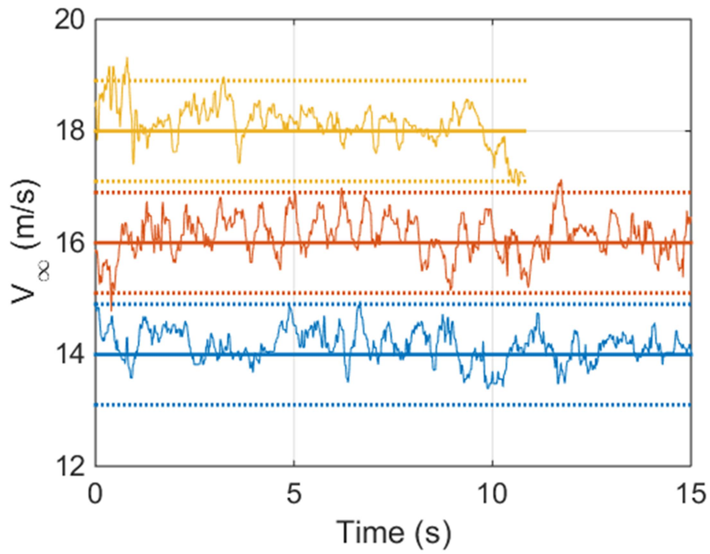


Figure 85: Velocity set point following performance for rate-of-climb tests. Dashed lines represent 3σ uncertainty bounds.

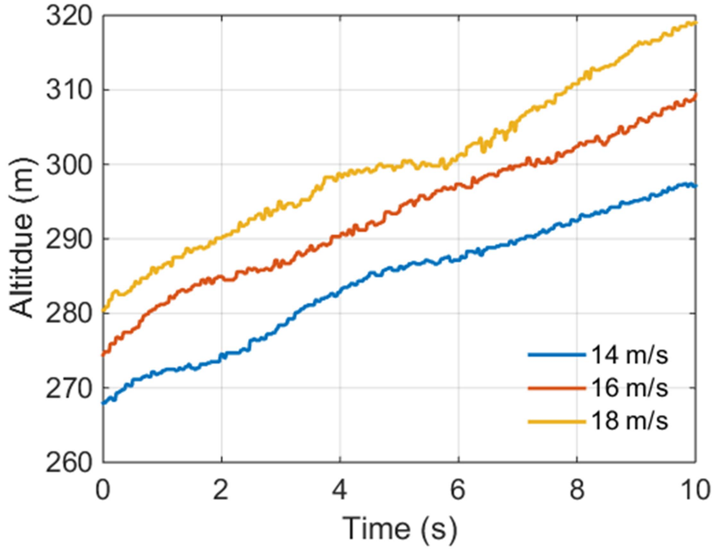


Figure 86: Altitude following performance for rate-of-climb tests.

The rate-of-climb performance is shown in Figure 87 based on the vehicle parameters outlined in section 5.2. From this data, the measured rate-of-climb is in close agreement with the estimated rate-of-climb. Based on this data, the measured and estimated excess power for the rate-of-climb and level acceleration is shown in Figure 88. The predicted excess power is given by the difference between the estimated power required (Figure 66) and the power available found using the system efficiency surface fit outlined in section 2.5.4. It should be noted that the power available corresponds to the full power performance outlined in section 2.5.1 as no attempt was made to limit or control the RPM of the motor (shown in Figure 89). The electrical power was monitored using the onboard voltage and current sensors (Figure 90). This allows for estimation of the propulsive efficiency necessary to establish the power available. Based on this analysis, the resulting C_{D_0} and e for the rate of climb tests are detailed in Table 23

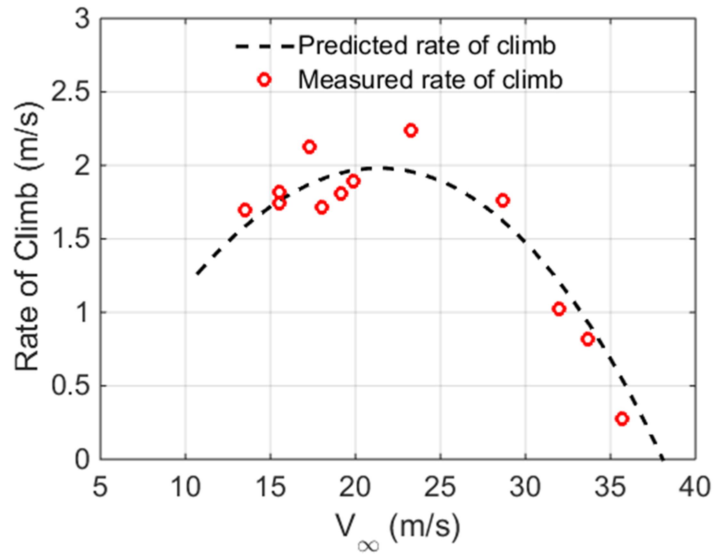


Figure 87: Rate-of-climb versus airspeed for constant airspeed climb tests

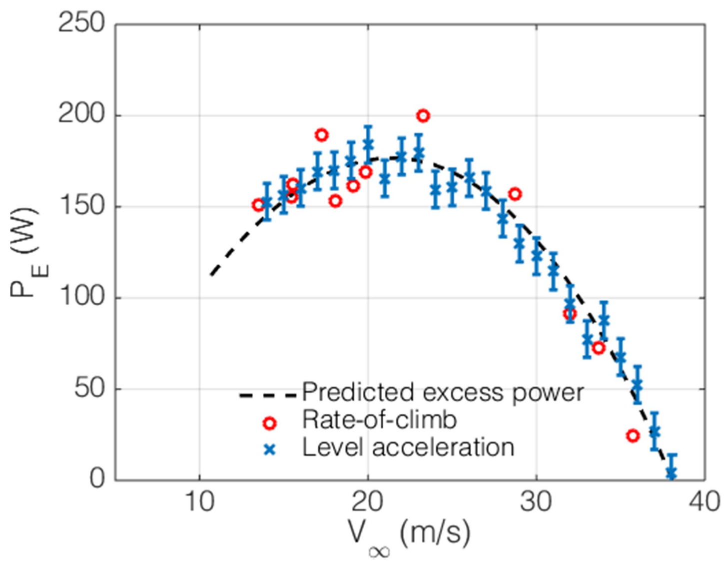


Figure 88: Excess power estimated by rate-of-climb test (red circles) and level acceleration (blue x).

Several salient features in the RPM plots and electrical power plots should be explained. The first is related to the operational nature of a fixed pitch propeller operated at full power. In this condition, the RPM increases with airspeed as the propeller unloads. This is visible in Figure 89 where the average RPM for the 18 m/s test is higher than the 14 m/s test. This trending is matched by a corresponding reduction in the electrical power required to drive the propeller, as outlined in section 2.5.1. Using the measured electrical power, RPM, and airspeed, the propulsive efficiency can be approximated using the system efficiency curves presented in section 2.5.4.

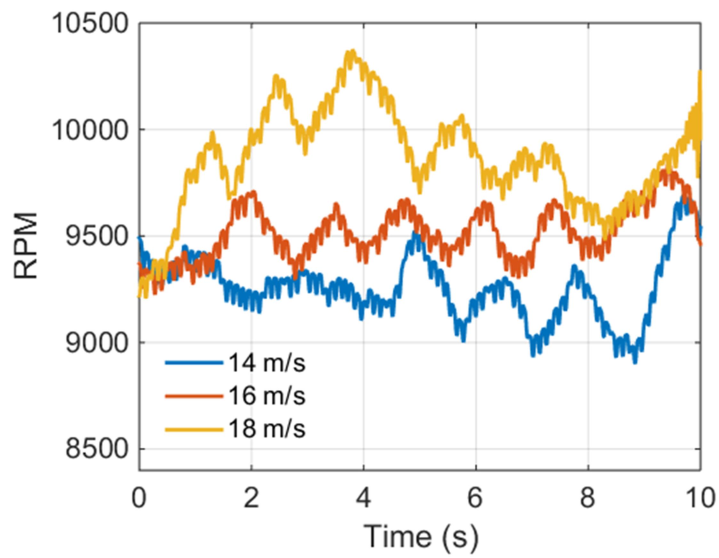


Figure 89: Full power RPM variation during rate-of-climb test.

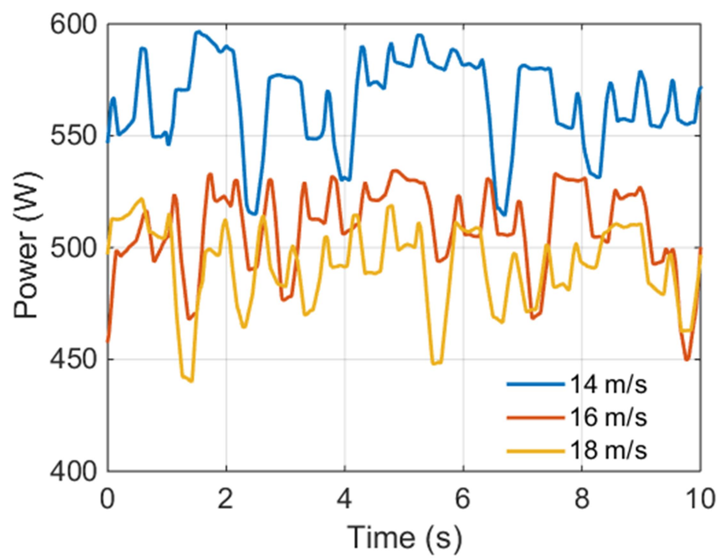


Figure 90: Full power electrical power variation during rate-of-climb test.

Table 23: Analytic and experimental drag estimation for rate of climb tests.

Parameter	Analytic	Experimental Level acceleration	Experimental Rate of climb
C_{D_o}	0.0261	0.0220	0.0205
e	0.90	0.85	0.86

Based on the predicted and estimated vehicle drag characteristics and Reynolds dependent propulsion models, an evaluation of the range and endurance of the vehicle can now be formulated.

5.6 Range and Endurance Estimation

Recent advances in electrical propulsion systems for small-scale aircraft necessitate revising the classical range and endurance equations to capture the unique operating characteristics of these vehicles. Unlike aircraft utilizing internal combustion engines for propulsion, electric aircraft maintain a constant weight during flight. In addition, the available power is related to the charge state of the battery. Quantifying these effects, and demonstrating the use of the Reynolds dependent propulsion model and flight-testing results is the goal of this section.

The power required for steady, level, unaccelerated flight is given as

$$P_R = DV_\infty, \tag{5.12}$$

where D is the drag acting on the vehicle, and V_∞ is the freestream velocity. From this expression, the drag term can be expanded using a first order approximation for an aircraft drag polar, given as[71]

$$D = qS\left(C_{D_o} + \frac{C_L^2}{\pi e AR}\right), \tag{5.13}$$

where q is the dynamic pressure given as $1/2\rho V_\infty^2$, C_{D_O} is the zero-lift drag coefficient, C_L is the lift coefficient, e is the wing aerodynamic efficiency factor, and AR is the wing aspect ratio. Equating the lift and weight of the vehicle, the lift coefficient is expressed as $C_L = W / qS$, where W is the vehicle weight. Substituting into (5.13) gives

$$D = qS\left(C_{D_O} + \frac{1}{\pi eAR}\left(\frac{W}{qS}\right)^2\right). \tag{5.14}$$

In general, the weight of the vehicle is a function of the empty weight and the capacity of the battery

$$W = W_{empty} + W_{battery} = W_{empty} + \frac{EgC}{\rho_{battery}}, \tag{5.15}$$

where the battery specific energy density, $\rho_{battery}$, is given by various manufacturers in units of W-hr/kg, E is the battery voltage, and C is the battery capacity in A-hr. For a given mission, C, will be constant. However, it also represents an initial design constraint that can be varied based on changing range or endurance requirements. Therefore, the trending between battery capacity and the resulting range and endurance calculations will be presented.

The battery capacity and performance directly affects the range and endurance of the vehicle. Therefore, a model relating these parameters to the power available is necessary. The power available relies on the propulsive system models presented in chapter 2. In general this model can be expressed as a function of the thrust required and Reynolds number of the propeller. Traub treated the propulsive efficiency as a constant with provisions for varying it in a piecewise fashion if adequate data is available.[94] The validity of this assumption will be assessed using the propulsion system model proposed in chapter 2. For the lithium-ion batteries used in this work, the capacity is also a function of the current draw rate. A model for this effect is given by a modified form of Peukert's equation,[95, 96]

$$t = \frac{R_t}{I^n}\left(\frac{C}{R_t}\right)^n, \tag{5.16}$$

where t is the total battery life in hours expressed as a function of the battery rating R_t (typically 1 hr), the discharge current, I, the total battery capacity C in A-hr, and the correlation factor n. For lithium-ion batteries, the correlation factor, n, is typically on the order of 1.3.[95] This expression can be used in conjunction with the battery voltage, E, and system efficiency, η_{total}, to determine the available power ,

$$P_A = \eta_{total} EI = \eta_{total} E \frac{C}{R_t}\left(\frac{R_t}{t}\right)^{1/n} . \tag{5.17}$$

In general, the system efficiency is a function of the aircraft velocity, and thrust power, TV_∞. For the baseline analysis, it is considered constant and equal to the full power average value outlined in chapter 2. This assumption is addressed in the forthcoming sections. The endurance is given by equating the power available with the power required, given by combining (5.14), (5.15), and (5.17)

$$\eta_{total} E \frac{C}{R_t}\left(\frac{R_t}{t}\right)^{1/n} = qV_\infty \left(C_{D_O} + \frac{1}{\pi eAR}\left(\frac{W_{empty} + \frac{EgC}{\rho_{battery}}}{q}\right)^2 \right) . \tag{5.18}$$

The endurance is found by solving for the time, t

$$t = R_t^{1-n}\left[\frac{\eta_{total} EC}{qV_\infty \left(C_{D_O} + \frac{1}{\pi eAR}\left(\frac{W_{empty} + \frac{EgC}{\rho_{battery}}}{q}\right)^2 \right)}\right]^n . \tag{5.19}$$

The range is simply the endurance multiplied by the flight speed, V_∞

$$R = V_\infty t = V_\infty R_t^{1-n}\left[\frac{\eta_{total}EC}{qV_\infty\left(C_{D_O}+\frac{1}{\pi eAR}\left(\frac{W_{empty}+\frac{EgC}{\rho_{battery}}}{q}\right)^2\right)}\right]^n. \tag{5.20}$$

It should be noted that the time and velocity associated with the best range and best endurance need not be the same.

5.6.1 Estimation Parameters

The remainder of this section will detail the predicted range and endurance using the methods outlined by Traub and Avanzini & Giulietti, and the effect of incorporating the variable system efficiency.[94, 96] The relevant aircraft parameters used in the range and endurance estimation are summarized in Table 24. Atmospheric properties are evaluated at standard temperature and pressure. The vehicle considered in this work is typically limited to flights below 400 ft agl. Therefore, the effects of altitude on the endurance and range are neglected for this work.

Table 24: Aircraft parameters for range and endurance estimation

Parameter	Value	Units
C_{D_O}	0.0261	-
e	0.95	-
AR	11.04	-
S	0.36	m^2
W_{empty}	2	kg
$\rho_{battery}$	150	W-hr/kg
R_t	1	hr
C	1-10	A-hr
n	1.3	-

5.6.2 Endurance Estimation

The endurance for the aircraft is given by (5.19) subject to the constants outlined in Table 24. From these relations, the endurance is computed for various battery capacities using a constant system efficiency, shown in Figure 91. From this figure, the maximum endurance increases quickly with increasing battery capacity. As the weight increases, the gains from adding additional battery capacity start to diminish. At higher gross weights, the maximum endurance velocity also increases. A common approximation for piston engine aircraft is that the maximum endurance occurs at the aircraft's minimum power required. This occurs for a propeller aircraft when the C_L term in (5.13) is replaced by $\sqrt{3C_{D_O}/k}$ where k is $1/\pi ARe$.[90] The analytic maximum endurance related to this approximation is shown in Figure 91 as red circles for each battery capacity.

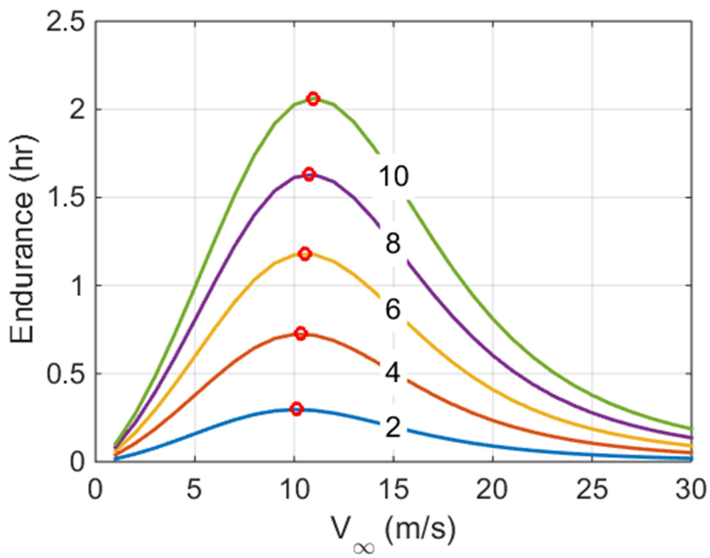


Figure 91: Endurance for the aircraft constraints listed in Table 24. Lines correspond to different battery capacities in A-hr and red circles represent analytic optimum.

A sensitivity analysis for the endurance estimation given in (5.19) is now developed. This is based on a single variable variance analysis to determine the effect of each parameter on endurance. For this analysis, each parameter in Table 15 is varied by ±10%. In practice, cross correlations exist between parameters. For a first order analysis, this is neglected to show the effect of each in isolation. The baseline configuration for comparison is chosen to be the maximum range condition for a 10 A-hr (C=10) battery pack. The resulting variation in endurance is listed in Table 25. From this table, the largest single driver of endurance is the empty weight of the vehicle. The importance of the wing aspect ratio and battery capacity are somewhat intuitive from a first principles perspective. Larger aspect ratios reduce the induced drag during cruise. Larger batteries lead to increased flight time as long as the gross weight is not exceeded. The effect of wing efficiency is largely overstated in that a ±10% variation is unlikely to occur without significant design change. Similarly, a large change in the battery discharge coefficient, n, is unlikely when using a given chemistry. Changing the wing area shows a smaller variation than the preceding coefficients and has a similar physical explanation to a change in aspect ratio. Lastly, changing the energy density of the battery and zero-lift drag coefficient gives the smallest overall impact on the endurance. As will be shown, the effect of C_{D_O} has a larger effect on the range due to a lower optimal cruising velocity. The low impact of C_{D_O} is related to the low cruising velocity of the aircraft in which the drag is dominated by the induced drag component (large effect of AR).

Table 25: Endurance change through parameter variation

Parameter	Nominal Value	Units	Percent difference (±%)
Endurance	2.05	hr	
C_{D_O}	0.0261	-	3.5/-3.0
e	0.95	-	-14.7/4.2
AR	11.04	-	-9.7/9.8
S	0.36	m^2	-6.5/6.5
W_{empty}	2	kg	18.0/-14.0
$\rho_{battery}$	150	W-hr/kg	-4.0/3.6
R_t	1	hr	-
C	10	A-hr	-9.3/9.1
n	1.3	-	-6.3/6.9

5.6.3 Range Estimation

The constant propulsive efficiency range of the aircraft is given by (5.20). For a propeller driven aircraft, the best range occurs when flying at the optimal L/D for the airframe. This corresponds to a C_L approximated by $\sqrt{C_{D_O}/k}$.[90] The range estimates provided by these relations are shown in Figure 92 for different battery capacities. The red circles correspond to the best range estimate using the $C_L = \sqrt{C_{D_O}/k}$

approximation. The discrepancy between the analytic optimum for a standard propeller driven aircraft and the electric aircraft is related to the discharge coefficient, *n*, appearing in the endurance and range equation.[96]

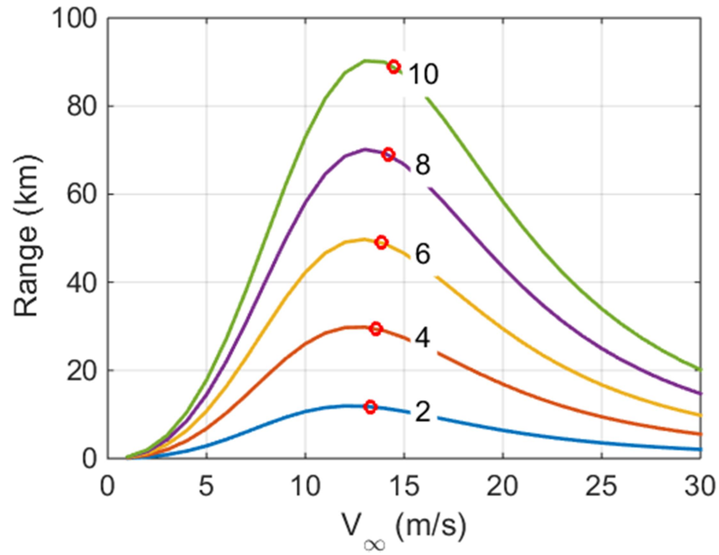


Figure 92: Range estimates aircraft constraints listed in Table 15. Lines correspond to different battery capacities in A-hr and red circles represent the analytic optimum.

As noted, the airspeed for maximum endurance and range differ. The maximum range airspeed occurs at the aircraft's maximum L/D, which is typically higher than the minimum power velocity. The results from a similar sensitivity analysis to the endurance test is given in Table 26. As in the endurance estimation, the empty weight has the largest overall impact on the resulting range. The trends between the range and endurance are similar for the remaining parameters. Of note is the reduced impact of the wing area. This is due to the overall decrease in angle of attack at the higher flight speeds required for extended range. Following this reasoning, C_{D_0} shows a larger overall influence given the higher flight speed.

Table 26: Endurance change through parameter variation

Parameter	Nominal Value	Units	Percent difference (±%)
Range	122.2	km	
C_{D_O}	0.026	-	6.3/-5.3
e	0.95	-	-10.9/3.0
AR	11.04	-	-7.3/7.1
S	0.36	m^2	-1.5/1.4
W_{empty}	2	kg	13.0/-10.6
$\rho_{battery}$	150	W-hr/kg	-3/2.6
R_t	1	hr	-
C	10	A-hr	-10.3/10.1
n	1.3	-	-4.8/5.0

5.6.4 Variable System Efficiency

The preceding analysis has treated the propulsive efficiency as a constant to identify the effect of each parameter on the endurance/range estimates. The propulsion testing presented in chapter 2 indicates that this assumption is likely problematic for the scale of the vehicles considered in this work. Two competing factors exist which must be taken into account to verify the validity of the range and endurance models. The first is the variation in system performance as a function of airspeed. For this analysis, the system efficiency is based on the high Reynolds number propeller testing data. This represents the best-case performance and limits the scale effects on the system efficiency. The second consideration includes the variation in efficiency as a function local propeller Reynolds number. As shown in chapter 2, the effect of propeller Reynolds number is of first-order importance in predicting the overall system efficiency. Additional efficiency reductions occur due to losses in the controlling electronics, which occur in tandem with low Reynolds propeller effects. To capture these effects, the propulsion model proposed in chapter 2 is used to estimate the system efficiency as a function of thrust power and airspeed.

An example of this Reynolds influence on system efficiency is given in Figure 93 for the APC 10x6E propeller. In this figure, the model predictions are represented by the ruled surface and the experimental data by the blue points. This variation in system efficiency versus freestream velocity and applied thrust power is due primarily to Reynolds scaling effects as outlined in previous chapters.

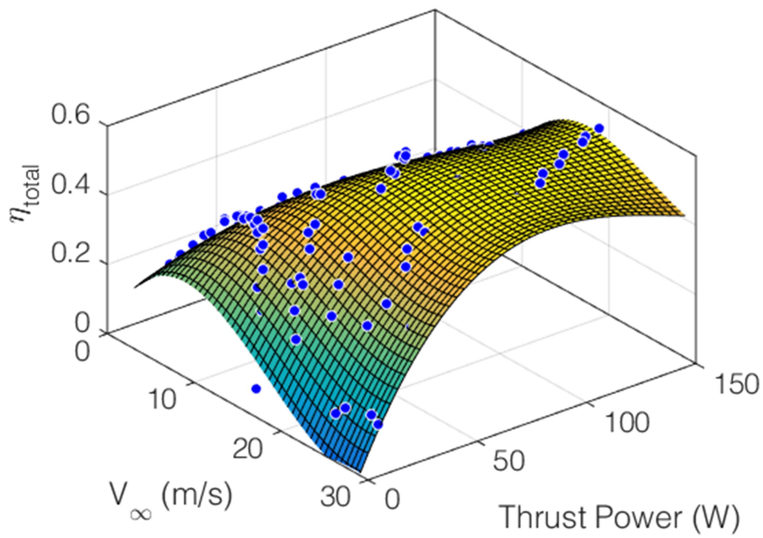


Figure 93: Propulsive efficiency for APC 10x6 versus freestream velocity and electrical power.

For a fixed airspeed and altitude, the power required is equal to the thrust power (TV_∞). Therefore, the system efficiency is formulated in terms of the airspeed and thrust power, as shown in Figure 93. It is evident from this figure that if the power required varies, the system efficiency can change dramatically. For the range and endurance optimization, the variation in system efficiency as a function of the power required should be included in the first-order analysis.

5.6.4.1 Variable Efficiency Endurance

This analysis shows the effect of including a variable system efficiency expressed as a singular function of the airspeed. The specific data used represents the full power testing case for the APC 10x6E outlined in chapter 2. The system efficiency as a function of airspeed is shown in Figure 94. From this figure, the system efficiency is interpolated based on the model data over the range of airspeeds and battery capacities outlined in Table 24. The resulting endurance predictions are shown in Figure 95 with the baseline data for the constant system efficiency case. Higher overall system efficiencies at higher airspeeds cause the shift in the endurance curve. For the constant efficiency case, the average propulsive efficiency was chosen. As evidenced in Figure 95, this dramatically affects the resulting performance predictions.

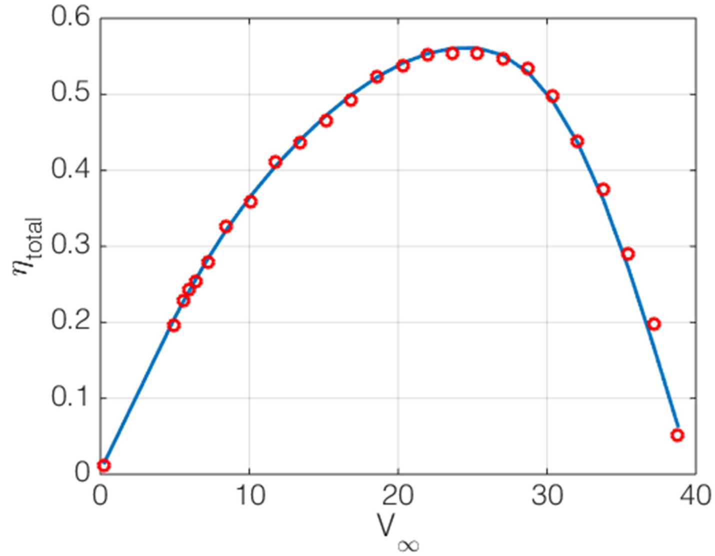


Figure 94: System efficiency versus airspeed for full power APC 10x6 testing. Blue line is model prediction and red circles are experimental data.

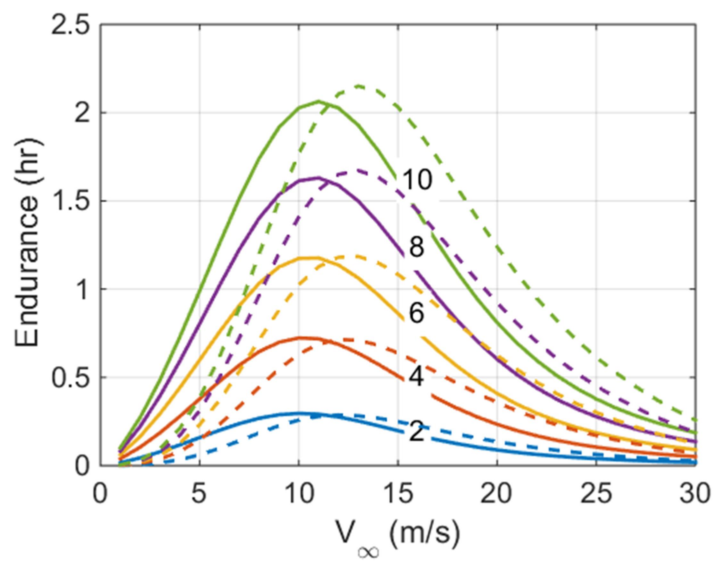


Figure 95: Endurance for variable propulsive efficiency model. Constant efficiency curves shown as solid lines and variable efficiency as dashed lines.

5.6.4.2 Variable Efficiency Range

The variable efficiency range (Figure 96) shows a similar trend to the endurance plot shown in Figure 95. In this case, the maximum range is significantly larger and occurs at a higher airspeed than the constant system efficiency case. From this analysis, the variation caused by the system efficiency term shows a greater overall impact on the resulting range and endurance predictions than any single variation in the model parameters.

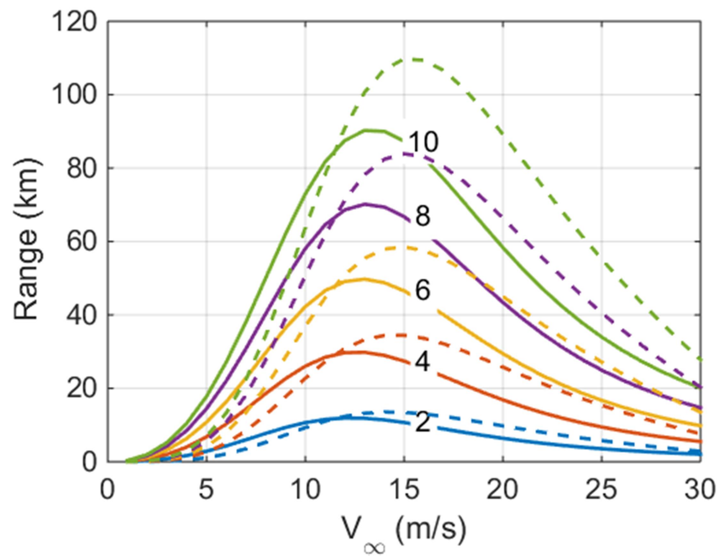


Figure 96: Range for variable propulsive efficiency model. Constant efficiency curves shown as solid lines and variable efficiency as dashed lines.

5.6.4.3 Reynolds-Based Efficiency, Endurance

The effect of including a variable system efficiency term on the endurance and range predictions was outlined in the previous section. Based on this data, the inclusion of a variable efficiency caused a shift in the magnitude of the range and endurance, as well as an increase in the associated optimum velocities. However, to complete this analysis, the effect of Reynolds number must also be taken into account. As illustrated in Figure 93 and Figure 94, the system efficiency shows a variation with both airspeed and thrust power. This implies that for any given airspeed, the system efficiency is directly related to the power required. To assess this effect on the endurance and range, the model surface outlined in Figure 93 is used to approximate the system efficiency as a function of airspeed and power required as outlined in (5.12) and (5.14). The resulting endurance for a variable battery capacity is shown in Figure 97. The dotted line in this figure represents the endurance prediction using the Reynolds-based propulsion model. The small increase in endurance compared to the variable efficiency is caused by higher propulsive efficiency at lower airspeeds. The additional degree of freedom in determining the system efficiency causes an overall slight increase in the maximum endurance, and a shift in optimum velocity towards the original constant parameter curve. From this analysis, and the previous variable model case, the overall effect on the aircraft's endurance is largely confined to shifting the optimum velocity for maximum endurance. The effect on the range is discussed in the next section.

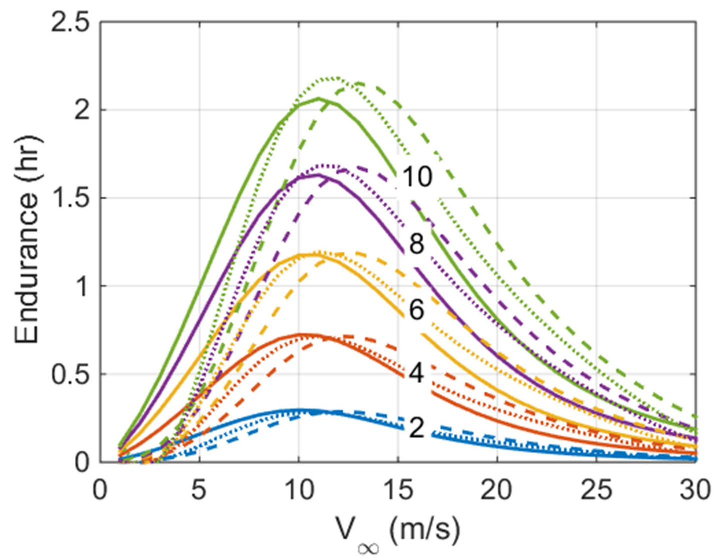


Figure 97: Endurance for Reynolds-based propulsive efficiency model. Constant efficiency curves shown as solid lines, variable efficiency as dashed lines, and Reynolds dependent as dotted lines.

5.6.4.4 Reynolds-Based Efficiency, Range

The Reynolds-based range predictions are shown as the dotted lines in Figure 98. From this figure, the Reynolds-based propulsion model shows a marked reduction in maximum range compared to the variable model (dashed line). The airspeed for maximum range is also reduced as in the endurance case. The reduction in range is due in part to a lower overall system efficiency resulting from the additional Reynolds-based interpolating parameter, shown in Figure 99. In this figure, the Reynolds-based efficiency estimates are lower at moderate flight velocities. For range optimization, a propeller should be chosen to maximize the efficiency at the necessary propeller Reynolds number, rather than simply choosing a propeller based on full power testing.

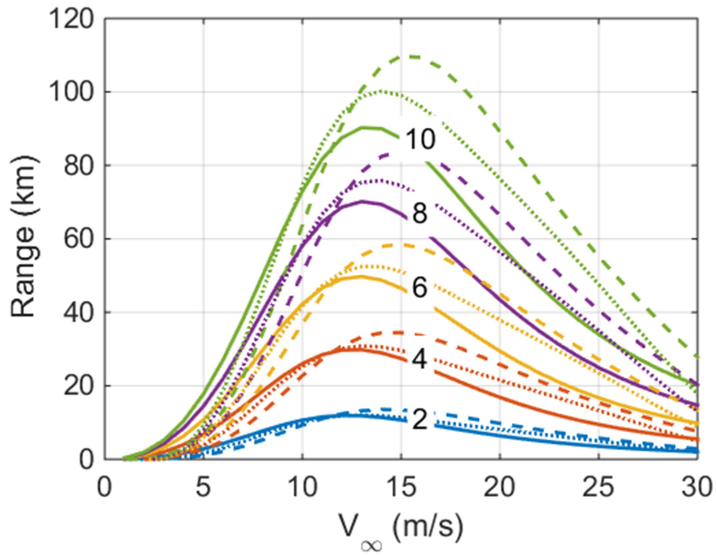


Figure 98: Range for Reynolds-based propulsive efficiency model. Constant efficiency curves shown as solid lines, variable efficiency as dashed lines, and Reynolds dependent as dotted lines.

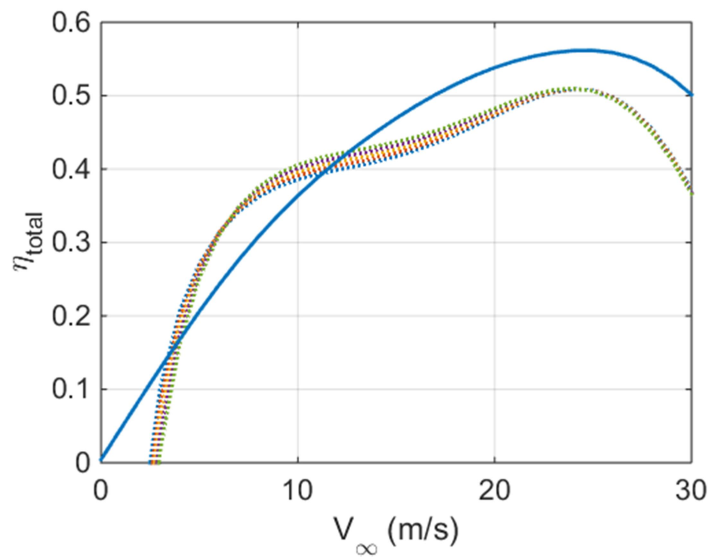


Figure 99: System efficiency versus airspeed for variable (solid) and Reynolds dependent models (dotted). Colors correspond to battery capacity.

5.7 Summary

This chapter details the flight-test performance analysis of a fixed-wing sUAS. Analytic performance predictions were presented for an electric powered sUAS named the Peregrine. A brief overview of the aircraft design and optimization was presented to highlight the parameter estimation validated through flight-testing. Based on this overview, flight-tests were performed to estimate the aircraft *L/D*, zero-lift drag coefficient, C_{D_0} and efficiency factor e.

Data reduction methods based on different sensor and INS system characteristics were outlined. Based on this analysis, the *L/D* estimates were shown to be sensitive to errors in the airspeed estimates and external wind gusts. A new method based on integration of the airspeed to determine the overall descent distance was proposed. This method provided increased accuracy as the wind speed increased. A GPS based method was also demonstrated, but ultimately proved inadequate for predicting the *L/D* in anything but the calmest flying conditions. The GPS data was augmented with wind estimates provided by the onboard INS, which reduced the overall uncertainty at higher wind speeds. The advantage of this method is that no connections and calibrations with the aircraft air data system are required.

The zero-lift drag coefficient, C_{D_0}, and efficiency factor e were estimated using a series of level accelerations and rate-of-climb tests. The variance in estimated C_{D_O} was on the order of 15% and 5.5% for e using the level acceleration test data. The variance in C_{D_O} increased to 22% for the rate-of-climb tests. This is due primarily to the larger variance in rate-of-climb data compared to the level acceleration data. Whereas the level accelerations supply data at the native data rate of the controller, the rate-of-climb provides information only at a set airspeed. This requires many more tests to populate the excess power curve. Along with an increased testing time, the rate-of-climb also requires a higher flight ceiling in order to capture the steady-state climb rate.

Finally, the drag polar from the aircraft is used in conjunction with the power available model presented in chapter 2 to estimate the range and endurance of the vehicle. The classical range and endurance equations were reformulated to match the characteristics of the electrically powered sUAS tested in this work. This included allowances for a constant weight and variable capacity battery. When combined with the Reynolds-based power available model, significant deviations from the baseline range and endurance estimates were noted. This demonstrates the necessity of using a high-fidelity power available model for performance estimation for sUAS applications.

Chapter 6: Conclusion

A framework for analyzing sUAS aircraft was developed. This effort focused on generating models for the three sub-systems common to sUAS aircraft, the propulsion system, navigation system, and airframe. This work tied together these three components utilizing features unique to the scale size of the vehicle to estimate the effect on the range and endurance. Insights into each of the three components are now discussed.

The propulsion system for the sUAS considered in this work is comprised of a propeller, electric motor, speed controller and battery as presented in chapter 2. While somewhat simple in construction, the lack of published performance data necessitated a detailed investigation into the performance of the system. This included an analytic and experimental investigation to assess the power and efficiency of each component in the system as a function of advance ratio. The proposed analytic model is based on the BEM methodology for predicting the thrust and power requirements for a fixed pitch propeller. The basic BEM model was augmented with a Reynolds appropriate aerodynamic database and three correction factors to account for limitations implicit in the basic model. These include, tip losses, Mach effects, and spanwise flow components.

Model predictions for three different propellers common to sUAS were analyzed with the proposed BEM model. From these predictions, the following trends were observed. First, the thrust and torque coefficients for the full power testing cases aligned closely with data published in the literature for similar propellers. Experimental testing of the propellers at full power across a range of advance ratios further reinforced these trends. Secondly, when operated at lower angular speeds, the performance of each blade was markedly reduced. Both BEM predictions and experimental measurements indicate the likely cause being a reduction in aerodynamic efficiency related to a reduction in overall operational Reynolds number. This claim is supported by other research efforts aimed at measuring the performance of other small-scale propellers.

The work presented here provides a methodical investigation into models necessary to predict this reduction in performance. The efficacy of these models in predicting the performance reduction at low Reynolds numbers was demonstrated by comparison with experimental data. During the experimental measurements of the propellers characteristics, data was also gathered to quantify the performance of the speed controller. This allowed for measurements of the motor RPM, voltage, and current to be mapped to the propulsive efficiency and thrust power. These measurements were critical in estimating the vehicle performance as presented in the subsequent chapters. Before this could take place, the second system , the navigation system needed to be characterized in order to estimate the error in flight-test data used for vehicle performance estimation.

The navigation computer used for vehicle state estimation and control consisted of an IMU and other aiding sensors such as a GPS, barometer, and magnetometer, outlined in chapter 3. Combined, these sensors formed an INS that allowed for precise vehicle control, and the attitude/velocity/position estimates required for vehicle performance estimation. The navigation equations were presented with a focus on tailoring them for sUAS use. INSs have been used in a myriad of applications, but scaling them down to the size required by sUAS presents special challenges. These challenges include decreased sensor capabilities, and computational limitations imposed by the onboard processor. To estimate the capabilities of the onboard INS, a systematic approach to analyzing the sources of error in the final state estimates was adopted. This started with an analysis of the onboard IMU sensors, the gyroscope and accelerometer. Sensor error models were proposed which capture the error sources common to MEMs sensors. Calibration methods were then developed to provide the alignment data necessary to minimize error growth in the inertial solution. These methods centered on optimization schemes that limit the need for high precision calibration devices. Following this work, the calibration methods and capabilities of the aiding sensors were detailed. A novel filter was developed for the magnetometer to alleviate interference induced by onboard EMI sources such as the motor and telemetry devices. Finally, the sensor fusion algorithm for correcting the onboard state estimates using the data provided by the aiding sensors was detailed. The mechanization of the filter is introduced with the tuning parameters necessary for accurate and stable operation. Quasi-static and HIL testing, discussed next, determined the data required for tuning the filters.

Tuning of the sensor fusion algorithm required error estimation from the sensors when operated in a quasi-static environment, presented in chapter 4. Therefore, testing was introduced which allowed for quantification of the attitude, velocity, and position error. This testing used a precision rate table to provide feedback for assessment of the calibration and algorithmic accuracy of the navigation equations presented. Error estimates furnished by the quasi-static testing were used to assess the in-situ performance of the INS using a HIL simulator. The interface and capabilities of the simulator were addressed and an interface program was created to provide inertial quality data from a soft real-time simulator. The software interface allowed for assessment of the sensor induced state errors. Based on this analysis, the sensor noise characteristics were determined to be the prime contributors to the overall uncertainty in state estimation. While this information is not new, the magnitude of the state errors and quality of the calibration are key components necessary for the flight-test performance evaluation presented next.

The flight-test performance evaluation of the Peregrine fixed-wing sUAS is presented in chapter 5. This assessment utilizes the propulsion system and state error models to validate the aircraft drag polar, zero-lift drag coefficient, and span efficiency. The flight-tests required to provide this data include, power-off constant airspeed glides, full-power constant airspeed climbs, and constant altitude full power accelerations. The methodology and required capabilities for performing these maneuvers are presented in their respective sections.

Based on the outlined tests, the design estimates of the *L/D* were in close agreement with the flight-test measurements. This assessment was made using three different methods for assessing the *L/D* ratio. A method uniquely suited for sUAS use was proposed in which the average airspeed was replaced with the integrated displacement along the descent path. This method provided better results when the vehicle was operated in windy/gusty conditions. The inadequacy of a purely GPS based solution was demonstrated. However, when combined with the wind data provided by the INS, the resulting uncertainty decreased in moderate wind conditions. This method allows for assessment of the *L/D* for a vehicle without an air data system.

Full power accelerations and rate-of-climb tests were performed to assess the zero-lift drag coefficient and span efficiency factors. The details of the flight testing methods tailored to sUAS are presented in their respective sections. The two methods provided similar results in estimating C_{D_0} and e . However, the full power acceleration was more effective at providing a better understanding of the power required as it provided data at all airspeeds from the stall to maximum velocity. In contrast, the rate-of-climb tests required one climb for each airspeed set point. This limitation combined with decreasing battery performance over long flights makes the analysis more intensive. If however, reasonable quality inertial sensors are available, the full power test requires one flight-test lasting approximately 10-15 seconds. In this time, the entire flight envelope is measured and an accurate determination of C_{D_0} and e can be found. However, a fundamental limitation of this method is the requirement for accurate attitude estimates to determine the linear acceleration in the locally level frame of reference. The effect of neglecting this transformation was outlined and shown to significantly skew the data at low airspeeds.

Finally, the performance estimates were combined with the Reynolds dependent propulsion model to demonstrate their effect on range and endurance estimates for sUAS. With the performance estimates and power available models, a battery model was introduced to capture the performance of the entire power system. The effect of increasing battery capacity was demonstrated for the case of constant propulsive efficiency and a 1-D propulsive efficiency curve. The large difference between these estimates demonstrated the need for a more complete propulsion model, provided in this case by the Reynolds dependent model outlined in chapter 2. Incorporation of this model into the range and endurance estimates provided a significant shift (~10-20%) to the estimates furnished using the more simplistic models. This highlights the necessity of accurate vehicle performance estimates and high-fidelity component models for both sUAS design and flight-test evaluation.

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Appendix A: Drag buildup

The drag buildup for the Peregrine aircraft is presented. The data presented herein is taken from Hoerner and Raymer.[90, 91] Several assumptions are made in the analysis. First, the motor can be stopped in flight and the propeller retracted by aerodynamic forces. It is assumed that the folded and stopped propeller acts as a minor protuberance and therefore does not contribute significantly to the overall vehicle drag. Secondly, the control surfaces and wing junction are sealed. This prevents spillage around the control surfaces which can result in a significant drag penalty. Thirdly, no cooling drag or thrust is considered. The cooling inlet surrounding the motor and exhaust ports have a net area ratio of 1/1.5, and are small relative to the overall vehicle wetted area. No internal baffles are used for cooling. Therefore, any small thrust and drag components are assumed to act in opposition and cancel.

The geometric parameters necessary for estimating the total zero-lift drag of the vehicle are given in Table 27. The total zero-lift drag coefficient is estimated using the wing area as a reference area. The resulting expression for the zero-lift drag is therefore the combination of the wing, fuselage, stabilizer, and miscellaneous components given as

$$C_{D_O} = C_{D_{O,wing}} + \frac{C_{D_{O,fuse}} S_{fuse} + C_{D_{O,stab}} S_{stab} + C_{D_{O,misc}} S_{misc}}{S_{wing}} = 0.0261 \tag{5.21}$$

where S is the reference area for the noted component, and $C_{D_{O,x}}$ is the zero-lift component drag. The estimated zero-lift drag coefficient for the Peregrine is 0.0261. The span efficiency factor, e, is 0.90.

Table 27: Aircraft geometric parameters.

Parameter	Value
Fuselage wetted area (S_{wet})	0.035 m^2
Fuselage maximum cross sectional area (S_{fuse})	0.016 m^2
Fuselage length (L)	1.5 m
Fuselage equivalent diameter (D)	0.127 m
Wing area (S_{wing})	0.36 m^2
Wing taper ratio	2.67
Wing aspect ratio	11.04
Wing tip washout	-1.24 deg
Stabilizer area (S_{stab})	0.08 m^2
Stabilizer wetted area	0.085 m^2
Stabilizer aspect ratio	4.8
Stabilizer washout	0 deg
Miscellaneous drag reference area	0.002 m

A.1 Fuselage

The fuselage drag is determined based on skin friction, fineness ratio, and wetted area ratio given as

$$C_{D_{O,fuse}} = C_f f_{LD} \frac{S_{wet}}{S_{fuse}} = 0.0153 \tag{5.22}$$

where

$$f_{LD} = 1 + \frac{60}{\left(L/D\right)^3} + 0.0025\left(\frac{L}{D}\right) = 1.06 \tag{5.23}$$

and L and D are the fuselage length and diameters respectively. For the Peregrine, the step between the nosecone and cooling scoop, and retracted propeller protrusion is assumed to trip the boundary layer over the remainder of the fuselage. Therefore, the skin friction coefficient, C_f is given as

$$C_{f,t} = \frac{0.455}{\left[\log_{10}\left(\mathrm{Re}\right)\right]^{2.58}} = 0.0041\ . \tag{5.24}$$

In this expression, the Reynolds number is estimated based on the total length of the vehicle, the cruise velocity of 17 m/s and standard temperature and pressure.

A.2 Wing and Stabilizer Surfaces

The zero-lift wing and stabilizer drag is estimated using the Matlab and XFOIL panel code. The wing uses the SD7043 and the stabilizer uses the E168 airfoil sections.[88] The relevant parameters are given in Table 27.

$$C_{D_{O,wing}} = 0.018;\ C_{D_{O,stab}} = 0.012 \tag{5.25}$$

A.3 Miscellaneous Drag

The Peregrine is a relatively clean airframe, with few protuberances and a molded composite skin. The two main parasitic drag components come from two exposed antennas for the GPS sensor and telemetry transceiver. The total drag from each component is estimated from Raymer and Hoerner[90, 91], given as

$$C_{D_{O,misc}} = \left(2\right)0.1 \tag{5.26}$$

where the (2) represents the total number of antennas. The total reference area is given in Table 27.

Appendix B: Power Available Uncertainty

B.1 Sensor Uncertainty

This section details the determination of the bias and precision uncertainty terms for each measurand. The resulting uncertainties are given in Table 28. Total uncertainty is formulated as a root-sum-squared of the bias and precision terms.

Table 28: Uncertainty for propulsion testing paper

Quantity	Units	Bias uncertainty, σ_b	Precision uncertainty, σ_p	Total uncertainty
ΔP	Pa	24.91 Pa	0.725 Pa	24.92 Pa
T_{total}	C	1°C	0.06° C	1.002° C
P_{static}	Pa	271.14 Pa	0.725 Pa	271.15 Pa
n	rev/s	-	-	-
D	m	$2.54e^{-5}$m	$1.31e^{-4}$ m	$1.33e^{-4}$ m
$Thrust\left(T\right)$	N	1.1 N	$3.22e^{-3}$ N	1.1 N
$Torque\left(Q\right)$	N-m	0.028 Nm	$8.22e^{-5}$ N-m	0.028 N-m
E	V	0.01V	0.014 V	0.017 V
I	A	0.6A	0.1 A	0.61 A

B.1.1 Differential pressure transducer

The differential pressure transducer is an Omega PX2650-10D5V. As per the data sheet, the bias uncertainty is quoted as 1% best fit straight line (BSFL) at constant temperature. This includes linearity, repeatability, and hysteresis. It is temperature compensated from -10 to 65 °C. For a 10” H_2O sensor, this gives an effective bias error of σ_b = 24.91 Pa.

The sensor output is an analog signal between 0-5V. This is digitized by a Ni-6009 ADC with 13 bits of resolution. This gives an effective resolution of

$$2^{13} = 8192 bits \rightarrow 5V / 8192 bits = 0.6 mV / bit$$
$$\left(249.1 \frac{Pa}{in - H_2O}\right)\left(\frac{10}{5} \frac{in - H_2O}{V}\right)\left(0.6 \frac{mV}{bit}\right) = 0.3 Pa \tag{5.27}$$

In practice, the ADC cannot fully resolve the last bit, thereby decreasing the effective resolution by a factor of 2. This gives a final resolution of $0.6 Pa$. The 1σ noise is 1.2 Pa from a 10 minute tunnel-off test. Thirteen data sets were used to estimate the uncertainty in this term. The t-method gives the 1σ data from each day,

$$\begin{aligned} &n = 13, v = 12 \\ &t = 2.179 \\ &\sigma_p = \pm t \frac{\sigma}{\sqrt{n}} = \pm 2.179 \frac{1.2 Pa}{\sqrt{13}} = \pm 0.725 Pa \end{aligned} \tag{5.28}$$

B.1.2 Temperature Transducer

A K-type thermocouple and a NI-9213 thermocouple amplifier was used to measure the test section temperature. The thermocouple is placed in close proximity to the testing apparatus and oriented to measure the total pressure. From the NI-9213 datasheet, the total accuracy is 1.0 °C over the operating range. This includes all scale factor, linearity, and cold-junction bias terms. Temperature was logged with differential and static pressure during the morning power up time. 10 minute logs were created for each day. From this data, and a similar t-method analysis is used.

$$\begin{aligned} &n = 13, v = 12 \\ &t = 2.179 \\ &\sigma_p = \pm t \frac{\sigma}{\sqrt{n}} = \pm 2.179 \frac{0.1C}{\sqrt{13}} = \pm 0.06C \end{aligned} \tag{5.29}$$

B.1.3 Static Pressure transducer

The static transducer uses the same differential transducer as the tunnel velocity sensor. However, in this case, one line is vented to the room. Therefore, the bias and uncertainty terms for the transducer are equivalent to the tunnel velocity sensor. To convert from a room differential static measurement to an actual static pressure, an additional weather monitoring station was used. From the datasheet, this device is accurate to within 270Pa. This bias was combined with the static pressure sensor bias using the root-sum-squared technique. No precision information was found for the desktop weather unit. It does display to 1Pa, though the validity of this is unknown.

B.1.4 Rotational velocity

The rotational velocity of the propeller is monitored by a digital back-EMF commutation pulse. This pulse is digital in nature and limited by the time resolution of the onboard gain amplifier and digital counter resolution. A back-EMF circuit, Schmitt gain amplifier, and digital counter were used to measure the RPM signal. The Schmitt amplifier specifications give it an effective bandwidth of 1GHz, but was only tested to 100 MHz. This is well above the maximum commutation frequency of 1.5kHz. The timing resolution of the digital counter is 50 ns. The counter is interrupt driven with an interrupt delay of 4ns.

The only major source of error is the delay error. At nominal RPM values of say 5,000, the signal transitions 1,300 times per second. Between these transitions, the RPM is considered constant. This is equivalent to a zero-order-hold. The impact on the resulting uncertainty is unknown, but is likely much lower than other error sources.

B.1.5 Diameter

The diameter of the propeller was measured 20 times to estimate the precision error. This estimate results in a standard deviation of 11 mil (2.79e-4m). A sample size of 20 gives the following uncertainty

$$\begin{aligned} &n = 20, v = 19 \\ &t = 2.093 \\ &\sigma_p = \pm t\frac{\sigma}{\sqrt{n}} = \pm 2.093\frac{2.79e^{-4}m}{\sqrt{20}} = \pm 1.31e^{-4}m \end{aligned} \tag{5.30}$$

where the resulting uncertainty is approximately ±5.15 mil. The bias is determined by the resolution of the calipers and set at 1 mil or 2.54e-5 m.

B.1.6 Thrust and Torque

Thrust and torque were measured directly by a Futek MBA500 multi-axis load cell. The load cell specifications give an error band of ±0.5% FS. The full-scale range of the thrust sensor is 222 N and the torque sensor is 5.65 Nm. This gives a bias error of 1.1N for the thrust sensor, and 0.028 Nm for the torque sensor. The sensors were digitized using the same NI-6009 ADC as the pressure sensors. A gain of 10 was used in the amplifier stage to increase the working resolution

$$\begin{aligned} &2^{13} = 8192 bits \rightarrow 5V / 8192 bits = 0.6 mV / bit \\ &\left(\frac{22.2}{5}\frac{N}{V}\right)\left(0.6\frac{mV}{bit}\right) = 2.64e^{-3}\frac{N}{bit} \end{aligned} \tag{5.31}$$

$$\begin{aligned} &2^{13} = 8192 bits \rightarrow 5V / 8192 bits = 0.6 mV / bit \\ &\left(\frac{0.565}{5}\frac{Nm}{V}\right)\left(0.6\frac{mV}{bit}\right) = 6.78e^{-5}\frac{Nm}{bit} \end{aligned} \tag{5.32}$$

As in the pressure sensor case, the effective resolution needs to be decreased by a factor of two to account for electrical noise in the system. Precision is based on logged noise and calculated the same way as the pressure sensors

$$\begin{aligned} &n = 13, v = 12 \\ &t = 2.179 \\ &\sigma_p = \pm t\frac{\sigma}{\sqrt{n}} = \pm 2.179\frac{5.328^{-3} N}{\sqrt{13}} = \pm 3.22e^{-3} N \end{aligned} \tag{5.33}$$

$$\begin{aligned} &n = 13, v = 12 \\ &t = 2.179 \\ &\sigma_p = \pm t\frac{\sigma}{\sqrt{n}} = \pm 2.179\frac{1.36^{-4} Nm}{\sqrt{13}} = \pm 8.22e^{-5} Nm \end{aligned} \tag{5.34}$$

B.1.7 Voltage and Current

Voltage was measured using a buffered voltage divider and impedance matching tracker circuit. In effect, it matches, scales, and isolates the voltage readings from the propulsion system. (Think isolated multi-meter that scales the output to match the 5V range of the NI-6009). The full-scale accuracy of the tracker circuit is 0.1% FS. For this particular application, the full-scale range is 10V. This gives an accuracy of 10mV. The accuracy of this circuit was verified using a function generator and oscilloscope. Its bandwidth is above 10MHz which limits aliasing.

Current was monitored using a similar buffer circuit and an Allegro low-loss current sensor. This sensor uses a precision magnetometer and Ferrite core to determine the current magnitude and direction. It has an effective bandwidth of 120kHz and an accuracy of 1.2% FS. The particular sensor used in this testing has a FS range of 50A, giving an overall accuracy of 0.6A.

No specific tests were done to estimate the precision error outside of monitoring the noise characteristics of the sensor. From this data, the 1σ variation is ±14mV for the voltage sensor, and ±0.1A for the current sensor.

B.2 Uncertainty plots

The uncertainty estimates generated in the previous section were used with a Monte-Carlo method to determine the total uncertainty in the propulsion metrics. Through this analysis, the largest error sources stem from the airspeed and torque sensors. These effects will be explained in the following sections.

B.2.1 Thrust coefficient versus advance ratio

The thrust coefficient is a function of the measured thrust, density, rotation rate, and diameter. J is a function of the airspeed (density, static and dynamic pressures, and temperature), rotation rate, and diameter.

$$C_T = \frac{T}{\rho n^2 D^4}, \tag{5.35}$$

The decrease in the C_T uncertainty at high advance ratios is due primarily to the increase in RPM (T decreases, n increases, T/n gets smaller). The uncertainty associated with density and diameter are small and do not change appreciably with increasing J.

The large variation in J at lower advance ratios is due primarily to the large influence of the static and dynamic pressure uncertainties. As the airspeed increases, this effect is somewhat diminished by the increasing RPM. The variation in *J* is common to all plots in the remainder of this document.

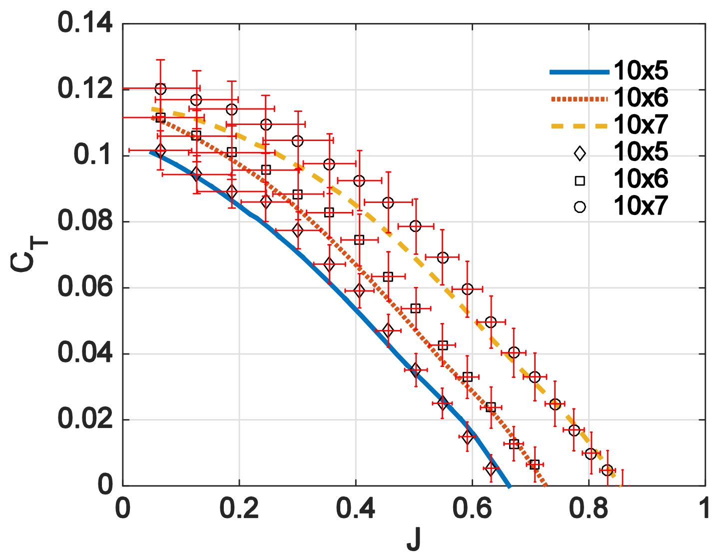

Figure 100: Thrust coefficient versus advance ratio

B.2.2 Torque coefficient versus advance ratio

The torque coefficient is formulated in a manner similar to the thrust coefficient. In this case, the trending and reasoning are equivalent. The decrease in torque uncertainty at high advance ratios is due to the increasing influence of the RPM.

$$C_Q = \frac{Q}{\rho n^2 D^5}, \tag{5.36}$$

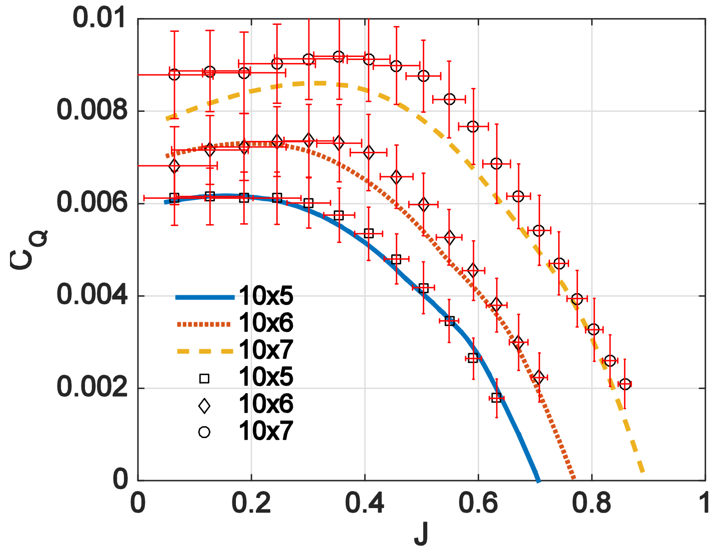

Figure 101: Torque coefficient versus advance ratio

B.2.3 Power coefficient versus advance ratio

The power coefficient is directly related to the moment coefficient and shares the same trending.

$$C_P = \frac{P}{\rho n^3 D^5} = \frac{Q\omega}{\rho n^3 D^5} = C_Q \frac{\omega}{n} = 2\pi C_Q, \tag{5.37}$$

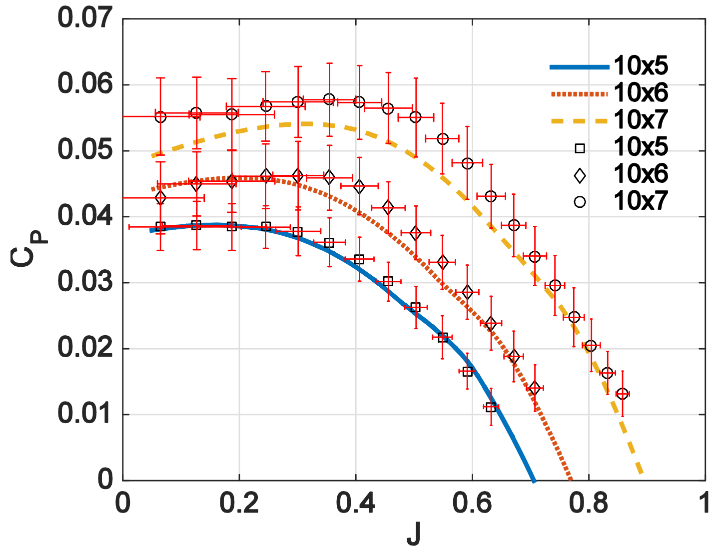

Figure 102: Power coefficient versus advance ratio

B.2.4 Propulsive efficiency versus advance ratio

The propulsive efficiency uncertainty at low advance ratios is dictated primarily by the uncertainty in the airspeed. As the advance ratio increases, the ratio of uncertainty in the thrust and moment is relatively constant. As in the thrust and moment coefficients, the increasing RPM at higher advance ratio decreases the total uncertainty.

$$\eta_{propulsive} = \frac{TV_\infty}{2\pi n Q} = \frac{1}{2\pi}\frac{C_T}{C_Q} J \,. \tag{5.38}$$

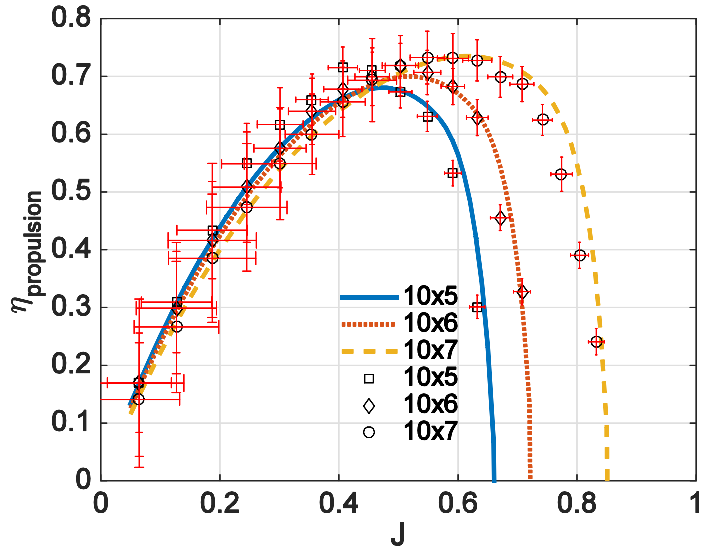

Figure 103: Propulsive efficiency versus advance ratio

B.2.5 Electrical efficiency versus advance ratio

The electrical efficiency is formulated as the ratio of motor power to electrical power, given as

$$\eta_{electrical} = \frac{2\pi nQ}{EI}, \tag{5.39}$$

The electrical efficiency is formulated somewhat differently in that the RPM signal appears in the numerator. This tends to amplify the total uncertainty at increasing advance ratios. Variation in the electrical uncertainty terms resulted in a very small overall impact. This is due to low uncertainty, and high signal to noise ratio with the electrical terms.

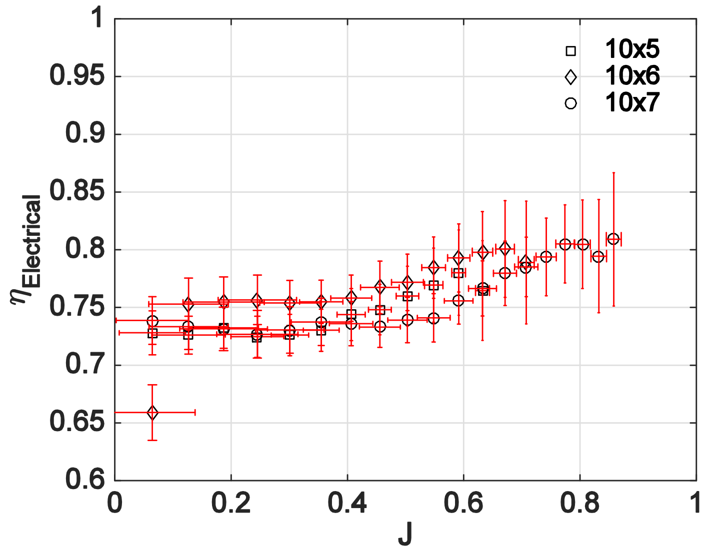

Figure 104: Electrical efficiency versus advance ratio

B.2.6 Total efficiency versus advance ratio

The total efficiency is given as

$$\eta_{total} = \frac{TV}{EI}, \tag{5.40}$$

The uncertainty in total efficiency is somewhat more complex as the ratio of uncertainties is not dominated by a single term. In this case, the total uncertainty increases as the advance ratio increases. If one term were chosen as the leader, it would likely be the airspeed.

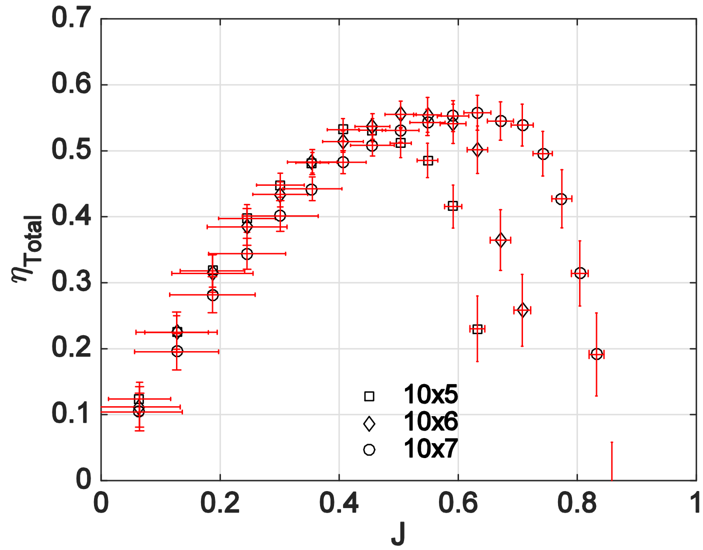

Figure 105: System efficiency versus advance ratio

B.3 Airfoil data

The NACA 4412 data from XFLR5 and Ostawari is shown in the following figure. Plots for RE = 2.5,5,7.5,10x^5. DataThief was used to extract the data for the NACA plot.

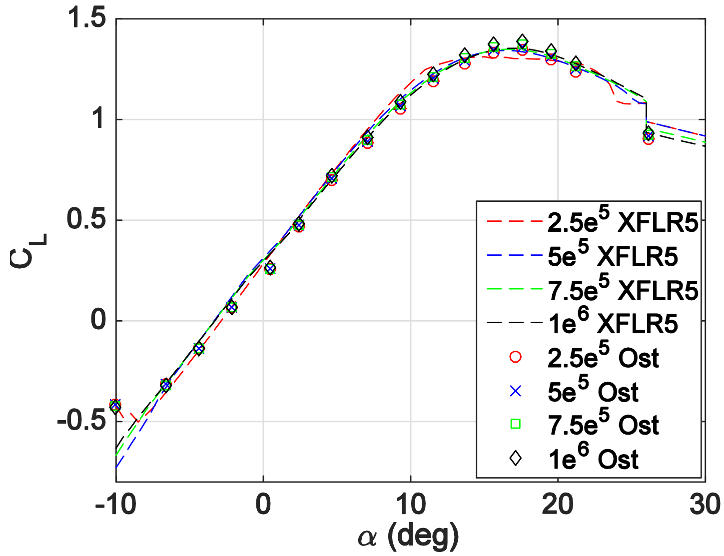

Figure 106: Lift coefficient versus angle of attack for NACA 4412